

Dealer Risk Limits and Currency Returns*

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Abstract

We leverage supervisory microdata to uncover the role of global banks' risk limits in driving exchange rate dynamics. Consistent with a model of currency intermediation under risk constraints, shocks to dealers' risk limits lead to price and quantity adjustments in the foreign exchange market. We show that dealers adjust their net position and increase the bid-ask spread in response to granularly identified limit shocks, leading to lower turnover and an adjustment in currency returns. These shocks exacerbate the effects of net currency demand on exchange rate movements, as predicted by theory, and trigger deviations from covered interest parity.

Keywords: exchange rates, currency returns, market making, risk constraints, financial intermediation

JEL Codes: F31, G15, G21

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Introduction

To what extent are exchange rates determined by the ability of market intermediaries to facilitate flows? An important recent theoretical literature highlights the role of financial intermediaries in foreign exchange markets (Gabaix and Maggiori, 2015; Itskhoki and Mukhin, 2021*a*). A common theme in this literature is that financial frictions, such as risk-bearing constraints of financial institutions, can affect exchange rate determination and dynamics. However, validating this theory empirically has proved elusive due to difficulties in observing the constraints of the specific actors. Validation is further encumbered by the fact that risk-bearing constraints are endogenous to the broader market conditions that may govern exchange rates.

Our paper provides empirical support for the role of financial intermediaries in driving exchange rate dynamics by focusing on the constraints of the currency trading desks of global banks. Exploiting novel supervisory dataset on risk limits at the desk level, we show that movements in market-maker constraints do in fact impact exchange rates when there is a currency flow to be mediated.¹ For identification, we construct idiosyncratic shocks to intermediary risk-bearing constraints following an approach similar to the granular instrumental variable (GIV) method in Gabaix and Koijen (2020).

Why focus on foreign exchange trading desks at global banks? These trading desks are the central market makers in the dealer-centric over-the-counter foreign exchange market (Chaboud, Rime and Sushko, 2023). Any cross-currency flow, which could be related to international trade in goods or financial assets, will ultimately be reflected in transactions with a few large currency dealers that intermediate supply and demand for foreign exchange. Regardless of the size of the residual risk held by dealers, shocks to their risk limits can undermine their currency mediation function, thereby impairing risk reallocation and triggering exchange rate adjustments.

We use this insight—that global banks’ trading desks are critical intermediaries of currency flows—to build a stylized economic model that guides our empirical analysis. We nest a monopolistically competitive dealer market that intermediates currency flows between net demand from the real sector and net supply from financiers into a Gabaix and Maggiori (2015) model. Dealers charge bid–ask spread and earn currency returns on their net open positions, but such risky positions are subject to heterogeneous risk limits. The equilibrium analysis shows that an exogenous tightening in risk limits—our observable empirical object

¹These risk limits refer to the risk-bearing constraints of trading desks imposed by internal managers for both risk-management and regulatory purposes. From a legal standpoint, risk-bearing constraints “define the amount of risk and the positions that a trading desk is permitted to take at a point in time, as defined by the banking entity for a specific trading desk” (17 CFR Appendix A to Part 255).

and core contribution of this paper—leads to higher exchange rate volatility, a reduction in dealers’ net positions, and an increase in bid–ask spreads.

The size and sign of the exchange rate effect intuitively depends on the shift in aggregate demand for dollars as well as dealers’ idiosyncratic positions. Indeed, limit shocks not only lead to a currency demand shift through adjustments in dealer positions, but, importantly, they also amplify the exchange rate effects of net-dollar demand shocks that originate either outside or inside the dealer sector.

We exploit two novel, supervisory micro datasets to construct the limit shocks and trace their impact on exchange rates.² First, we use the FR VV-1 data, collected by the Federal Reserve, which contain risk limits and usage of the FX trading desks of all large US banks. To understand the effects of these limits on individual currencies, we also leverage the supervisory FR Y-14Q data, schedule F, which contain quarterly information on net currency positions of the trading desks of large US banks, broken down by currency.³ We are thus able to tackle the question of how shocks to these risk limits impact currency returns via financial intermediation. To our knowledge, this paper is among the first to use either dataset and the first to combine them.

Looking at the raw data, we find substantial variation in daily risk limit usage and risk limits. Specifically, limits are decreasing in market-wide and currency volatility. At the bank-level, limits are positively correlated with banks’ supplementary leverage ratios and negatively correlated with stock prices. Indeed, a core part of our empirical strategy and the key contribution of our paper is to focus exactly on overcoming such endogeneity of risk limits to market and bank conditions by constructing an exogenous limit shock and tracing its impact on the foreign exchange market.

We identify exogenous limit shocks building on a striking institutional feature of the foreign exchange market: It is a highly concentrated dealer market, in which a few large global banking institutions intermediate the vast share of supply and demand of foreign currency.⁴

²Both these datasets are collected by the Federal Reserve to support the stress testing exercise mandated by the 2010 Dodd-Frank Act after the Great Financial Crisis. All information in these datasets refers to activities consolidated at the bank holding company (BHCs). Thus, our data capture activity by foreign branches and subsidiaries of US BHCs. For brevity, however, we use the term “bank” instead of BHC throughout the rest of this paper.

³BHCs with average gross trading assets and liabilities over the previous four calendar quarters equal to \$20 billion or more need to file VV-1. The Y-14Q respondent panel comprises BHCs with \$100 billion or more in total consolidated assets. Both datasets also cover the activity of foreign banking operations (at the intermediate holding company level). Full reporting requirements are available on the corresponding Federal Reserve Board websites. Our final sample includes 11 BHCs observed in both datasets.

⁴For example, the top eight dealer banks have a 92 percent market share in spot transactions, an 81 percent market share in outright forwards, a 74 percent market share in FX swaps, and a 92 percent market share in options; see the Survey of North American Foreign Exchange Volume Market Share, Federal Reserve Bank of New York, October 2022.

This institutional feature allows us to leverage the insights of the granular instrumental variable method (Gabaix and Koijen, 2020) to construct an aggregate limit shock from idiosyncratic changes in large dealers’ risk limits as observed in supervisory microdata. The idea is that, due to their size, idiosyncratic shocks to individual dealers matter in the aggregate. We provide an empirical method to identify such idiosyncratic limit changes and provide supportive narrative evidence of their idiosyncratic nature. We then use the identified limit shocks to US dealers to estimate the effects on their aggregate net positions, exchange rates, and other key quantities. Our main sample covers quarterly data for the period from 2016 through 2022 for 32 currencies.

Our empirical results are robust and consistent with our model predictions. First, we show that granularly identified limit shocks are significantly related to exchange rate realized volatility. Unconditionally, our limit shocks explain about 1.7% of the quarterly exchange rate realized volatility, a sizable share given its tight identification. The economic effects of these shocks are also meaningful. Our baseline estimate suggests that a one-standard-deviation tighter limit shock increases realized volatility by 38 basis points. On the other hand, we do not find an exchange rate response when using (endogenous) raw limit changes. Using instrumental variable (IV) estimation, we find that limit shocks are significantly related to raw limit changes (first stage), with a one-standard-deviation tighter limit shock decreasing limits by 3.9%. Second-stage results indicate that a 1% decrease in dealers’ limits increases exchange rate volatility by about 9.8 basis points, about 2% of the standard deviation of volatility (4.54 percentage points). While the effects of limit shocks on foreign exchange markets are sizable, they are transitory, with the effects dissipating after one quarter, suggesting that other dealer banks step in over time and bring the exchange market back to the previous equilibrium. Indeed, we confirm that dealers that do not experience tighter limits when other banks face such constraints take up some of the slack by increasing their positions.

Second, consistent with the model, we show that the direction of the exchange rate response to limit shocks depends crucially on demand shifts, both net demand shifts outside the dealer sector as well as idiosyncratic dealer shifts. Specifically, we find that when dealers absorb more foreign currency exposure and experience tighter limits, the foreign currency depreciates by more. Our results also support the model prediction that limit shocks amplify the exchange rate response to aggregate demand shifts. In our baseline results, we proxy external net dollar demand shifts using idiosyncratic innovations in dollar-denominated

sovereign credit default swap (CDS) spreads.⁵ We find that a negative standard-deviation shock in net demand (CDS innovation) for a foreign currency implies a 1 percent depreciation of that currency within the quarter. If, simultaneously, US dealers experience a one-standard-deviation tighter risk-limit shock, the currency depreciates by an additional 33 basis points (an increase of about one-third), thereby exacerbating effects from shifts in net currency demand.

We further zoom in on the mechanism by testing the model predictions about the effect of limit shocks on dealers' net position and intermediation spread. Using supervisory information on dealer positions and aggregate bid-ask spreads from Bloomberg, we find sizable effects: in response to a one-standard-deviation tighter limit shocks, dealers' (absolute) net position decreases by about 9.6% and bid-ask spread increases by 6.9%.

We provide evidence of intermediation impairment by examining the factors behind banks' profits from currency intermediation. This intermediation profit can be thought of as a combination of volume and spread. Our analysis shows that currency turnover, measured at either the currency-time level or the bank-time level, decreases as a result of limit shocks. Additionally, consistent with our earlier finding that bid-ask spreads increase at the currency level, intermediation spreads (trading margins) across all currencies at the bank level also increase.⁶ Our estimates are economically relevant, pointing to a reduction in turnover of as much as 3 percent and an increase in margins of about 8 percent in response to a one-standard-deviation limit shock. The risk-limit shocks also cause changes in covered interest parity (CIP) deviations, further supporting the proposed mechanism of impediments to a dealer's intermediation activity due to risk-limit tightening. We estimate a substantial widening of CIP deviations of 3.8 basis points in response to a one-standard-deviation tightening limit shock.

Finally, we estimate heterogeneous exchange rate responses to limit shocks for each currency in our sample. The results highlight that for the majority of currencies, limit shocks have significant and sizable effects, consistent with the strong average effect uncovered in our baseline analysis, although there is variation in the magnitude of the estimate response coefficient. In fact, we find that the strength of the exchange rate response is closely related to the volatility of the currency return (as measured by the average quarterly standard

⁵In our baseline analysis, we estimate innovations in the log CDS spread as the residual of an autoregressive model with currency fixed effects and principal component analysis to account for global risk factors, estimated for each country separately. We show robustness to different demand shift proxies. These spreads capture sovereign default risk, an increase of which generally implies selloffs of domestic assets such as local-currency-denominated government bonds (Augustin et al., 2016; Hébert and Schreger, 2017). This selloff constitutes a currency net demand, which is intermediated by the dealer banks in our sample.

⁶We measure bank-level spread as foreign exchange trading revenue over gross notional in foreign exchange trading instruments.

deviation): More volatile currencies exhibit stronger responses to limit shocks. This pattern is consistent with predictions from a multi-currency extension of our baseline model, in which—as in the case of the value-at-risk (VaR) limits exploited in our analysis—risk-sensitive constraints govern the heterogeneous adjustment of position and spread. However, a causal interpretation of this empirical relationship is not possible because limit shocks themselves exacerbate exchange rate movements, thereby contributing to volatility.

Related Literature Overall, our study underscores the importance of financial frictions in understanding exchange rate movements. In doing so, it fits into a few strands of the literature. First, it contributes to the literature that focuses on the role of market segmentation in exchange rate determination (Hau and Rey, 2006; Alvarez, Atkeson and Kehoe, 2009; Bruno and Shin, 2014). Specifically, this paper provides empirical support for theories that center on the role of financiers who mediate currency transactions (Gabaix and Maggiori, 2015; Greenwood et al., 2020; Gourinchas, Ray and Vayanos, 2022; Itskhoki and Mukhin, 2021*a,b*). These theories emphasize the role of financiers’ risk-bearing constraints in determining currency returns. We contribute to this literature in three ways. First, we observe and test the importance of dealers’ constraints in the FX market. We point out that they are time varying and endogenous, and we find an identification strategy to test their effects in pricing risk. Second, we find that dealers’ risk constraints are also connected to the liquidity-provision function of such dealers. Finally, we verify empirically that the effects of changes in risk limits are consistent with (short-term) market segmentation.⁷

Recent empirical papers use currency demand flows as determinants of exchange rate fluctuations (Froot and Ramadorai, 2005; Hau, Massa and Peress, 2010; Corte, Riddiough and Sarno, 2016; Pandolfi and Williams, 2019; Dao, Gourinchas and Itskhoki, 2025; Hacıoglu Hoke et al., 2026). Our main focus is on observed dealers’ constraints rather than market-driven demand flows. Second, we emphasize the causal role of single dealers in determining aggregate market risk, as opposed to explaining the total explained FX variation. Third, our data allows us to focus on all global assets related to the same currency (spots and all derivatives instruments) while most empirical papers focus on a specific geographic market or asset class.

A small strand of the literature relates bank-level information or speculator activity to currency returns. Adrian, Etula and Groen (2011) and Adrian, Etula and Shin (2015) use publicly available bank balance sheet data to understand how US bank funding liquidity can help forecast US dollar (USD) exchange rates. By contrast, our paper is the first that

⁷This is the first paper demonstrating that risk limit shocks independently generate currency volatility and enable model mechanism testing. However, quantifying their explanatory power exceeds our scope, as we primarily examine idiosyncratic variation of dealers. Moreover, even stable risk limits can amplify exchange rate volatility in the presence of noise traders.

we know of to utilize trading desk-level limits and positions in specific currencies. A couple papers focus on futures market positions in specific currencies: Hong and Yogo (2012) look at the impact of futures market contract open interest on exchange rate returns, and Kim, Liao and Tornell (2014) relate survey results on futures market positions to exchange rate returns. However, our paper considers all currency cash and derivative instruments (it is worth mentioning that futures constitute a tiny portion of currency derivatives). Additionally, in contrast to this literature, our paper uses granular shocks to achieve causal identification of the effects.

Our paper also relates to the literature studying the role of market micro-structure and order flows in asset pricing determination. In this literature, market makers influence the bid-ask spread either because of fixed costs of doing business (for example, inventory costs), information asymmetries (Glosten and Milgrom, 1985), or monopoly power (Duffie, Gârleanu and Pedersen, 2005). Our contribution to this literature is to test that dealers in the FX market demand currency risk compensation, on top of ‘rents’ for liquidity. In other words, we show how FX dealers also price the risk-premium. Evans and Lyons (2002) and Evans and Lyons (2007) use trading data of a large dealer in the Deutsche mark/US dollar market to show that order flows (net buying pressure on a particular currency) can explain as much as 60 percent of daily foreign exchange (FX) volatility. In their model, order flows from better informed investors generate price concessions for the dealers. Our paper differs from this literature in that we focus on the effect of exogenous changes in the risk capacity of dealers.⁸ We also focus on exchange rate returns over a quarterly horizon, rather than concentrating on intraday changes.

More broadly, our paper relates to the strand of literature studying the relationship between bank balance sheets and foreign exchange markets. In particular, several papers argue that post-financial-crisis regulation has made banks’ balance sheet capacity more costly, leading to deviations from CIP (e.g., Du, Tepper and Verdelhan, 2018; Abbassi and Bräuning, 2020; Cenedese, Della Corte and Wang, 2021).⁹ We provide support for desk-level risk limits as an additional bank constraint that impacts the foreign exchange market. While balance sheet capacity may inform risk limits, these limits are also informed by other factors such as internal risk aversion and regulatory bans on proprietary trading. Overall, however, our findings reinforce the idea that bank constraints are crucial for understanding the foreign exchange market.

⁸Huang et al. (2023) show how constraints, such as VaR, affect dealers’ liquidity provision in the foreign exchange market.

⁹Our work also relates to papers arguing more broadly that investors’ hedging of international portfolios is a driver of spot and forward exchange rates (e.g., Liao and Zhang, 2020).

The remainder of the paper is organized as follows. Section 1 presents a stylized model that guides the empirics. Section 2 discusses the data, and Section 3 focuses on the construction of limit shocks. Sections 4 present the empirical results. Section 5 concludes.

1 Model

To study how trading-desk limits shape the foreign currency market, we develop a model with heterogeneous foreign exchange dealers.¹⁰ A continuum of dealers $i \in [0, 1]$ supplies monopolistically competitive intermediation services to customers and charges a symmetric bid–ask spread $2 \log s_{it}$ around the mid-point exchange rate $\mathcal{E}_t$, defined as foreign currency units per US dollar.¹¹ Note how s_{it} represents one leg of the (gross) bid–ask spread and it is charged both on demand and supply for dollars. That is, customers give dealer i $s_{it}\mathcal{E}_t$ units of foreign currency for every dollar demanded and give $s_{it}/\mathcal{E}_t$ dollars for every unit of foreign currency demanded.

Time is indexed by $t \in \{0, 1\}$. At each time t , dealer i faces intermediation demand for dollars by foreign country importers x_{it}^ξ , demand for foreign currency by US importers x_{it}^ϵ , and demand for foreign currency by financiers x_{it}^f .¹² We distinguish financiers from dealers to highlight that dealer risk limits can affect equilibrium even when dealers hold only a small share of the aggregate stock of currency risk.

For simplicity, all agents have discount factor $\beta = 1$, and face safe domestic and foreign returns $R^* = R = 1$. Therefore, think of agents in this model as being risk-neutral and exchanging spot positions at high frequency.

Besides providing liquidity, dealers hold a risky position in foreign currency by supplying dollars to the real sector at $t = 0$ and unwinding positions at $t = 1$. Dealers choose an amount of foreign currency δ_i (specified in dollar terms) to invest.¹³ At $t = 0$ dealers borrow δ_i dollars, exchange for $\mathcal{E}_0\delta_i$ units of foreign currency, and at $t = 1$ convert back to dollars,

¹⁰Our model builds on Gabaix and Maggiori (2015), but introduces dealers as intermediaries.

¹¹The FX dealer market is concentrated, with a small number of large players. Our continuous monopolistic-competition environment is adopted to deliver transparent, tractable expressions. We follow the tradition in Amihud and Mendelson (1980); Ho and Stoll (1981); Biais, Martimort and Rochet (2000); Duffie, Gârleanu and Pedersen (2005); Duffie (2023) of modeling dealers as a monopolist facing a downward sloping demand curve for liquidity and subject to internal operational costs. Alternatively, Glosten and Milgrom (1985) and Evans and Lyons (2002) considers spread as a compensation for adverse selection in order flows under near-perfect competition. Potentially, both such micro-foundations will deliver our qualitative comparative statics under volatile risk limits. Our model, however wants to stress that dealers banks can affect the FX market both by charging a spread for the dealing activity and by actively absorbing marginal currency risk.

¹²Prices are flexible in this model but price rigidities (or the currency of invoicing of goods under rigid prices) do not affect the main conclusions of this model, as in Gabaix and Maggiori (2015).

¹³ δ_i can also be interpreted as the residual risky position that the dealer does not hedge. δ_i can be negative indicating a short position.

earning a net return of $\mathcal{E}_0/\mathcal{E}_1 - 1$.

Dealers face a VaR-type constraint limiting their ability to bear currency risk between $t = 0$ and $t = 1$. As in Gabaix and Maggiori (2015), we model such a risk limit constraint by assuming that in each period, after taking positions but before shocks are realized, the dealers can divert a portion of the funds they intermediate. If the dealer diverts the fund, creditors can recover a portion $1 - \gamma_i \delta_i$ of their credit position δ_i .¹⁴ We will derive comparative statics of key market outcomes with respect to the parameter γ_i .

We model the dealer's problem as:

$$\max_{\{s_{it}\}_{t=0,1}, \delta_i} \sum_{t=0}^1 [(s_{it} - 1)x_{it} - \frac{c_i}{2}x_{it}^2] + \mathbb{E}_0 [\mathcal{E}_0/\mathcal{E}_1 - 1] \delta_i, \quad (1)$$

$$\text{such that: } \mathbb{E}_0 [\mathcal{E}_0/\mathcal{E}_1 - 1] \delta_i \mathcal{E}_0 \geq \gamma_i (\mathcal{E}_0 \delta_i)^2. \quad (2)$$

The first term of the dealers' problem represents profits from the intermediation activity. The second term represents profits from holding the risky position δ_i between time 0 and 1. Note how operating costs of providing liquidity, $\frac{1}{2}c_i x_{it}^2$, are proportional to a dealer-specific parameter $c_i > 0$ and to the square of transacted flow.¹⁵ We leave c_i as unspecified but we later show how our main propositions remain unchanged if c_i is a function of the risk-limit parameter γ_i . For now, consider c_i as purely operational costs, separate from risk-taking costs.¹⁶

Appendix C shows that the optimal solutions of the dealer maximization problem in (1) are:

$$\delta_i = \frac{1}{\gamma_i} \mathbb{E} \left[\frac{1}{\mathcal{E}_1} - \frac{1}{\mathcal{E}_0} \right], \quad (3)$$

$$s_{it} = \frac{\sigma}{\sigma - 1} c_i x_{it}. \quad (4)$$

Intuitively, dealers' currency holdings increase with expected appreciation and decline as risk limits become tighter (higher γ_i). Spreads are determined as markup over marginal costs,

¹⁴This is a common assumption in the macroeconomic literature to maximize tractability (Gertler and Kiyotaki, 2010; Maggiori, 2017). The constraint could be also due to regulation, but not necessarily. For instance, Oskolkov (2025) provides a foundation for the value-at-risk constraint through robustness concerns, where creditors fear model misspecification. It is beyond the scope of this paper to provide a micro-foundation for the risk constraint.

¹⁵Operating costs are not essential for our conclusions. For instance, under linear demand and zero marginal operating costs, decreasing marginal revenues would also deliver a spread increase with higher volumes. Alternatively, a Kimball aggregator (rather than CES) would yield similar qualitative results.

¹⁶For example, such operational cost could be (risk-insensitive) balance sheet costs or general administrative expenses including personnel costs, which are likely convex in the short run. We introduce convex operating costs as it delivers diminishing returns to spread-charging activity.

where marginal costs increase with volumes.

To provide a straightforward aggregation, importers and financiers have CES preferences over dealers generating the following demand functions:

$$x_{it}^{\iota} = \left(\frac{s_{it}}{S_t} \right)^{-\sigma} \iota_t, \quad (5)$$

$$x_{it}^{\xi} = \left(\frac{s_{it}}{S_t} \right)^{-\sigma} \frac{\xi_t}{\mathcal{E}_t}, \quad (6)$$

$$x_{it}^f = \left(\frac{s_{it}}{S_t} \right)^{-\sigma} \frac{1}{\Gamma} \mathbb{E}_0 \left[\frac{1}{\mathcal{E}_1} - \frac{1}{\mathcal{E}_0} \right], \quad (7)$$

$$x_{it} = x_{it}^{\iota} + x_{it}^{\xi} + x_{it}^f, \quad (8)$$

where x_{it}^{ι} is US importers' demand for foreign currency intermediated by dealer i (In effect, US importers are demanding foreign currency from all dealers in order to buy the aggregate value of ι_t foreign goods to supply US households), x_{it}^{ξ} is the foreign importers' demand for US dollars from dealer i , and x_{it}^f is the financiers' demand for foreign currency from dealer i . S_t is the aggregate spread, defined below. As in Gabaix and Maggiori (2015), ξ_t is the exogenous ex-spread aggregate US goods demand from foreign consumption preferences, ι_t is the exogenous ex-spread aggregate demand for foreign goods from US consumption preferences, and Γ is the financiers' risk-constraint parameter.¹⁷ Appendix C presents the full derivation of the demand functions in (5)–(7).

CES preferences of importers and financiers across dealers imply the corresponding aggregate spread index:

$$S_t = \left(\int s_{it}^{1-\sigma} \right)^{\frac{1}{1-\sigma}}. \quad (9)$$

Each dealer balance sheet can potentially have a dollar imbalance δ_{it}^I , which is absorbed by the inter-dealer market at time 0 and 1:

$$s_{i0}(x_{i0}^{\xi} - x_{i0}^{\iota} - x_{i0}^f) - \delta_i = \delta_{i0}^I \quad \forall i, \quad (10)$$

$$s_{i1}(x_{i1}^{\xi} - x_{i1}^{\iota} + x_{i1}^f) + \delta_i = \delta_{i1}^I \quad \forall i. \quad (11)$$

¹⁷We make simplifying assumptions on the endowment process of households to keep marginal utilities (and ex-spread relative prices) constant.

Finally, we impose a frictionless inter-dealer market to clear¹⁸:

$$\int \delta_{i0}^I di = 0, \quad (12)$$

$$\int \delta_{i1}^I di = 0. \quad (13)$$

Substituting (10)–(11), the demand functions (5)–(7), aggregate spread (9), and the solution for δ_i into (12)–(13), gives the aggregate market clearing:

$$\overbrace{S_0 \left(\frac{\xi_0}{\mathcal{E}_0} - \iota_0 \right)}^{\text{Cum-spread Net Exports}} - \overbrace{\left(\frac{S_0}{\Gamma} + \int \frac{1}{\gamma_i} di \right)}^{\text{Aggregate Risk Limit}} \overbrace{\mathbb{E}_0 \left[\frac{1}{\mathcal{E}_1} - \frac{1}{\mathcal{E}_0} \right]}^{\text{Expected Appreciation}} = 0, \quad (14)$$

$$S_1 \left(\frac{\xi_1}{\mathcal{E}_1} - \iota_1 \right) + \left(\frac{S_1}{\Gamma} + \int \frac{1}{\gamma_i} di \right) \mathbb{E}_0 \left[\frac{1}{\mathcal{E}_1} - \frac{1}{\mathcal{E}_0} \right] = 0. \quad (15)$$

Appendix C derives the closed-form equilibrium equations of this model. Proposition 1 presents the log-linearized solution to highlight the core intuition behind our empirical results. Log deviations are denoted $\hat{X} = d \log X$, while deviations are denoted dX . Bars denote steady-state values.

Proposition 1. *Given the profit maximization of dealers in (1), demand and supply functions (5)–(7), market clearings (10), (11), (12) and (13), and assuming all variables at time 1 remain at steady state, the general equilibrium solutions for variables at time 0 can be expressed as follows, to a first-order approximation:*

$$\hat{\mathcal{E}}_0 = \frac{1 + \bar{S}\bar{\xi}\bar{\gamma}}{2 + \bar{S}\bar{\xi}\bar{\gamma}} (\hat{\xi}_0 - \hat{\iota}_0) \quad \text{where} \quad \bar{\gamma} = \left(\int \gamma_i^{-1} di + \bar{S}/\Gamma \right)^{-1}, \quad (16)$$

$$d\delta = \frac{1}{\bar{\mathcal{E}}} \frac{1 + \bar{S}\bar{\xi}\bar{\gamma}}{2 + \bar{S}\bar{\xi}\bar{\gamma}} (\hat{\xi}_0 - \hat{\iota}_0) \int \gamma_i^{-1} di, \quad (17)$$

$$\hat{S}_0 = \frac{\psi_\sigma}{1 - \sigma\psi_\sigma} \frac{1}{2} \left[\hat{\xi}_0 + \hat{\iota}_0 + \left(\frac{\zeta}{\Gamma\bar{\xi}} - 1 \right) \hat{\mathcal{E}}_0 \right] \quad , \text{where} \quad \zeta = \text{sign}(\hat{\xi}_0 - \hat{\iota}_0). \quad (18)$$

All variables are defined as changes relative to steady state. $\hat{\mathcal{E}}_0$ is the time-0 foreign-currency depreciation (USD appreciation); $d\delta$ is the change in aggregate dealers exposure; and $\hat{S}_0$ is the time-0 spread change. The net flow $\hat{\xi}_0 - \hat{\iota}_0$ captures the change in net dollar demand imbalance from the real economy. The parameter $\bar{\gamma}$ represents the curvature of the

¹⁸We choose the inter-dealer market with no spread for simplicity. As long as the inter-dealer market spread is lower than the client-to-bank spread, all propositions remain qualitatively the same.

aggregate dollar supply function, as also evident in equations (14)-(15). Moreover, dealers add the spread wedge to all transactions. Appendix D shows a graphical interpretation of the equilibrium in this model.

If $\hat{\xi}_0 - \hat{\iota}_0 = 0$, the exchange rate and dealers' net positions do not move. Spreads can still change because unchanged net positions need not imply unchanged gross positions $\hat{\xi}_0 + \hat{\iota}_0$. Exchange rate, net positions, and spreads all increase with $\hat{\xi}_0 - \hat{\iota}_0$: stronger net dollar demand implies a larger USD appreciation (to compensate constrained intermediaries), greater dollar supply by banks, and wider spreads. The exchange-rate equation (16) mirrors the solution in Gabaix and Maggiori (2015), except with the addition of a heterogeneous dealer market generating spread and risk-limit variation.

In the baseline, dealers' constraints do not affect equilibrium variables via market shares of the affected dealers. What matters for aggregate outcomes is the relative size of the risk taking limit of dealer i relative to all other dealers and financiers in the economy, i.e., how it affects the object $\tilde{\gamma}$. It is only via $\tilde{\gamma}$ that spread and exchange rate move.¹⁹ While this is a simplifying assumption, it has the intuitive feature that individual risk-limits can only directly affect aggregate risk-limits. Aggregate risk-limit shocks then affect spreads and mid-price levels via general equilibrium forces, i.e., by moving the exchange rate level and risk positions.

The basic structure of the proposition would remain unchanged if we depart from the assumption of exogenous dealer costs c_i or atomistic dealers. Specifically, Appendix C demonstrates that dealers' costs c_i only affect the term ψ_σ , which is proportional to the market-share-weighted average of all dealers' total costs $c_i \bar{x}_i$. If costs for providing FX liquidity were proportional to currency risk limits —as, for instance, in Duffie (2023)— the spread would widen further after risk tightening. If we further departed from our model assumptions and allowed for non-atomistic dealers, large dealers would absorb some of the latter forces in their mark-up s_{it} after a cost change, but average spreads would still increase with risk tightening.

1.1 Testable Hypotheses

In equilibrium, both net demand and risk limits influence prices. Risk limits are assumed to be exogenous in the model, while we allow for endogeneity in positions δ_i . In reality both observed risk positions and risk limits are endogenous (see Section 2). Therefore, the first step of our empirical strategy will be to identify idiosyncratic shocks to γ_i using the richness of the data. This section focuses on developing three complementary tests that also address

¹⁹Proposition 2 will relax this assumption.

the endogeneity of observed bank positions δ_i , conditional on exogenous shocks to γ_i .

First, we exploit the following corollary to proposition 1, which allows us to test one aspect of the model without relying on the identification of any net demand proxies:

Corollary 1. *Under the conditions of Proposition 1, the following comparative statics with respect to bank's i risk limit, γ_i , holds:*

$$\frac{\partial |\mathcal{E}_0|}{\partial \gamma_i} > 0, \quad (19)$$

$$\frac{\partial |d\delta|}{\partial \gamma_i} = \frac{\partial |\int d\delta_i di|}{\partial \gamma_i} < 0, \quad (20)$$

$$\frac{\partial \hat{S}_0}{\partial \gamma_i} > 0, \quad \text{if } \frac{1}{\Gamma} > \bar{\xi}. \quad (21)$$

Corollary 1 implies that tightening risk limits raises exchange-rate volatility, compresses dealers' positions toward zero, and increases spreads. Intuitively, tighter limits amplify price movements by restricting average intermediation capacity. Thus, risk-limit fluctuations are themselves a source of exchange-rate variation under dollar demand imbalance, not merely an amplifier of other shocks (Gabaix and Maggiori, 2015; Itskhoki and Mukhin, 2021a). Moreover, as long as dollar demand is more inelastic than financiers' supply ($1 - \bar{\xi}\Gamma > 0$), the spread always increases with tighter limits. Why is this elasticity condition required only for spreads but not for the appreciation? The key is that the spread is proportional to gross flows. Tighter risk limits moves $\hat{\mathcal{E}}_0$ in the direction of net demand flows, i.e., under positive dollar demand, tighter limits imply dollar appreciation. Gross flows will move in the same direction as net flows only if net demand is sufficiently inelastic. The low elasticity of FX demand compared to supply is supported empirically in Bippus, Lloyd and Ostry (2023).²⁰

Second, we use aggregate proxies for shocks to net dollar demand $\hat{\xi}_0 - \hat{\iota}_0$ and exploit the following corollary to Proposition 1:

Corollary 2. *Under the conditions of Proposition 1, the following comparative statics with*

²⁰In alternative to assuming low demand elasticity, any micro foundation where higher imbalances constraints imply larger dealers costs (and therefore spread) for the dealer would imply the same spread sign response as in Corollary 1.

respect to bank's i risk limit, γ_i , holds:

$$\frac{\partial \hat{\mathcal{E}}_0}{\partial \gamma_i} \propto w_{\gamma_i}^2 (\hat{\xi}_0 - \hat{\iota}_0), \quad \frac{\partial \hat{\mathcal{E}}^2}{\partial \gamma_i \partial (\hat{\xi}_0 - \hat{\iota}_0)} > 0, \quad (22)$$

$$\frac{\partial d\delta_i}{\partial \gamma_i} \propto - \left(\frac{w_{\gamma_i}}{\gamma_i} \right)^2 (\hat{\xi}_0 - \hat{\iota}_0), \quad \frac{\partial d\delta_i^2}{\partial \gamma_i \partial (\hat{\xi}_0 - \hat{\iota}_0)} < 0, \quad (23)$$

$$\text{where: } w_{\gamma_i} = \frac{\gamma_i^{-1}}{\int \gamma_j^{-1} dj + (\bar{S}/\Gamma)^{-1}}. \quad (24)$$

These comparative statics show that the sign of a tighter risk limit's effect on the exchange rate and on net positions depends on the dollar-demand shifter. If $\hat{\xi}_0 - \hat{\iota}_0 > 0$, tighter limits raise foreign-currency depreciation (USD appreciation) and reduce dealers' net foreign-currency positions; the opposite holds when $\hat{\xi}_0 - \hat{\iota}_0 < 0$. The magnitude of dealer i 's effect scales with its relative weight w_{γ_i} . Recall from Corollary 1 that the spread does not depend on position direction.

For our third empirical strategy, we examine idiosyncratic changes in observable dealer positions and their interaction with risk limits. To understand the role of idiosyncratic dealer demand shifts, assume dealers have idiosyncratic beliefs about the expected currency appreciation.²¹ Each dealer's expectation deviates from the rational expectation by an amount ν_i .²² The dealer's problem becomes:

$$\max_{\{s_{it}\}_{t=0,1}, \delta_i} \sum_{t=0}^1 \left[(s_{it} - 1)x_{it} - \frac{c_i}{2}x_{it}^2 \right] + \mathbb{E}_0^i [\mathcal{E}_0/\mathcal{E}_1 - 1] \delta_i, \quad (25)$$

$$\text{such that: } \mathbb{E}_0^i [\mathcal{E}_0/\mathcal{E}_1 - 1] \delta_i \mathcal{E}_0 \geq \gamma_i (\mathcal{E}_0 \delta_i)^2, \quad (26)$$

$$\text{where: } \mathbb{E}_0^i \left[\frac{1}{\mathcal{E}_1} \right] = \mathbb{E}_0 \left[\frac{1}{\mathcal{E}_1} \right] + \nu_i. \quad (27)$$

In this case, the solution for the spread determination will remain unchanged, and the (linearized) solution for bank i 's position (assuming no shocks at time 1) as well as the

²¹Similarly, other mechanisms could lead to idiosyncratic demand, and we focus on heterogeneous expectation simply for expositional reasons.

²²Appendix C shows that this formulation is isomorphic to any model with a dealer-specific wedge linear in position δ .

aggregate dealer position will be:

$$d\delta_i = \frac{1}{\gamma_i} \frac{\hat{\mathcal{E}}_0}{\bar{\mathcal{E}}} + \frac{d\nu_i}{\gamma_i}, \quad (28)$$

$$d\delta = \frac{\hat{\mathcal{E}}_0}{\bar{\mathcal{E}}} \int \gamma_i^{-1} di + \int \frac{d\nu_i}{\gamma_i}. \quad (29)$$

Each bank's risk position is the sum of two terms. The first familiar term is represented by the depreciation of the foreign currency at time zero, in turn the result of aggregate demand shifters. The second term represents idiosyncratic exogenous shifts in bank's exposure demand, due to heterogeneous beliefs, introducing an intercept to dealer's currency demand which decreases when γ_i increases. Equation (28) shows how the raw risk position δ_i is partly correlated with the exchange rate movement. The following proposition instead expresses exchange rates as a function of primitives of the model, after a risk limit shift.

Proposition 2. *Given the profit maximization of dealers in (25), demand and supply functions (5) and (7), market clearings (10), (11),(12) and (13), and assuming all variables at time 1 remain at steady state, the following derivative holds at steady state:*

$$\frac{\partial \hat{\mathcal{E}}_0}{\partial \gamma_i} \propto \left(\frac{\tilde{\gamma}}{\gamma_i} \right)^2 \overbrace{\left(\bar{S}\bar{\xi}/\bar{\mathcal{E}}(\hat{\xi}_0 - \hat{i}_0) - 2d\tilde{\nu} \right)}^{\text{Aggregate Demand Flow}} + \left(\frac{2\tilde{\gamma} + \bar{S}\bar{\xi}\tilde{\gamma}^2}{\gamma_i^2} \right) \underbrace{\left(\frac{d\nu_i}{\gamma_i} \right)}_{\text{Idiosyncratic Flow}}, \quad (30)$$

$$\text{where } \tilde{\gamma} = \left(\int \gamma_i^{-1} di + \bar{S}/\Gamma \right)^{-1}, \quad (31)$$

$$d\tilde{\nu} = \int d\nu_i/\gamma_i di. \quad (32)$$

Proposition 2 shows that exchange-rate movements load on the interaction of dealer-specific constraints with (i) aggregate and (ii) dealer-specific net dollar demand. It follows that the interaction between a limit change and the idiosyncratic position of the bank experiencing the limit change contains unambiguous information on the direction of the exchange rate depreciation.

2 Data

Our analysis utilizes two supervisory datasets: the Regulation VV Quantitative Measurements (henceforth VV-1) data and the FR Y-14Q (henceforth Y-14) data.²³ The 2010 Dodd–Frank

²³The reported data are confidential supervisory information, but the reporting forms and instructions, including the list of variables collected by the Federal Reserve, are publicly available at <https://www.federalreserve.gov/apps/reportforms/Default.aspx>.

Act mandated that the Federal Reserve System collect both these supervisory data, with the aim of building a more resilient financial system against the backdrop of the 2008–2009 Great Recession. Both datasets contain information about the consolidated global activity of the largest banking institutions domiciled in the United States, in particular US bank holding companies (BHCs). Therefore, we capture the banks’ activities from both foreign and domestic offices (including branches and subsidiaries). Moreover, both datasets cover the activities of foreign-owned intermediate holding companies (IHCs) with substantial presence in the United States. As discussed, for simplicity, we refer to these institutions collectively as “banks.”

VV-1 data The VV-1 dataset originated under the 2013 Volcker rule of the Dodd–Frank Act to monitor compliance with restrictions on proprietary trading. Banks with average gross trading assets and liabilities over the previous four calendar quarters equal to or more than \$20 billion are required to submit trading-desk names and descriptions and report a variety of metrics for each desk, including daily desk-specific internal risk limits and usage.²⁴ Risk limits are primarily of two types: sensitivity (for example, delta position, or cash equivalent position) and value-at-risk.²⁵ We focus on VaR-based limits in our main analysis.²⁶ Because limits are desk-specific and not product-specific (for example, foreign exchange), we limit our analysis to desks that trade currencies as one of their primary products. We do so by selecting desks that match specific strings in their desk names or desk descriptions (for example, FX, currency, foreign exchange).²⁷

It is worth noting that not all foreign exchange products were included in the original

²⁴According to the Volcker rule, a “desk” is a unit of organization that purchases or sells financial instruments for a bank’s trading account. Desks are structured according to a business strategy and to set and monitor trading limits, losses, and strategies. See 12 CFR § 248.3(e)(14)(ii).

²⁵The “cash equivalent” or the “delta position” of a derivative is the expected change in the value of a derivative for each dollar change in the price of the underlying asset, multiplied by the face value of the derivative. In exchange rate markets, it is also called the “spot equivalent position” and is used as a summary measure of the overall position to a given exchange rate across spot and derivative instruments.

²⁶Often, a single desk has multiple limit types; for example, the desk may report both VaR-based and delta-based limits. VaR limits and delta limits are highly correlated. The final authority to set internal VaR limits sits with independent market risk group, which is part of the internal risk management group and reports to chief risk officer (CRO). Trading desk limits are often proposed by trading desks and their management. These proposals are then subject to review, potential amendments, and approval by independent market risk.

²⁷Banks record desk names and descriptions along with their internal desk IDs. We select desks whose names include one of the following strings: “FX,” “currency,” or “foreign exchange.” We also include desks for review whose description includes “FX” or “currency,” regardless of their desk name. We hand-verify all selected desks; that is, we verify that a desk is trading FX as one of its primary activities, not using FX just to hedge positions in other primary activities (such as equities or commodities). We do not include interest rate trading desks that trade cross-currency swaps as one of their products if they do not trade other FX products as one of their primary products.

Volcker rule that restricted proprietary trading and other activities.²⁸ However, banks still limit exposure to both FX cash and derivative products based on internal risk-management practices, and the Volcker rule reporting requirement gives us a unique window into these limits and their usage regardless of the specific regulation on FX trading.

Our final data include 167 FX desks for 11 banks. Of these, 120 desks have delta limits, and 165 have VaR limits. One hundred eighteen desks report both delta and VaR limits. In this paper, we focus on VaR limits. Since the reporting entity is the holding company, our dataset covers trading desks affiliated with large US banks outside of the United States. For example, we capture FX desks in London. We also capture the US trading desks of foreign banks, and the data includes the foreign desks of some foreign banks as well. All desks declare the limit and limit use, in dollars, for each limit measure. Desks may also declare the maximum and minimum of two-sided limits, which are most relevant for delta limits.

Figure 1 displays aggregate information about the percent limit use of VaR limits. Panel (b) shows the distribution of the percent limit use at the desk-day level. We can see that on most days and for most desks, limit usage is not close to the limit. Panel (a) confirms this each year—the median percent limit use at the desk-day level hovers between 15 and 30 percent. However, the tails of the distribution show that there are days on which desks’ limit usage approaches their limit. It is also important to note that dealer behavior can be affected before limits are hit and that target usage levels may differ from specified limits.²⁹

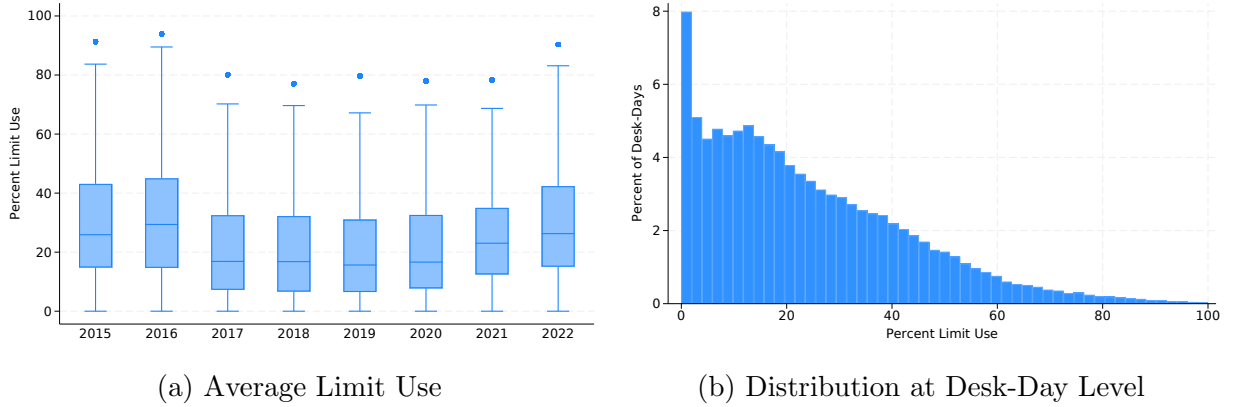
The observed changes in these trading desks’ daily risk limits are at the core of our identification strategy. Based on the total of 1,071 nonzero limit changes observed in our data, we find substantial variation in limit changes over time and in the cross section of desks. Figure 2, panel (a), depicts the monthly average of trading desks’ daily risk limit changes (measured in percentages). Large limit shocks stand out throughout the sample period exceeding ± 50 percent on average in some months. There is also a decline in limit increases throughout the sample. Panel (b) shows the histogram of corresponding desk-day level limit changes. The distribution exhibits substantial kurtosis. For confidentiality, the data in the graphs are winsorized at the top and bottom 2.5 percent.

The risk limits are endogenous to macroeconomic conditions and subject to bank-level

²⁸While FX derivatives were included, FX cash was exempt. Furthermore, FX derivatives became exempt with the 2020 amendment to the Volcker rule, which became effective in January 2021. Nevertheless, the rule states, “A banking entity is required to report the Internal Limits and Usage quantitative measurement for all trading desks engaged in covered trading activities.” Thus, for all desks in the dataset, we see a comprehensive picture of their internal risk limits, even if some of the activities they engage in are not covered under the Volcker rule.

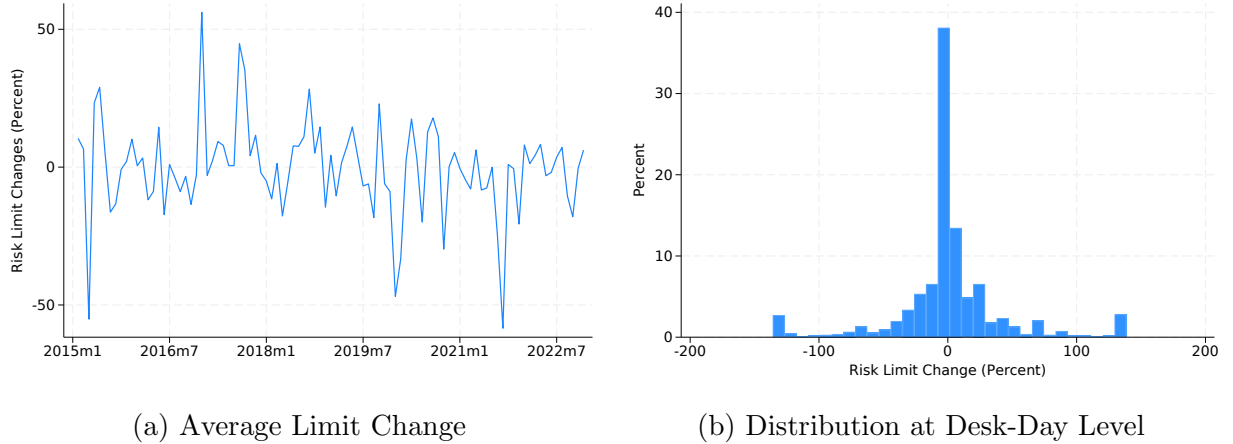
²⁹Anderson, McArthur and Wang (2023) and Li, Petrask and Tian (2024) both provide evidence consistent with binding VaR limits by documenting how dealers manage their inventories as their limit utilization increases.

Figure 1: Limit Usage of Trading Desks



Notes: Panel (a) shows the monthly average of trading desks' percent limit use. For confidentiality reasons, the dots represent the mean of the outside values (rather than individual data points). Panel (b) shows the histogram of percentage limit use at the desk-day level. *Source:* FR VV-1, authors' calculations.

Figure 2: Risk-limit Changes of Trading Desks



Notes: Panel (a) shows the monthly average of trading desks' percentage changes (log difference times 100) in risk limits. Panel (b) shows the histogram of percentage changes (log difference times 100) in risk limits at the desk-day level. Both data series are winsorized at the top and bottom 2.5 percent. *Source:* FR VV-1, authors' calculations.

variation in the way they are declared (see Anderson, McArthur and Wang, 2023). Table 1 documents how limits are decreasing in market-wide volatility metrics and how percent limit usage is increasing in these metrics. Additionally, Table 2 documents how own limits decrease and limit usage increases when the bank's stock price rises (though this is not necessarily the case when looking at stock returns computed on a daily basis). The limit itself is positively correlated with the supplementary leverage ratio, which indicates that increased balance

sheet space is partially distributed to FX trading desks. Meanwhile, the percent limit usage is negatively correlated with the T1 risk-based capital ratio, which makes sense given that VaR is a component of this ratio. Due to this clear endogeneity of risk limits, Section 3 develops a method for isolating the idiosyncratic risk-limit shifts.

Table 1: Market-Wide Covariates of Limits and Limit Usage

	Percent Limit Use				Log Limit			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
VIX	0.015*** (0.006)			-0.047*** (0.007)	-0.005*** (0.001)			-0.001* (0.001)
EUR/USD Vol.		0.619*** (0.023)		0.494*** (0.032)		-0.040*** (0.002)		-0.019*** (0.002)
JPY/USD Vol.			0.440*** (0.027)	0.203*** (0.028)			-0.034*** (0.002)	-0.021*** (0.002)
Bank-Desk FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Within R-squared	0.000	0.010	0.008	0.011	0.006	0.026	0.028	0.031
N	139,201	142,306	142,306	139,201	139,201	142,306	142,306	139,201

Notes: This table shows regressions of either percent limit use or the log limit on various market-wide measures. The volatility refers to the one-month implied volatility of the referenced exchange rate. Standard errors are clustered at the day level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. *Sources:* FR VV-1, Haver, Bloomberg, authors’ calculations.

FR Y-14 data Our second data source is the FR Y-14. The Y-14 dataset includes detailed information on the consolidated portfolio holdings of the largest US banks (and US affiliates of foreign banks). Banks with more than \$100 billion in total consolidated assets are subject to the mandated Dodd–Frank Act Stress Test (DFAST) and are required to file the Y-14 for stress-testing and supervisory purposes. Specifically, we exploit Y-14Q, Schedule F, on trading that contains information on banks’ FX trading activity. For each bank and quarter, the schedule reports the delta positions, that is, the cash equivalent net exposures, in each currency. This currency exposure reflects both spot and derivative positions. In what follows, for simplicity, we refer to this delta position as “net position” or “net exposure.” Again, because the reporting entities are BHCs, we capture all consolidated positions, irrespective of

Table 2: Bank-Specific Covariates of Limits and Limit Usage

	Percent Limit Use					Log Limit				
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Log(Stock Price)	8.323*** (0.209)					-0.415*** (0.009)				
Stock Return		2.249 (4.337)					-0.104 (0.279)			
T1 Risk-Based capital ratio (%)			-0.324* (0.173)		-0.160 (0.405)			0.003 (0.005)		-0.044** (0.017)
Supplementary leverage ratio (%)				0.643 (0.494)	0.824 (0.615)				0.043** (0.017)	0.093*** (0.032)
Bank-Desk FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Within R-squared	0.022	0.000	0.003	0.001	0.001	0.039	0.000	0.000	0.005	0.016
N	138,991	132,893	1,936	1,517	1,517	139,557	133,432	1,936	1,517	1,517

Notes: This table shows regressions of either percent limit use or the log limit on bank-specific measures. Columns (1), (2), (6) and (7) are at the daily level. The remaining columns are at the quarterly level and specifically show regressions of either last quarterly percent limit use (columns 3-5) or last quarterly log limit (columns 8-10) on bank balance sheet utilization measures. Standard errors are clustered at the time-period level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. *Sources:* FR VV-1, FR Y-9C, Capital IQ, authors' calculations.

whether they come from a trading desk in the United States or abroad.³⁰

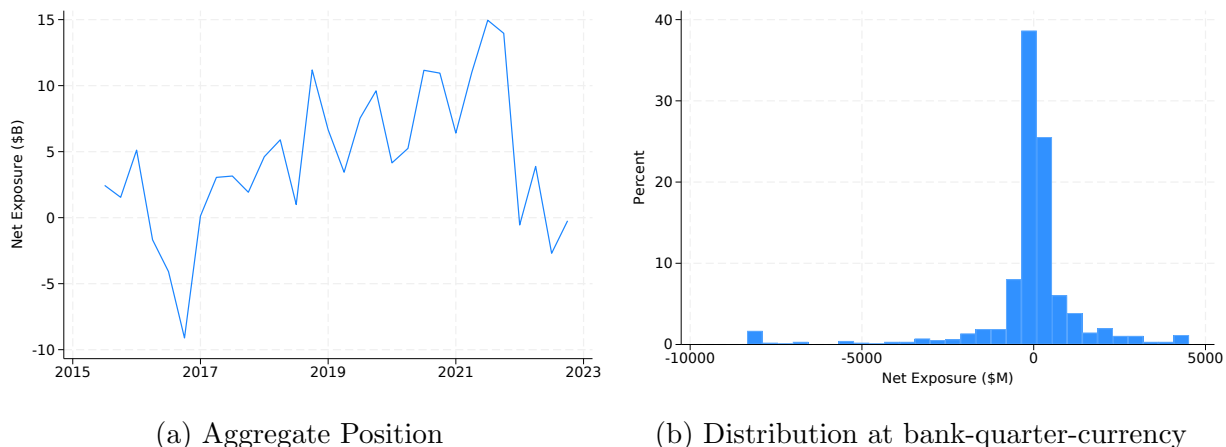
Figure 3, panel (a), shows the evolution of aggregate net positions (total delta positions across banks and currencies) over time. The series reveals that the banks are generally long in foreign currencies, thus providing dollars to the rest of their trading counterparties. The figure also reveals that positions in foreign currency increased from about 2017 through 2021 but then fell. Moreover, positions vary from $-\$10$ billion to $\$15$ billion. Panel (b) zooms in on the distribution of net positions at the bank-quarter-currency level, revealing substantial cross-currency variation and fat tails, which indicate that some large positions exceed $\$1$ billion. Appendix Figure A.3 shows the evolution of the aggregate net positions by currency over time and demonstrates additional time variation, which is significant for several currencies.³¹

Appendix Figure A.4, panel (a), zooms in on the distribution of positions by currency, across time and banks. The majority of net positions are rather small, in the order of tens to

³⁰Submission types are further broken down into the following: (1) FVO hedges are positions that are used to hedge loan assets that are held for sale (HFS) or held under fair value option (FVO) accounting; (2) AL hedges are positions that are used to hedge held-for-investment (HFI) accrual loans; (3) Credit valuation adjustment (CVA) refers to the market value of the credit risk due to any failure of the counterparty to deliver; and (4) trading. In most of our analysis, we focus on the trading submission, given our research question.

³¹The figure also reveals large short positions in the South Korean won (KRW), which is why we have excluded the KRW exposure from panel (a) of Figure 3.

Figure 3: Trading Desks’ Net Position in Foreign Currencies



Notes: Panel (a) shows the sum of net position across all banks and currencies, for each quarter (in \$ billion). Positive values correspond to long position in foreign currency. Panel (b) shows the histogram of the net positions at the bank-currency-quarter level (in \$ million), winsorized at the top and bottom by 1 percent. We exclude Korean won exposure from panel (a) because some desks have large net short positions with an outsized effect on aggregate exposure. Appendix Figure A.4 shows full details on the net exposures for each currency. *Source:* FR Y-14Q, BIS, authors’ calculations.

hundreds of millions, with maximum net position being \$1 billion. (Note that to maintain confidentiality, the data are winsorized at the top and bottom 5 percent.) The relatively small net positions held by trading desks contrast starkly with the average daily FX market turnover from the BIS triennial survey, which is in the order of trillions. This insight is consistent with trading desks focusing on intermediation rather than proprietary trading.³²

Additional data In addition to these supervisory micro datasets, we use banks’ income and balance sheet information at the quarterly frequency as reported in the publicly available FR Y-9C data. These data also contain information on income from FX trading as well as gross notional in FX trading. Moreover, we collect bilateral exchange rates, bid–ask spreads, implied volatility, CDS spreads, forward exchange rates and settlement dates, and interest rates from Bloomberg. Additionally, we use turnover data at the currency level from the Federal Reserve Bank of New York’s Foreign Exchange Volume Survey.W

³²However, Hacıoglu Hoke et al. (2026) shows how considering intra-dealer net currency exposure in the London derivatives markets shrinks the relative size of the market from tens of trillions into hundreds of billions at most. It is plausible that observing the consolidated global net exposure of each dealer to all FX assets (as we do) shrinks aggregate net positions even further. Nevertheless, our model rationalizes how changes in limits have significant and sizable effects on exchange rates at the margin—even when risk limits and net positions are small.

3 Identification

While risk limits are assumed to be exogenous in our stylized model, in reality they may respond endogenously to market conditions, including exchange rates, dealers' net positions, and bid-ask spread. Our results will show that neglecting the endogeneity of risk limits generally leads to biased estimates of the effect of risk limits on exchange markets. Section 1.1 also outlined how raw dealer positions are endogenous and how, to test the model, we should consider either aggregate or dealer-specific net-demand shocks.

For the identification of the risk limit shocks, we combine two insights from Gabaix and Koijen (2020). First, we identify desk-level idiosyncratic risk-limit shocks, that is, changes in risk limits that are exogenous to the overall evolution of the foreign exchange market and its own bank. To bridge the model and the data discussed in Section 2, we assume now that risk limits of bank i are defined at the desk level $d \in \mathcal{D}_i$, where $\mathcal{D}_i$ is the set of desks of bank i . Desk risk limits are composed of an endogenous and an idiosyncratic component (as in the model section, we omit the currency subscript):

$$\gamma_{it} = \sum_{d \in \mathcal{D}_i} \gamma_{dt} = \sum_{d \in \mathcal{D}_i} \left(g_d(\hat{\mathcal{E}}_t, X_t, X_i) + \eta_{dt} \right), \quad (33)$$

where $g_d(\cdot)$ is an unobserved function of expected depreciation $\hat{\mathcal{E}}_t$, other market characteristics X_t , and institutional characteristics X_i . In line with the method of Gabaix and Koijen (2020), we assume that observable changes in risk limits as in equation (33) can be decomposed into

$$\Delta \gamma_{dt} = \Lambda_d^\top F_t + \Delta \eta_{dt}, \quad (34)$$

where F_t are estimated common factors, time, bank, and other control effects absorbing $g_d(\cdot)$ parametrically. Then

$$\mathbb{E} \left[\widehat{\Delta \boldsymbol{\eta}_t} \cdot \mathbf{e}_t \right] = 0 \quad \forall t, \quad (35)$$

where $\mathbf{e}_t$ is the vector of aggregate structural shocks determining the exchange rate in general equilibrium and $\widehat{\Delta \boldsymbol{\eta}_t}$ is the estimated idiosyncratic residual vector of all desks. In other words, the methodology assumes that there is an idiosyncratic risk limit component (uncorrelated to the exchange rate and across time) and that we can recover it. This assumption ensures the validity of our instrument. We provide supporting evidence of this exogeneity in subsections 3.1 and 3.4.

The second assumption in Gabaix and Koijen (2020) concerns the relevance of the instrument, and deals with aggregation. In our model, currency appreciation responds to

aggregate risk aversion defined as $\tilde{\gamma}_t = (\int \gamma_{it}^{-1} + \bar{S}/\Gamma_t)^{-1}$. A first order linearization around a steady state gives:

$$\Delta \tilde{\gamma}_t \approx \sum_d \underbrace{\bar{\gamma}_d^{-2}}_{w_d} \Delta \gamma_{dt} + \underbrace{\bar{S}/\bar{\Gamma}^2}_{w_\Gamma} \Delta \Gamma_t. \quad (36)$$

The weights w_d and w_Γ represent the amount of currency risk absorbed by each desk and the representative financier around the approximation point. Intuitively, it shows how the aggregation weight should be proportional to the relative size of each desk’s marginal risk-allocation in the FX market. The granularity of the dealer-centric market—a few large dealers account for a large share of FX intermediation—allows exogenous idiosyncratic limit changes to affect aggregate quantities.

In practice, the currency-level amount of risk allocated to each currency at any given time by all risk takers in the market is unobserved for two reasons. First, we observe risk-limits at the desk level, but not at the currency-desk level. We can however observe proxies for w_d , discussed in section 3.2. Second, we also do not observe the quantity of risk allocated by non-dealers in the market. This does not imply that our estimates are biased. It does however imply that our estimates identify the effect of dealer risk limits through the dealer channel, not the total effect of changes in aggregate market risk aversion (which would include both dealer and non-dealer channels).

3.1 Risk-limit Shocks

Our granular approach to isolating risk-limit shocks involves two steps. First, we recover dealer-day-level idiosyncratic limit shocks as the residuals from a saturated regression of dealer-day-level limits, controlling for time trends and dealer-specific effects. Specifically, in our baseline specification, we model the log limit changes at the desk-day (d, τ) level using a saturated fixed-effects model:

$$\Delta \log \text{Limit}_{d,\tau} = \sum_{h=1}^{20} \gamma_h \log \text{Limit}_{d,\tau-h} + \sum_{h=1}^{20} \omega_h \text{Usage}_{d,\tau-h} + \alpha_d + \alpha_{b,\tau} + e_{d,\tau}, \quad (37)$$

where d indexes desks, τ indexes days, and $\alpha_{b,\tau}$ is a bank-time effect to control for common changes of limits across different desks (for example, New York and London desks) within a given bank. For instance, the fixed effects would control for changes in the bank’s equity capital or applicable banking regulation. We also include desk fixed effects to account for heterogeneity in limits across desks. In addition to these fixed effects, we control for 20 lags of

the limit and the percentage usage of the limit to isolate surprise changes in limits orthogonal to limit and usage history. This is important because limits may adjust endogenously with usage; for example, when expected currency returns are high, limit usage may be high, and banks may increase risk limits to benefit from the expected exchange rate change.

To understand the variation in changes in (log) limits, Table 3 shows the R^2 of the different sets of fixed effects included in our baseline model, as well as the explanatory power of the full model. Column (3) shows that time fixed effects alone can explain only about 2 percent of the variation. On the other hand, column (4) shows that bank-day fixed effects alone can explain about 13 percent of the variation in limit changes, meaning that common bank factors, such as changes in total bank equity, have considerable explanatory power. The most saturated model in column (5) can explain about 44 percent of the variation. Moreover, the reported coefficients (sum of lags) on the control variables suggest a negative correlation between current and past limit changes—if limits increased in the recent past, they tend to decline again. The positive coefficient on the lagged limit usage shows that risk limits respond endogenously to usage, with higher usage triggering increases in risk limits.

Table 3: Variation in $\Delta \text{Log Limit}$

	$\Delta \text{Log Limit}$					
	(1)	(2)	(3)	(4)	(5)	(6)
Past Limits ($\sum_{h=1}^{20} \gamma_h$)					-0.022*** (0.001)	-0.002*** (0.000)
Past Usage ($\sum_{i=1}^{20} \omega_h$)					1.930*** (0.522)	1.943*** (0.284)
Bank FE	Yes	-	-	-	Yes	-
Desk FE	-	Yes	-	-	Yes	-
Time FE	-	-	Yes	-	Yes	-
Bank $\times$ Time FE	-	-	-	Yes	Yes	-
R-squared	0.000	0.000	0.024	0.125	0.437	0.331
N	98956	98956	98956	98956	98956	98956

Notes: Explanatory power of different models for desk-day level (log) limit changes. Column (5) corresponds to equation (37). *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. *Sources:* FR VV-1, FR Y-14Q, authors' calculations.

We further allow for a common factor structure in the residuals of equation (37) that may not be captured by our extensive set of fixed effects and controls:

$$e_{d,\tau} = \lambda_d' f_\tau + \epsilon_{d,\tau}, \quad (38)$$

where f_τ is a vector of common factors, and λ_d are the associated loadings of desk d . The

residual $\epsilon_{d,\tau}$ represents the truly idiosyncratic components. We estimate the factor structure using principal component analysis of the residuals obtained from a least-squares regression of equation (37). In our baseline analysis we remove one common factor, but results are robust to removing more common factors.

The residual of equation (38) is our baseline measure of idiosyncratic (desk-level) limit innovations. In Appendix B, we show results from alternative models for constructing limit innovations.

3.2 Risk-limit Shock Aggregation

We aim at building the macroeconomic index equivalent of changes in (36), which is a currency-specific measure of aggregate risk tolerance. As predicted by our theory, changes in macro risk-tolerance should be the weighted average of dealer-by-currency specific risk tolerance changes, weighted by the size of their initial risk tolerance. Since our risk limits are desk-specific but not currency-specific, we first compute bank-level limit shocks and then we map bank-level limit shocks to currencies.

Specifically, first we roll up the innovations to the bank-quarter (b, t) level as:

$$\text{Bank Limit Shock}_{i,t} = - \sum_{\tau \in t} \sum_{d \in i} w_{d,t-1}^i \hat{\epsilon}_{d,\tau}, \quad (39)$$

where $w_{d,t-1}^i = \text{Limit}_{d,t-1} / \sum_{d \in i} \text{Limit}_{d,t-1}$ is the relative limit size of desk d (the summation occurs over all desks affiliated with a given bank, $d \in i$) on the last day of the previous quarter. The outer summation then aggregates these daily bank-level shocks to the quarterly frequency. Importantly, note the negative sign in equation (39); an increase in the limit shock variable thus corresponds to a tightening of limits, similar to an increase in γ in our model.

Finally, our benchmark aggregation to the currency-quarter-level follows the following formula:

$$\text{Limit Shock}_{c,t} = \sum_i w_{i,t-1}^c \text{Bank Limit Shock}_{i,t}, \quad (40)$$

where $w_{i,t-1}^c = \text{abs}(\delta_{i,c,t-1}) / \sum_i \text{abs}(\delta_{i,c,t-1})$ is the relative (absolute) position of each bank in the quarter prior to the shock. The relative position is an imperfect proxy for the unobservable fraction of risk limit allocated by each bank in any given currency. We take the absolute value of the position because both long and short positions (negative positions) are informative about risk taking. As Gabaix and Koijen (2020) show, a potential mis-measurement of the size weights in the aggregation affects the efficiency of the GIV estimator but not its

consistency.³³

Appendix Figure A.6 depicts our baseline limit shocks at the currency-quarter level. The shocks are symmetric and recurrent for most considered currencies. All shocks are transitory and not clustered around any specific quarter, in line with our identification of idiosyncratic innovations. For robust inference, all shocks are winsorized at the 5th and 95th percentiles. Finally, we standardize the limit shocks in each estimation sample to have mean zero and unit variance.

3.3 Risk-limit Shock Interaction

While the construction of this limit shock is enough to test the effects on exchange-rate volatility, corollary 2 and proposition 2 predict that the *directional* effects of limit shocks on the exchange rate depend on the interaction between limit shocks and either aggregate demand shifts or idiosyncratic dealer demand. We use two approaches to identify proxies of both demand components.

Our granular data allows us to construct the following interaction objects predicting a directional change in the exchange rate:

$$\text{Dealer Position by Limit Shock}_{c,t} = \frac{1}{\sum_i \text{abs}(\delta_{i,t}^c)} \sum_i \delta_{i,t}^c \times \text{Bank Limit Shock}_{i,t}, \quad (41)$$

where $\delta_{i,t}^c$ is the observed raw net position held by bank i at quarter t on currency c (negative values imply short positions), which we normalize by the total absolute values of net position of all banks in our sample. Equation (41) is a proxy for the partial equilibrium equation $\hat{\mathcal{E}}_0 = \int d\delta_i \gamma_i$ predicted by equation (28). If the risk-bearing capacity of a dealer bank is reduced (larger γ in our model) and a dealer bank has a positive net position (long position) in currency c , this will act as a negative demand shock for that currency and will cause its depreciation (given imperfectly elastic net demand from all other investors).

Our theory shows that dealer positions are generally endogenous to market conditions, and in particular, to movements in exchange rates. Therefore, using (41) could introduce bias in our estimation of exchange rate effects. A more valid identification strategy is to identify the empirical equivalent of the object $\int d\nu_i / \gamma_i di$ in equation (29), that is the aggregation of each dealer *idiosyncratic* currency demand, interacted with the dealers' risk limit. The following interaction factor aims at capturing the latter concept:

³³Appendix Figure A.5 shows the distribution of the weights used in equation (40) by currency, and Appendix Table B.2 shows that the weights are fairly persistent over time.

$$\text{Dealer Shock by Limit Shock}_{c,t} = \frac{1}{\sum_i \text{abs}(\delta_{i,t}^c)} \sum_i \delta_{i,t}^{c,\perp} \times \text{Bank Limit Shock}_{i,t}, \quad (42)$$

$$\delta_{i,t}^{c,\perp} := \hat{\epsilon}_{i,t}^c \quad \text{from} \quad \delta_{i,t}^c = \alpha_{c,t} + \alpha_{i,c} + \sum_{j=1}^4 \phi_j^c \delta_{i,t-j}^c + \epsilon_{i,t}^c. \quad (43)$$

We residualize the position $\delta_{i,t}^c$ with respect to fixed effects and common factors to isolate idiosyncratic position changes of dealer banks unrelated to common market factors. The residuals of this regression are idiosyncratic position innovations which are, by construction, orthogonal to common (currency-specific) market factors and hence represent a idiosyncratic position shift. We also show results when we remove a common factor structure in the residuals using principal component analysis.

Finally, corollary 2 states that the interaction of limit changes with aggregate demand shocks – that is, a proxy for $\hat{\xi}_t - \hat{l}_t$ – should also predict appreciations. Our preferred proxy for aggregate demand shocks is the spread of five-year credit default swaps written on dollar-denominated sovereign debt (but we validate our results with an additional proxy in the appendix). An increase of sovereign default risk generally implies foreign selloffs of domestic assets, such as local currency-denominated government bonds, putting downward pressure on the local currency (Augustin et al., 2016; Hébert and Schreger, 2017). Thus, such a shift represents an increase in (net) demand for dollars. To capture this we compute

$$\text{Agg. Demand Shock}_{c,t} \times \text{Limit Shock}_{c,t} = \log \text{CDS}_{c,t}^\perp \times \text{Limit Shock}_{c,t}, \quad (44)$$

$$\log \text{CDS}_{c,t}^\perp := \hat{u}_{c,t}, \quad \text{from} \quad \log \text{CDS}_{c,t} = \alpha_c + \sum_{j=1}^4 \phi_{c,j} \log \text{CDS}_{c,t-j} + u_{c,t}. \quad (45)$$

Because changes in sovereign credit risk are related to macroeconomic conditions more broadly, we compute surprise innovations to (log) CDS spreads, $\log \text{CDS}_{c,t}^\perp$, as our baseline analysis.³⁴ Moreover, to account for potential common (global) factors across currencies, we also clean the residuals using a principal component analysis in one specification.³⁵ By construction, limit shocks and the CDS demand shifter are likely unrelated, see also Appendix Figure A.11 which confirms a zero correlation between the two variables.

³⁴These CDS innovations are indeed negatively correlated with both changes and innovations in log total inflows, as computed from the IMF Balance of Payments data.

³⁵Further, in the robustness analysis, we also compute innovations to one-month at-the-money implied volatility instead of CDS changes. Implied volatility is a measure of the market's expectation of the forward risk of a currency. It is calculated using currently outstanding over-the-counter (OTC) options contracts on currency futures.

3.4 Narrative Analysis

The assumption that desk-level idiosyncratic risk-limit changes are exogenous to broader market conditions is key to our identification strategy. We provide support for this assumption in two ways: first, through conversations with traders and industry specialists, and second, through a quantitative textual analysis of bank earning calls. Note that due to the confidential nature of our microdata, we cannot describe individual shocks in our data.

Anecdotally, VaR limits indeed do not move in anticipation of specific future exchange rate changes. Rather, according to conversations, risk limits tend to change due to routine changes to the VaR methodology or recalibration, changes in business strategy (that is, top-down decision to reduce the activity of a set of desks), regulatory compliance, or occasionally, trading scandals that cause rethinking of the bank’s underlying risk models. For example, consider the Federal Reserve’s enforcement actions against Deutsche Bank in April 2017.³⁶ Deutsche Bank was found to have violated the Volcker rule, as it had failed to adequately monitor proprietary trading, and was required to improve oversight and controls. In general, traders view VaR risk metrics as slow-moving historical metrics, rather than forward-looking or news-sensitive metrics.

Additionally, a textual analysis of banks’ earnings calls finds that risk-limit shocks are not correlated with institutional or macro events in our sample. We focus on the 20 largest bank-quarter risk-limit shocks in our sample, as defined in equation (39), because they are more likely related to sizeable macro or bank-related events. Figures A.7 and A.8 plot the currency-level idiosyncratic shocks and the variation driven by the 20 largest bank-quarter shocks. The largest bank-quarter-level shocks impact all currencies throughout the sample period, and they correspond to most of the largest movements of our aggregate currency-quarter-level idiosyncratic limit shocks.³⁷

We use a large language model (LLM) to summarize and rank the most important topics discussed in each bank’s earnings call. Appendix E outlines the method used to summarize the most important topics. We compare topics discussed when we observe large idiosyncratic risk-limit changes and all other calls in the sample. Figure A.9 presents a word cloud of the most relevant topics, as perceived by the LLM, in the large-shock sample versus all other bank-quarter observations.³⁸ We do not find any evidence of global or bank-related events being prominent in the large-shock sample, although some words like Leverage ratio, CET1 ratio, and balance sheet appear potentially indicating that during limit shocks, banks are

³⁶These enforcement actions were publicly listed on the Federal Reserve website and reported in the news.

³⁷Note that we do not need our risk-limit shocks to have a granular distribution because all the banks in our sample are granular intermediaries in the foreign exchange market.

³⁸This approach is similar in spirit to Bippus, Lloyd and Ostry (2023), although we rely on the LLM to summarize the textual content and minimize researcher bias.

more concerned about regulatory constraints. We can formally test that there is no significant difference in topics being discussed across a wide range of topic categories. Figure A.10 shows the mean testing estimates in the two samples for the mention of 15 general topic clusters. The only (borderline) significant result is that there is less talk of future outlook under large risk-limit shocks. We find similar results when we explore the correlation between earnings calls sentiment and either raw limit shocks or idiosyncratic risk-limit shocks (see Appendix Table B.3). Overall, these results confirm the exogeneity of idiosyncratic risk-limit changes to market events, in line with our anecdotal evidence.

3.5 Summary Statistics

Our main analysis is based on a currency-quarter-level dataset, which covers data on 30 currencies observed for 30 quarters each. Table 4, Panel A reports summary statistics of the key variables in this datasets. To ensure confidentiality of supervisory information, we winsorize all shock variables, as well as our measure of aggregate banks' net currency positions, at the bottom 5 percent and top 5 percent. During our sample period, foreign currencies tended to depreciate slightly against the dollar, with an average depreciation rate of 1.08 percent. More importantly, fluctuations in exchange rates are sizable, with a standard deviation of 5.9 percent. In 5 percent of the observations, the foreign currency depreciates by more than 10 percent over the quarter, while for the bottom 5 percent of observations, the appreciation is larger than 6.5 percent. The table also reveals that the average net position is centered at zero (albeit recall that *aggregate* positions are long foreign currency for most of the sample, as shown in Figure 3). However, there is substantial variation in net positions.

Table 4: Summary Statistics

	(1) Mean	(2) Std. Dev.	(3) P5	(4) P25	(5) P50	(6) P75	(7) P95	(8) N
<i>Panel A: Currency-Quarter Data</i>								
Δ FX Rate	1.08	5.94	-6.50	-2.15	0.19	3.60	10.45	960
$ \Delta$ FX Rate	4.08	4.54	0.14	1.34	3.00	5.37	11.47	928
Norm. Net Position (Pct.)	-0.10	12.02	-34.06	-2.23	0.31	3.14	25.28	928
Δ Net Position	-0.01	8.08	-10.95	-2.02	0	2.18	9.76	896
Bid-Ask Spread (Pct.)	0.25	0.48	0.01	0.04	0.12	0.27	0.80	959
Δ Bid-Ask Spread	0.05	1.06	-1.70	-0.57	0.02	0.69	1.81	904
Limit Shock (Pct.)	0	0.99	-2.92	-0.15	0.24	0.43	1.61	960
Limit Change (Pct.)	0.01	0.94	-0.32	-0.04	-0.02	0.02	0.51	960
Agg. Demand Position	0	0.26	-0.34	-0.13	-0.01	0.08	0.43	731
Vola Demand Position	0	0.23	-0.31	-0.14	-0.03	0.09	0.40	832
CIP Deviation 1m (B.P.)	11.26	302.06	-162.41	-1.16	20.54	60.55	262.98	888
Δ Log(Turnover)	0.01	0.38	-0.39	-0.12	0.02	0.18	0.48	268
<i>Panel B: Bank-Quarter Data</i>								
Δ Log(Gross Notional)	0	0.23	-0.29	-0.08	0.01	0.11	0.30	256
Δ Margin	-0.16	6.84	-4.18	-0.32	0.09	0.31	1.92	198

Notes: Panel A presents summary statistics for key variables in our baseline currency-quarter level data. Log difference in exchange rate is multiplied by 100. Aggregate banks' net positions are normalized by the sum of limits across banks. Bid-ask spread is presented as a percentage of the mid-price. Limit Shocks and Changes as well as CDS and Volatility Innovations are measured in log differences. CIP deviations are measured in basis points. The sample covers data from 2015:Q3 through 2022:Q4 and 32 currencies. Panel B shows summary statistics for key variables at the bank-quarter level. The sample covers seven banks from 2014:Q3 through 2022:Q. *Sources:* FR VV-1, FR Y-14Q, FR Y-9C, Bloomberg, NY Fed FX Volume Survey, authors' calculations.

4 Empirical Results

In this section, we present our empirical model to test Corollary 1-2, and Proposition 2. We first focus on exchange rate responses as our main macroeconomic variable of interest. Second, we focus on bid-ask spread, net positions, and the evidence supporting the mechanism of intermediation impediment. We also discuss heterogeneous effects across different currencies.

4.1 Limit Shocks and FX Volatility

Empirical Model To test the model predictions in Corollary 1, we regress our measure of realized exchange rate volatility both in reduced form and with an IV strategy:

$$\text{Reduced Form: } |\Delta \text{FX Rate}|_{c,t} = \alpha_c + \alpha_t + \beta \text{Limit Shock}_{c,t} + \gamma' X_{c,t} + u_{c,t}, \quad (46)$$

$$\text{IV: } |\Delta \text{FX Rate}|_{c,t} = \alpha_c + \alpha_t + \beta^{\text{IV}} \widehat{\Delta \text{Limit}}_{c,t} + \gamma' X_{c,t} + u_{c,t}, \quad (47)$$

where $|\Delta \text{FX Rate}|_{c,t}$ is the absolute value of the quarterly change in the log spot exchange rate of currency c , multiplied by 100. The limit shock is the granular limit shock, as identified in equation (40). It represents standardized exogenous shifts in limit changes, with an increase corresponding to a tightening of limits. The $\widehat{\Delta \text{Limit}}$ variable represents the raw log-change of risk limits as observed in our data, and instrumented by the limit shock. An increase in the raw limit change represents instead a limit loosening. The vector of controls $X_{c,t}$ contains lags of the dependent variable.³⁹ Moreover, α_t is a quarter-year fixed effect, and α_c is a currency fixed effect. For statistical inference, we compute robust standard errors two-way clustered at the currency and quarter levels.

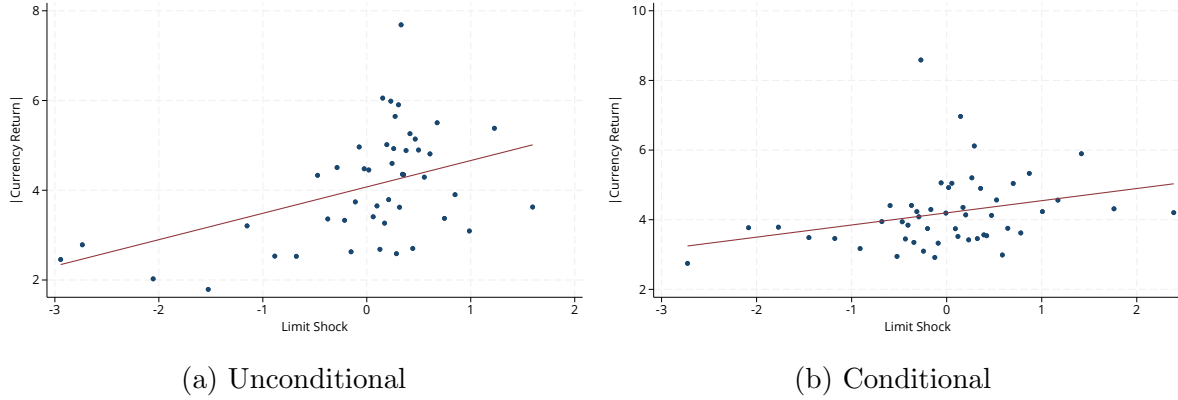
We use both reduced form and IV estimation because the first stage is near the conventional weak-instrument threshold. While there are gains from being able to interpret the IV coefficient in (47) in terms of percentage of raw limit change, the reduced form estimation in (46) is unbiased and consistent and is our preferred estimation throughout the paper.

Baseline Results Corollary 1 predicts that FX volatility increases with tighter limit shocks. Figure 4 visualizes the relationship between the two variables using a binned scatter plot. Panel (a) shows the unconditional relationship using the raw data, while Panel (b) partials out the controls and fixed effects as in our baseline model (equation 46). In both cases, we see a positive relationship—tightening limit shocks increase the exchange rate volatility. Importantly, while identification seem to come from the tails, the figure highlights that the positive relationship is not driven by only a few outliers.

In Table 5, we zoom in on the relationship between limit shocks and exchange rate volatility. Columns (1) and (3) present additional information on the relationship visualized in the binscatter plots. Column (1), which reports the bivariate regression results corresponding to Figure 4, Panel (a), shows that absolute exchange rate change increases by about 59 basis points after a one-standard-deviation tighter limit shocks. This represents a sizable effect of about 14% percent of the in-sample standard deviation of absolute exchange rate change (4.08 percentage points). In terms of explanatory power, limit shocks explain about 1.7% of

³⁹We use 4 lags for variables in log levels (position and spread) and 3 lags for variables in differences (i.e., realized volatility) to keep the number of observations constants. Note also that, because of Frisch–Waugh–Lovell, our parameter estimates on Limit Shock are the same if we, instead of $y_{c,t}$, model $\Delta y_{c,t}$ on the LHS of equation (46). Because the COVID-19 outbreak in 2020:Q1 triggered significant turmoil in financial markets, including the foreign exchange market, we interact the currency fixed effects in our baseline results with a dummy variable indicating 2020:Q1, thereby removing the impact of 2020:Q1 on our estimates.

Figure 4: Effect of Limit Shocks on Exchange Rate Volatility



Notes: This figure shows binned scatter plots of the relationship between exchange rate volatility (y-axis) and Limit Shocks (x-axis), along with a linear fit. Panel (a) shows the unconditional relationship, while panel (b) is based on the controls and fixed effects of our baseline model. The sample is at the quarterly frequency and includes data from 2016:Q3 through 2022:Q4 for 32 currencies. *Sources:* FR VV-1, FR Y-14Q, Bloomberg, authors' calculations.

the overall variation in quarterly exchange rate volatility.⁴⁰ This magnitude may not sound large at first, but considering our tight identification of limit shocks it represents a meaningful number.

Column (3) reports the coefficient estimate on the limit shock for our full model which includes time (quarter-by-year) fixed effects and currency fixed effects, as well as three lags of the dependent variable. (Note that we lose observations due to the lags and controls.) In the fully saturated model (Column 3), which is our preferred specification, the coefficient estimate shrinks somewhat, indicating a still sizable effect about 38 basis points larger exchange rate volatility per one-standard-deviation tighter limits. Column (3), at the bottom, also reports the within R^2 that we compute after partialling out *all* controls and fixed effects. The results show that also in the full model our tightly identified limit shocks account for about 1% of the quarterly exchange rate volatility.

One may also be interested in the effect of limit changes—rather than shocks—on exchange rate movements, for example, for model calibration. In Columns (2) and (4), we report least squares estimates from a regression of the absolute exchange rate change on the change in raw (log) limits, once without fixed effects and controls (column 2) and when including them

⁴⁰In the bivariate regression in column (1), the value shown in the “within R^2 ” row equals the standard R^2 given lack of covariates and fixed effects. Note that the explanatory power of our limit shocks for our outcome variables is small throughout all regressions. Within R^2 's are small and exclude the explanatory power of the control variables. This is not surprising because, by construction, we focus on idiosyncratic limit shocks and remove any common factors that explain the bulk of variation of our outcome variables. We sacrifice explanatory power to achieve narrow identification.

Table 5: The Effect of Limit Shocks on Exchange Rate Volatility

	Δ FX Rate				Δ Limit	
					IV	
	(1)	(2)	(3)	(4)	2nd Stage (5)	1st Stage (6)
Limit Shock	0.588*** (0.128)		0.382*** (0.115)			-0.039* (0.020)
Δ Limit		-0.031 (0.262)		-0.011 (0.134)	-9.789* (5.189)	
Dep. Variable Lags	No	No	Yes	Yes	Yes	Yes
Time FE	No	No	Yes	Yes	Yes	Yes
Currency FE	No	No	Yes	Yes	Yes	Yes
R^2 within (partial)	0.017	0.000	0.009	0.000	0.009	0.014
Anderson-Rubin CI					(-36.77, -4.47)	
Eff. F-Stat						4.444
N	928	928	800	800	800	800

Notes: This table shows the effect of (tightening) shocks to banks' risk limits on the realized exchange rate volatility, as measured by the absolute value of the currency return. In columns (1) through (5), the dependent variable is the absolute value of the quarterly log difference in the spot exchange rate, multiplied by 100. In columns (6), the dependent variable is the log difference in total limits. The independent variables Limit Shock is constructed as described in the main text, and it is standardized to have zero mean and unit variance in each regression sample. Controls include lags of the dependent variable. Quarter fixed effects and currency fixed effects are included. The sample is at the quarterly frequency and includes data from 2016:Q3 through 2022:Q4 for 32 currencies. Within R^2 is computed after partialling out all controls and fixed effects. Robust standard errors are two-way clustered at the currency and quarter levels and are shown in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. *Sources:* FR VV-1, FR Y-14Q, Bloomberg, authors' calculations.

(column 4). In these endogenous regressions, we find an attenuated coefficient estimate on the limit change that is close to zero and statistically insignificant. The small and insignificant estimate suggests that it is important to account for the endogeneity of limit *changes* to market conditions by focusing on identified limit *shocks*. Recall here that we rescaled the limit shocks such that a higher (more positive) value means more tightening, while an increase in Δ Limit means that limits increase (become less tight).

In the last two columns, we use our limit shocks in an instrumental variable (IV) estimation. Specifically, we instrument the endogenous limit change with our granularly identified limit shocks. This approach allows us to consistently estimate the effect of limit changes on exchange rate movements. Column (6) reports the first-stage regression, where we regress endogenous limits on the limit shock. The significant negative coefficient estimate shows

that tightening limit shocks do in fact decrease aggregate limits. The first-stage effective F-statistics of value 4.4 is reported at the bottom of the table. Using the weak instrument test of Montiel Olea and Pflueger (2013), we can reject the null hypothesis of a weak instrument at the 10% significance level. The partial within R^2 shows that limit shocks explain about 1.4% of the variation in limit changes, net of controls and fixed effects.

Column (5) reports the second stage results of the IV estimation. The coefficient estimate suggests that exchange rate volatility decreases (increases) by 9.8 basis points per one-percentage-point increase (decrease) in limits (again, recall that an increase in the limit represents a limit loosening). This key elasticity points to a sizable effect of limit changes on exchange rate volatility. At the bottom of the table, we also report 90% Anderson-Rubin confidence intervals given potential concerns about weak instruments. While relatively wide, the confidence interval does importantly not include the zero, indicating that the estimated elasticity is not only economically sizable but also statistically significant. The IV estimation implies a positive bias on the endogenous limit coefficient. The positive bias is likely due to the interlinked movement between limit usage (which generally increases with volatility) and changes in raw risk limits.

Appendix Table B.4 shows our baseline result using alternatively constructed limit shocks, where we remove, by principal component analysis, three (instead of one as in our baseline) common factors from desk-level limit innovations before aggregating them to the currency level. The results are robust. Consistent with idiosyncratic shocks to FX trading desks driving our limit shock, we find that the price movements are isolated to the foreign exchange market; Appendix Table B.6 shows that interest rate differentials do not seem to be affected by limit shocks, supporting the alternative interpretation of our FX volatility measure as the size of UIP violations.⁴¹

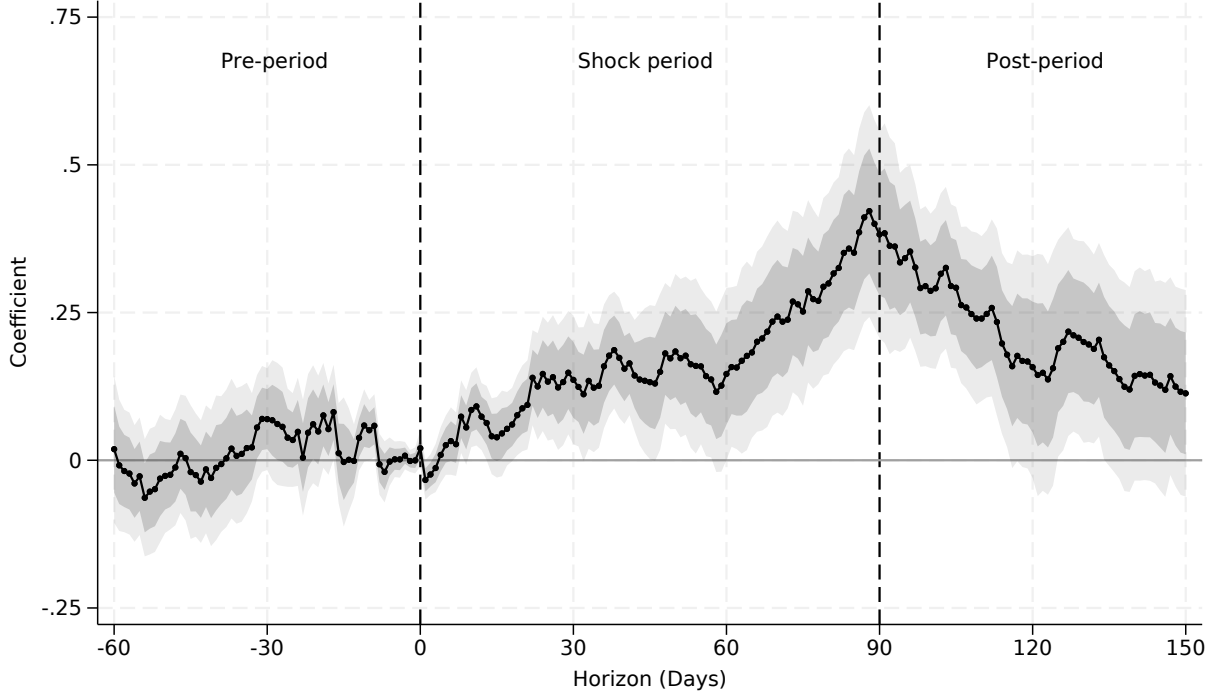
Dynamic Results Our limit shocks vary at the quarterly frequency, which is the baseline throughout our main analysis. However, we can collect data on exchange rates at the daily frequency to estimate the higher frequency response of exchange rates. We do this by estimating the following daily local projection:

$$|\Delta_{\tau-\tau_{\text{start}}(t)}\text{FX Rate}| = \alpha_c + \alpha_\tau + \beta_\tau \text{Limit Shock}_{c,t} + \gamma' X_{c,t} + u_{c\tau} \quad \text{for } \tau \in \{-60, \dots, 150\}.$$

⁴¹Remember that, by construction, limit shocks are orthogonal to bank-time factors, so changes in banks' interest rate trading desk activity that could move rates are accounted for. Using the Y-14 data, we confirm that interest rate sensitivity to a specific currency is not significantly correlated with delta sensitivity to that currency at the bank-currency-quarter level.

Days are represented as τ , and quarters are represented by t . $\Delta_{\tau-\tau_{\text{start}}(t)}$ represents the daily deviation between the day prior to the shock of quarter t , i.e. $\tau_{\text{start}}(t)$, and τ days after. Given the quarterly frequency of the Limit Shock and our definition of the reference period, $\tau_{\text{start}}(t)$, the ‘shock period’ can be any time between $\tau_{\text{start}}(\tau)$ and $\tau_{\text{start}}(\tau) + 90$. As before, our regression model controls for three (quarterly) lags as control variables.

Figure 5: Dynamic (Daily) Exchange Rate Response to Limit Shock



Notes: This figure depicts the dynamic response of the absolute value of the exchange rate to a Limit Shock happening during quarter t . Responses are reported at several daily horizons (on the horizontal axis). Horizon 0 indicates the first day of the quarter during which the Limit Shock occurred, assuming a 90-day quarter. Formally, we estimate equation (46) for the absolute value of the exchange rate change as the dependent variable using our dataset at the quarterly frequency and focus on the estimate of β , as before. However, we now shift the dependent variable (absolute value of the exchange rate change) from the first day of the quarter by τ days back or forward, with $\tau = -60, \dots, 150$. That is, $\tau = 90$ corresponds to the quarter end of the shock quarter, the pre-period corresponds to $\tau < 0$, and the post-period, or the period after the quarter in which we define the shock, corresponds to $\tau > 90$. Thus, the estimate reported at $\tau = 90$ corresponds to the estimate shown in Table 5, column 2 (see the caption of this table for additional information). For each τ , estimated coefficients are presented as black dots. The dark gray area represents a standard error band, and the light gray area represents the 90 percent confidence interval. Robust standard errors are clustered at the currency level. *Sources:* FR VV-1, FR Y-14Q, Bloomberg, authors' calculations.

Figure 5 depicts the exchange rate response for different daily horizons on the horizontal axis. The figure shows that the limit shock leads to a significant exchange rate movement during the shock quarter and for up to about a month and a half afterwards.

The dynamics of the exchange rate effect are consistent with segmented market frictions prevailing in the short run. Due to such frictions, granular limit shocks have an effect on the exchange rate on impact. However, this effect is transitory, likely because, over time, other dealers step in such that the exchange rate is pushed back to the previous equilibrium. Indeed, while Appendix Table B.9 confirms that banks reduce positions after experiencing a limit shock, Appendix Table B.8 shows that banks *increase* their positions when *other* banks' limits become tighter, with an increase in the absolute value of the net position by about 15 percent in response to a one-standard-deviation tighter limit at other banks. The increase in positions of unaffected banks may reduce the magnitude and persistence of granular limit shocks. Importantly, it is possible that aggregate limit shocks—shocks that tighten the limits of *all* dealer banks—have more persistent and larger effects on the exchange rate than those of our *granular* limit shocks identified from idiosyncratic limit changes.

4.2 Demand Shocks and FX Appreciation

Turning to our directional analysis, we study how limit shocks affect currency returns conditional on a shift in dealer demand (as in Proposition 2) and conditional on a shock in aggregate currency demand (as in Corollary 2). While this section relies on the identification of either dealer idiosyncratic flows or aggregate demand flows, the advantage is that such approach allow us to study whether the direction of the currency appreciation moves as in our theory.

We study the effects of shifts in dealer demand using the following regression model:

$$\Delta \text{FX Rate}_{c,t} = \alpha_i + \alpha_t + \beta \text{Dealer Shock by Limit Shock}_{c,t} + \gamma' X_{c,t} + u_{c,t}, \quad (48)$$

where $\Delta \text{FX Rate}_{c,t}$ is the currency return (log difference in the exchange rate), α_t is a quarter-year fixed effect, α_c is a currency fixed effect, and the vector of controls contains lags of FX Rate. We also present results where we use the endogenous variable Dealer Position by Limit Shock $_{c,t}$.

Similarly, we study interaction of aggregate demand shifts and dealer limit shocks, using the following regression model:

$$\Delta \text{FX Rate}_{c,t} = \alpha_i + \alpha_t + \beta \text{Agg. Demand Shock}_{c,t} \times \text{Limit Shock}_{c,t} + \gamma' X_{c,t} + u_{c,t}, \quad (49)$$

where the vector of controls now also contains the uninteracted limit shock and CDS demand shifter. We also present results where we use the endogenous version of the CDS demand shifter.

We first turn to the effects of dealer demand shifts. Table 6 reports the results of differently constructed dealer demand shifts on the exchange rate change. In column (1), we find a negative but insignificant effect when we use the raw (unconditional) net positions in the interaction with the limit shock. This would mean that the currency actually appreciates after dealers experience a negative demand shock. However, raw net positions are endogenous to market conditions and hence could introduce bias to our estimates. Column (2) shows results when we orthogonalize the net positions with respect to controls and fixed effects before interacting them with the limit shocks and aggregating across dealers. In this specification, we find a positive and significant coefficient, consistent with a negative dealer demand shift leading to a depreciation of the foreign currency.

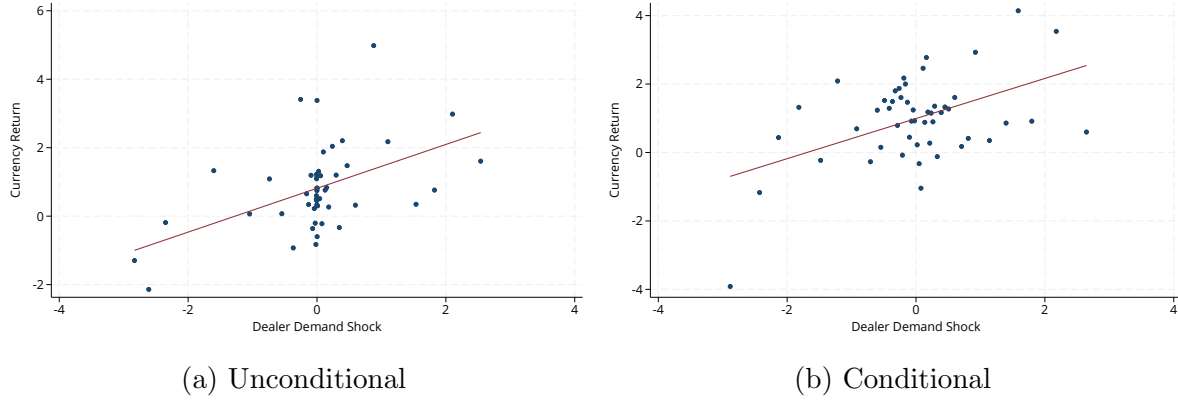
Table 6: Currency Returns, Limit Shocks, and Dealer Demand Shocks

	Δ FX Rate		
	(1)	(2)	(3)
Dealer Position by Limit Shock	-0.112 (0.113)		
Dealer Shock by Limit Shock		0.610** (0.246)	
Dealer Shock ^{PCA} by Limit Shock			0.232 (0.182)
Dep. Variable Lags	Yes	Yes	Yes
Time FE	Yes	Yes	Yes
Currency FE	Yes	Yes	Yes
R^2 within	0.036	0.053	0.038
N	800	800	800

Notes: This table shows the directional effect of (tightening) shocks to banks' risk limits on key endogenous variables in the foreign exchange market depending on net demand shifts. In columns (1) through (3), the dependent variable, Δ FX Rate, is the quarterly log difference in the spot exchange rate, multiplied by 100. The independent variable Limit Shock is constructed as described in the main text, and it is standardized to have zero mean and unit variance in each regression sample. Controls include four lags of the dependent variable. Quarter fixed effects and currency fixed effects are included. The sample is at the quarterly frequency and includes data from 2016:Q3 through 2022:Q4 for 32 currencies. Within R^2 is computed after partialling out all controls and fixed effects. Robust standard errors are two-way clustered at the currency and quarter levels and are shown in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. *Sources:* FR VV-1, FR Y-14Q, Bloomberg, authors' calculations.

Figure 6 visualizes this relationship between the currency return and the dealer demand shifts using binned scatter plots. Panel (a) reports the relationship without any fixed effects and controls, while Panel (b) depicts the relationship controlling for our baseline set of controls and fixed effects as in column (2) of Table 6. Both figures confirm a positive relationship that

Figure 6: Currency Returns, Limit Shocks, and Dealer Demand Shocks



Notes: This figure shows binned scatter plots of the relationship between the currency return (y-axis) and dealer demand shifts constructed from orthogonalized positions (x-axis), along with a linear fit. Panel (a) shows the unconditional relationship, while panel (b) is based on the controls and fixed effects of our baseline model (Table 6, Column 2). The sample is at the quarterly frequency and includes data from 2016:Q3 through 2022:Q4 for 32 currencies. *Sources:* FR VV-1, FR Y-14Q, Bloomberg, authors' calculations.

is not driven by outliers but which holds more broadly across the distribution of demand shocks.

In column (3), we construct the dealer limit shock using net positions that have been further orthogonalized with respect to one principal component (in addition to the controls and fixed effects of our baseline model). Results show that the coefficient estimate decreases but remains positive, while standard errors increase. As a result, we lose significance in this specification. The within R^2 remains meaningful with the demand shifter explaining 4% of the variation, net of controls and fixed effects.

We next turn to the analysis of the CDS demand shifter. Table 7 reports the results. When we investigate the average effect of limit shocks in columns (1), we find that, on their own, they have an insignificant effect on currency returns. As discussed in the model section, this result is theoretically justified: Because limit shocks prompt currency changes in the direction of net demand, theory predicts an ambiguous effect of limit shocks on exchange rates when net demand is not accounted for.

Crucially, our theory predicts that the response of exchange rates and banks' net positions depends on the demand shifter. We test this prediction in columns (2) through (4), which include the interaction term between limit shock and differently constructed demand shifters. Column (2) uses raw changes in CDS spread, column (3) shows our baseline specification that computes innovation in CDS spreads as described in Section 3, and column (4) also removes common factors from the CDS innovations using principle component analysis (Longstaff et al., 2011).

Table 7: Currency Returns, Limit Shocks, and Aggregate Demand Shocks

	Δ FX Rate			
	(1)	(2)	(3)	(4)
Limit Shock	0.045 (0.115)	0.123 (0.112)	0.112 (0.095)	0.072 (0.131)
Agg. Demand Position		0.818 (0.551)		
Limit Shock $\times$ Agg. Demand Position		0.262* (0.130)		
Agg. Demand Shock			1.022*** (0.294)	
Limit Shock $\times$ Agg. Demand Shock			0.331** (0.138)	
Agg. Demand Shock ^{PCA}				1.076*** (0.224)
Limit Shock $\times$ Agg. Demand Shock ^{PCA}				0.290* (0.163)
Dep. Variable Lags	Yes	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes
Currency FE	Yes	Yes	Yes	Yes
R^2	0.470	0.476	0.483	0.496
R^2 within	0.000	0.004	0.007	0.004
N	800	713	702	702

Notes: This table shows the effect of (tightening) shocks to banks' risk limits on currency returns depending on net demand shifts. The dependent variable, Δ FX Rate, is the quarterly log difference in the spot exchange rate, multiplied by 100. The independent variables Limit Shock and the Aggregate Demand Position and Shock are constructed using information from CDS spreads as described in the main text, and they are standardized to have zero mean and unit variance in each regression sample. Controls include four lags of the dependent variable. Quarter fixed effects and currency fixed effects are included. The sample is at the quarterly frequency and includes data from 2016:Q3 through 2022:Q4 for 32 currencies. Within R^2 is computed after partialling out all controls and fixed effects and the uninteracted Aggregate Demand Position/Shock. Robust standard errors are two-way clustered at the currency and quarter levels and are shown in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. *Sources:* FR VV-1, FR Y-14Q, Bloomberg, authors' calculations.

Focusing on our baseline results in column (3), for the average effect of the demand shifter, we find that an increase in net dollar demand (decrease in foreign currency demand) leads to a depreciation of the foreign currency by about 1 percent.⁴² Focusing on the interaction

⁴²Given our standardized limit shock variable, the coefficient estimate on Demand Shifter represents the average effect of a demand shift in our sample.

term, we find that the foreign currency depreciates even more after a demand shock when banks' risk limits become tighter. Specifically, relative to the average effect of a demand shift of about 1 percentage point, this implies a one-third larger effect on exchange rates after a limit shock (a differential effect of 33.1 basis points). This effect is statistically significant and in line with our theoretical predictions. In Appendix Table B.5, we show that results are qualitatively similar when we use implied volatility as a proxy for the aggregate demand shifter instead of CDS spreads.

4.3 Positions, Spreads, and Turnover

Limit shocks affect exchange rates because they impair banks' ability to hold and intermediate currency risk, triggering larger adjustments of the exchange rate in response to demand shifts. We next provide more evidence on the mechanism by analyzing the response of dealers' net positions, bid-ask spreads, and turnover.

In Table 8, we report the effect of limit shocks on the logarithm of dealers' aggregate net positions and the logarithm of bid-ask spreads. Because the sign of the effect on net positions is a function of the underlying demand shift, we look at the (logarithm of the) absolute value of the net position instead, for which our theory predicts a reduction irrespective of the sign of the demand shift. Consistent with our theoretical predictions from Corollary 1, we find that tightening limit shocks lead to a reduction in banks' foreign currency net positions (column 1). The coefficient estimate suggests a sizable 9.6 percent decline in net positions per one-standard-deviation of tightening limit shock.

Column (2) shows that limit shocks also have a significant impact on bid-ask spreads. The estimated coefficient indicates that spreads increase by 6.9 percent in response to a one-standard-deviation limit shock (recall that bid-ask spreads are measured in logs). This response is also consistent with our theoretical predictions. Intuitively, a wider bid-ask spread around the mid-point (which changes depending on net demand shifters) allows the bank to manage its order flow in a way that its net currency position decreases in response to a tighter limit.

Limit shocks affect exchange rates because they impair banks' ability to hold and intermediate currency risk, and we have shown the effect of limit shocks on banks' net positions (delta position) and bid-ask spreads. Because our model focuses on net dollar demand, the notion of total turnover is not defined in our model. However, if dealers' ability to intermediate currency risk is diminished due to a limit shock, it intuitively follows that turnover would be affected.

Unfortunately, turnover data at the bank level by currency are not available in either the

Table 8: Effect of Limit Shock on Net Positions and bid–ask Spreads

	Δ Net Position	Δ Bid Ask Spread
	(1)	(2)
Limit Shock	-0.096** (0.045)	0.069*** (0.023)
Dep. Variable Lags	Yes	Yes
Time FE	Yes	Yes
Currency FE	Yes	Yes
R^2	0.599	0.719
R^2 within	0.006	0.008
N	800	758

Notes: This table shows the effect of (tightening) shocks to banks’ risk limits on currency returns depending on positions and bid–ask spreads. In column (1), the dependent variable is the logarithm of the absolute value of dealers’ aggregate net position. In column (2), the dependent variable is the logarithm of the relative bid–ask spread calculated as a percent of midpoint. The independent variable Limit Shock is constructed as described in the main text, and it is standardized to have zero mean and unit variance in each regression sample. Controls include four lags of the dependent variable. Quarter fixed effects and currency fixed effects are included. The sample is at the quarterly frequency and includes data from 2016:Q3 through 2022:Q4 for 32 currencies. Within R^2 is computed after partialling out all controls and fixed effects. Robust standard errors are two-way clustered at the currency and quarter levels and are shown in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. *Sources:* FR VV-1, FR Y-14Q, Bloomberg, authors’ calculations.

VV-1 or the Y-14Q data we use in our main analysis. We therefore resort to additional, more aggregate data sources. To start, we examine turnover at the *currency level* using information provided in the (publicly available) biannual Federal Reserve Bank of New York’s FX Volume Survey. This survey captures the size and structure of foreign exchange activity in North America, and we use these data from 2016 onward, in line with the availability of our other variables. The most recent survey builds on responses from 21 leading institutions active in the North American market.⁴³ Our key response variable is the log difference in dealer turnover. Turnover is based on total interdealer turnover in all foreign exchange instruments (spot, forward, swap, options). Limit shocks are quarterly and survey dates are matched to the corresponding quarter of the limit shock.

We use the same regression model of equation (46) to study turnover responses in this dataset. Table 9, columns 1 and 2, presents the estimation results. Given the small time series sample, we present results without (column 1) and with (column 2) currency fixed effects. The results show that tighter limit shocks indeed reduce turnover: The coefficient estimates

⁴³These data are compiled by the NY Fed in collaboration with the Foreign Exchange Joint Standing Committee in London. The FXC agreed to collect one month’s foreign exchange turnover data covering customer, product, currency pair, and execution data in April and October. See <https://www.newyorkfed.org/fxc/fx-volume-survey> for full details.

indicate that turnover falls by about 3 percent in response to a one-standard-deviation risk-limit shock. To put this number into perspective, the average daily turnover in our sample equals about \$19 billion; hence, the estimated reduction of 3 percent translates into a sizable reduction in average daily turnover by about \$573 million.⁴⁴

Table 9: Response of Turnover, Gross Notional, and Margin to Limit Shocks

	Currency-Time		Bank-Time	
	(1)	(2)	(3)	(4)
	Δ Turnover	Δ Turnover	Δ Gross Notional	Δ Margin
Limit Shock	-0.038** (0.016)	-0.028* (0.013)		
Bank Limit Shock			-0.011* (0.006)	0.057* (0.029)
Dep. Variable Lags	No	No	Yes	Yes
Time FE	Yes	Yes	Yes	Yes
Currency FE	No	Yes	-	-
Bank FE	-	-	Yes	Yes
Within R-squared	0.009	0.004	0.112	0.223
N	268	268	144	144

Notes: This table shows the response of turnover, gross notional, and margin to Limit Shocks at the currency level and bank level. In columns (1) and (2), the dependent variable is the log difference in dealer turnover reported in the biannual public NY Fed FX Volume Survey from 2016 through 2022. Estimates are based on total turnover across all instruments of reporting dealers. Limit Shocks are quarterly, and survey dates are matched to the corresponding quarter of the Limit Shock. Columns (3) and (4) are based on Y-9C bank-quarter-level data. In column (3), the dependent variable is the change in the (logarithm of) adjusted gross notional outstanding. In column (4), the dependent variable is the change in total FX trading revenue (from cash instruments and derivative instruments) relative to the adjusted gross notional (in basis points). The sample includes a constant set of seven banks from 2014:Q3 through 2022:Q4. Limit shocks are standardized to have mean zero and unit variance in each regression. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. *Sources:* NY Fed FX Volume Survey, FR Y-9C, FR VV-1, FR Y-14Q, authors' calculations.

Next, we study turnover and margin at the *bank level* using the public Y-9C data. These data are at the BHC-quarter level (but consistent with the rest of the paper, we use the term “banks” to refer to BHCs), and they contain information on banks’ foreign exchange trading activity. Unfortunately, the data do not break down foreign exchange activity by currency. We use two key variables from the Y-9C. First, we proxy turnover using gross notional values

⁴⁴The turnover response does not depend on the net demand shifter, thereby resembling the bid-ask spread response, for which we also find an average effect of limit shocks but no differential effect depending on demand shifters.

of foreign exchange trading.⁴⁵ Second, we compute margins from foreign exchange trading as total trading revenue (from cash instruments and derivative instruments) relative to adjusted gross notional values. This income includes intermediation income from bid–ask spreads as well as gains or losses from net positions.⁴⁶

We use the following linear regression model to estimate the effect of limit shocks on bank-level turnover and margin:

$$\Delta y_{i,t} = \alpha_i + \alpha_t + \beta \text{ Bank Limit Shock}_{i,t} + \gamma' X_{i,t} + u_{i,t},$$

where $y_{i,t}$ is either the (log) adjusted gross notional value or the margin measure (in basis points). Bank Limit Shock $_{i,t}$ is defined in equation 39. The regression also includes bank fixed effects (α_i) and quarter fixed effects (α_t) as well as four lags of the dependent variable. Summary statistics of the bank-quarter variables are presented in Panel B of Table 4.

The estimation results are presented in Table 9, columns (3) and (4). Column (3) shows that, in response to tighter limit shocks, total gross notionals at the bank level decrease, consistent with our findings from the cross-currency analysis in columns (1) and (2). The estimated coefficient suggests that gross notionals decrease by about 1.1 percent in response to a one-standard-deviation limit shock. Column (4) shows that the total foreign exchange margin increases, consistent with the findings on currency-level bid–ask spreads. The estimate implies an increase of about 7.5 per 100 basis points in response to a one-standard-deviation limit shock, which equals about 13 percent relative to the average trading margin of 0.59 basis points. Together, both the quantity decline and the price increase in response to limit shocks are evidence that the adjustments in currency returns are driven by a contraction of global banks’ currency intermediation.

4.4 Deviations from Covered Interest Parity

To provide additional evidence of the mechanism driving our results, we next examine covered interest parity (CIP) deviations. CIP states that the forward premium (that is, the forward exchange rate relative to the spot exchange rate) is equal to the interest rate differential. Because CIP constitutes a riskless arbitrage opportunity, deviations are associated with limits to arbitrage. The risk-limit shocks that we identify constitute constraints to the

⁴⁵These gross notional values capture all contracts (spot and all derivatives) that are outstanding at the quarter-end reporting date. We adjust the reported raw quarter-end gross notional value for cross-bank differences in the shares in different instruments that differ in maturity to estimate the implied turnover; see Figures A.1 and A.2 for details.

⁴⁶Foreign exchange trading income information is available only for holding companies with \$5 billion or more in total assets that reported total trading assets of \$10 million or more for any quarter of the preceding calendar year.

intermediation capacity of banks in the foreign exchange market; thus, we should expect these shocks to result in CIP deviations.

For example, one mechanism that would lead to such CIP deviations could be the differential VaR contribution of forwards vs spot contracts. Dealers might not be involved in the full CIP trade, for instance they might only quote forwards and provide spot. If forwards contribute more to the VaR utilization, tightening limit shocks would lead to a larger spread increase and position adjustment for forwards. Indeed, a forward contract's VaR is often larger than a spot position because the former have more risk factors, most notably the longer time horizon. Another mechanism, similar to De Leo, Keller and Zou (2024), is that tighter risk limits on UIP-exposed dealer banks trigger more pressure on (risk-limited) arbitrageurs in the forward market.

We define the (log approximate) n -month CIP deviation between USD LIBOR and country c 's XIBOR on day t as:

$$\Phi_{c,n,t}^{CIP} = r_{c,n,t} - r_{USD,n,t} - \rho_{c,n,t}^{CIP},$$

where $\rho_{c,n,t}$ is the n -month market-implied forward premium for hedging currency c against the US dollar. Specifically, the forward premium is constructed as

$$\rho_{c,n,t}^{CIP} = \frac{360}{d_{c,n,t}} (\ln F_{c,n,t} - \ln \mathcal{E}_{c,t}),$$

where $\mathcal{E}_{c,t}$ is again the spot exchange rate at time t , $F_{c,n,t}$ is the forward foreign exchange rate for tenor n (both expressed in units of foreign currency per units of US dollars), and $d_{c,n,t}$ is the number of days until maturity of the forward rate contract of tenor n for the given currency on date t .⁴⁷

Table 10 confirms that limit shocks trigger CIP deviations. Specifically, one-month CIP deviations widen by 6.02 basis points in response to a one-standard-deviation limit shock, with similar effects for longer tenors. This widening of CIP deviations in response to our shocks is evidence that bank intermediation ability is a key mechanism behind our baseline findings. Importantly, these price effects are isolated to the foreign exchange market; Appendix Table B.6 shows that interest rate differentials do not seem to be affected by limit shocks.

4.5 Currency Heterogeneity

Our baseline analysis focuses on testing predictions derived from a two-currency model. How would the dealer bank choose the spread and net position in a multi-currency setup? In such

⁴⁷Overall, the CIP deviation expressed in this way gives the difference between the synthetic borrowing cost in foreign currency and the direct borrowing cost in US dollars. Thus, a positive CIP basis means that the synthetic US dollar rate is more expensive than the direct USD rate.

Table 10: Effect of Limit Shocks on CIP deviations

	$\Delta \text{CIP Deviation} $		
	(1) One Month	(2) Three Month	(3) Six Month
Limit Shock	2.877** (1.189)	1.283* (0.707)	1.626* (0.876)
Dep. Variable Lags	Yes	Yes	Yes
Time FE	Yes	Yes	Yes
Currency FE	Yes	Yes	Yes
R^2	0.559	0.619	0.688
R^2 within	0.002	0.001	0.002
N	727	714	628

Notes: This table shows the effect of shocks to banks' risk limits on CIP deviations. The dependent variable is the change in the absolute value of the CIP deviation, measured in basis points. Interest differential are computed using XIBOR rates. The independent variable Limit Shock is constructed as described in the main text, and it is standardized to have zero mean and unit variance in each regression sample. Controls include lags of the dependent variable. Quarter fixed effects and currency fixed effects are included. Robust standard errors are two-way clustered at the currency and quarter level and are shown in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. *Sources:* FR VV-1, FR Y-14Q, Bloomberg, authors' calculations.

a setting, bank i has to decide for each currency c , the spread s_{ic} and net position δ_{ic} to maximize expected income from intermediation services and returns on net positions, net of holdings. We can express equation (1) in vectorized form, where vectors are represented in bold and where each vector contains currency- c specific prices and/or quantities. :

$$\max_{\{\mathbf{s}_{it}\}_{t=0,1}, \boldsymbol{\delta}_i} \sum_{t=0}^1 \left[(\mathbf{s}_{it} - \mathbf{1})^\top \mathbf{x}_{it} - \frac{1}{2} \mathbf{x}_{it}^\top \mathbf{C}_i \mathbf{x}_{it} \right] + \mathbb{E}_0 \left[\boldsymbol{\varepsilon}_0 \oslash \boldsymbol{\varepsilon}_1 - \mathbf{1} \right]^\top \boldsymbol{\delta}_i, \quad (50)$$

$$\text{s.t. } \mathbb{E}_0 \left[\boldsymbol{\varepsilon}_0 \oslash \boldsymbol{\varepsilon}_1 - \mathbf{1} \right]^\top \boldsymbol{\delta}_i \geq \gamma_i \boldsymbol{\delta}_i^\top \Sigma \boldsymbol{\delta}_i, \quad (51)$$

where the holding cost is a scaled version of the net positions' return variance-covariance, Σ . As before, a larger γ_i means holding costs increase, for a given covariance of currency returns.

Focusing on the first-order conditions for banks' positions gives the the vector of optimal net positions

$$\boldsymbol{\delta}_i = \frac{1}{\gamma_i} \Sigma^{-1} \mathbb{E}_0 \left[\boldsymbol{\varepsilon}_0 \oslash \boldsymbol{\varepsilon}_1 - \mathbf{1} \right], \quad (52)$$

where the $\oslash$ operator is the element-wise ratio. Thus, the optimal net positions depend on the inverse of the return covariance, the expected return, and the scaling parameter capturing the risk tolerance of the bank. While this equation (and the general equilibrium) can be solved

analytically to the first-order, the solutions become algebraically involved. Yet, we can gain some understanding of the basic additional forces that come into play in the multi-currency setting. For example, suppose that the covariance terms are zero. Then, each net positions would be:

$$\delta_{ic} = \frac{\mathbb{E}_0[\mathcal{E}_{c,0}/\mathcal{E}_{c,1} - 1]}{\gamma_i \sigma_c^2},$$

such that a higher variance decreases the holding. In fact, with zero covariances, the risk-tolerance parameter γ_i enters multiplicatively with σ_c^2 , meaning that larger variances exacerbate the effects of limit shocks. More intuitively, in the general model, expected returns on positions that have higher variance and/or are highly correlated with other positions are down weighted by more in the optimal allocation.

We estimate currency-specific exchange rate effects using the following extension of our baseline regression model for the realized exchange rate volatility (equation 46):

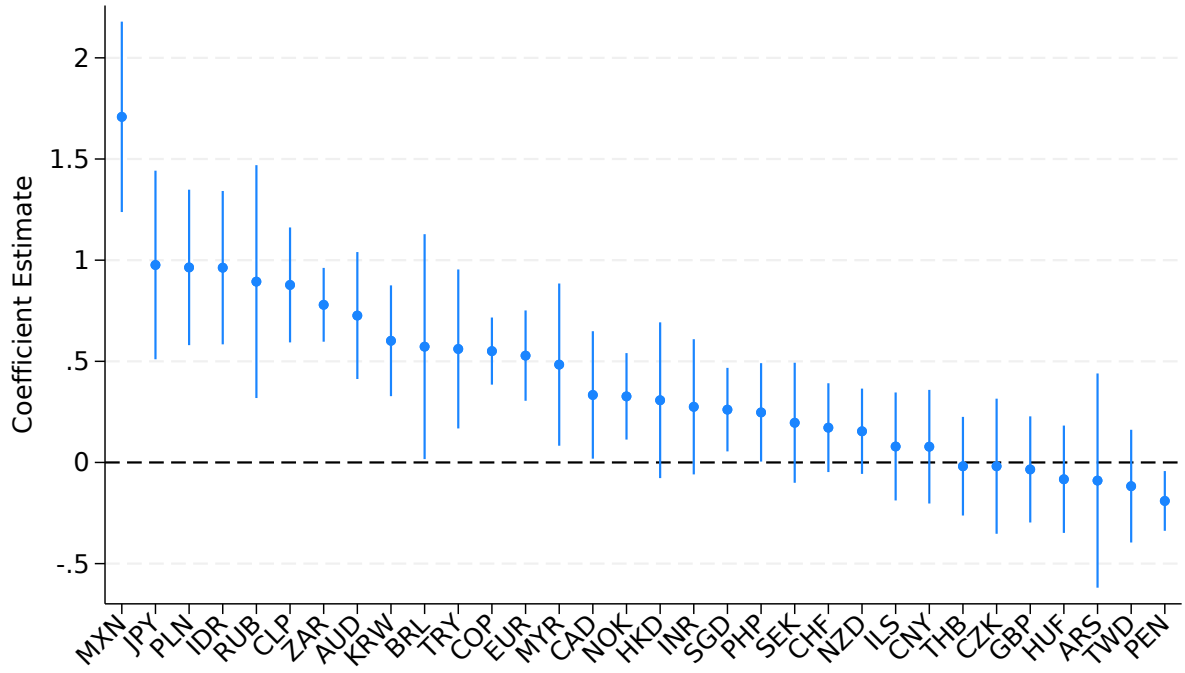
$$|\Delta \text{FX Rate}_{c,t}| = \alpha_c + \alpha_t + \beta_c \text{Limit Shock}_{c,t} + \gamma' X_{c,t} + u_{c,t};$$

that is, we allow the slope coefficient on the limit shock to be currency-specific, while pooling all other parameters. The controls include four lags of the dependent variable.

Figure 7 reports the country-specific estimates along with 95 percent confidence intervals. For the large majority of currencies, we estimate significant and positive coefficients, confirming that the average effect reported in the previous section holds across a wide variety of currencies. In Appendix Table A.13, we report country-by-country results when we regress the currency return on the interaction between CDS innovations and limits shock (compare the pooled results in Table 7).

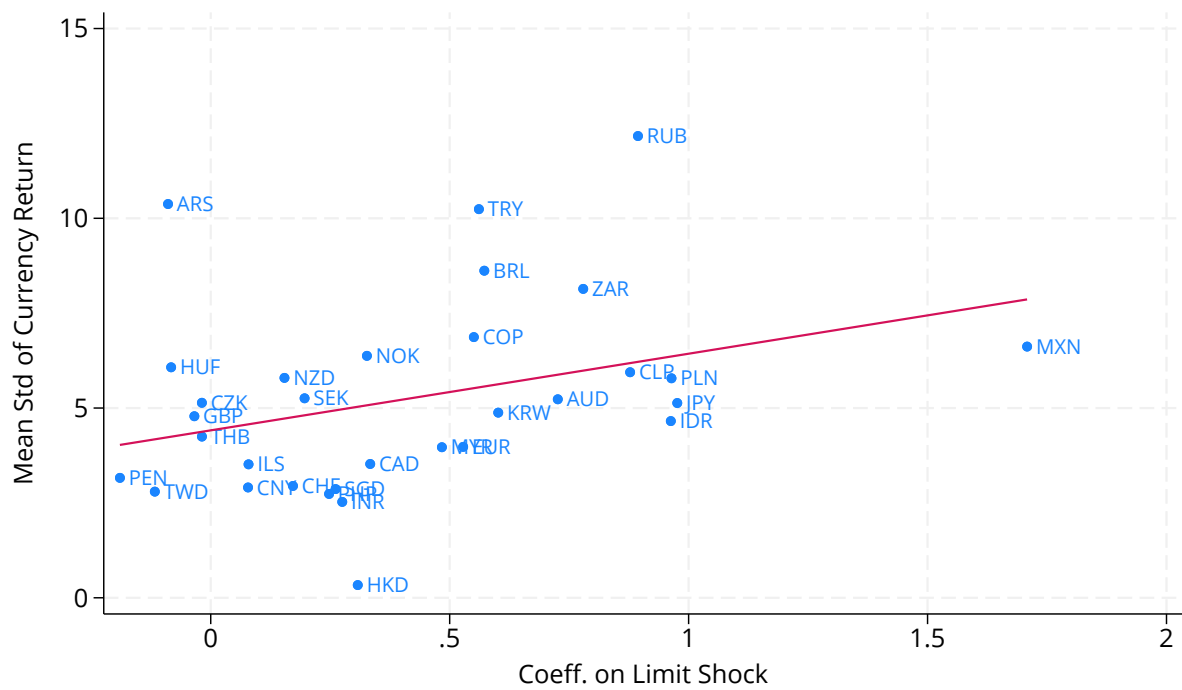
To better understand the drivers of the currency heterogeneity, we use the insights from our model extension suggesting that limits shocks should be larger for more volatile currencies. Figure 8 presents a scatter plot of the estimated country-specific limit shock effect against the standard deviation of the quarterly currency return during our sample period. The results show that indeed the limit shock effects are larger for currencies that are more volatile (a correlation of about 0.5). That said, this result cannot be interpreted causally due to a lack of exogenous variation in foreign exchange volatility. In fact, reverse causality could be at play—at least to some degree—as a large sensitivity to limit shocks could lead to higher FX volatility, as suggested by our earlier findings.

Figure 7: Effects of Limit Shock on Exchange Rate Volatility by Currency



Notes: This figure depicts currency-specific estimates of the effect of Limit Shocks on the absolute value of the currency return. The regression equation is $|\Delta e_{c,t}| = \beta_{1,c} \text{Limit Shock}_{c,t} + \alpha_i + \alpha_t + \gamma' X_{c,t} + u_{c,t}$. Limit Shock is standardized by currency to have mean zero and unit variance within each currency. Point estimates are shown as dots, and the bars represent 95 percent confidence intervals. *Sources:* FR VV-1, FR Y-14Q, Bloomberg, authors' calculations.

Figure 8: Heterogeneous Limit Shock Effects and FX Rate Volatility



Notes: This scatter plot shows the relationship between the Limit Shock effects reported in Figure 7 and the standard deviation of quarterly currency returns during our sample period. *Sources:* FR VV-1, FR Y-14Q, Bloomberg, authors' calculations.

5 Conclusion

Understanding exchange rate dynamics is one of the major research topics in international macroeconomics and finance. In line with recent theoretical papers highlighting the potential role of financial intermediaries in currency returns, this paper shows that changes in global banks' risk-taking capacity have a statistically and economically sizable effect on exchange rates. We exploit an array of detailed supervisory microdata, which provide a unique window into global banks' currency trading activities and allow us to identify exogenous risk-limit shocks.

Specifically, using desk-level risk-limit information from supervisory data, we construct exogenous risk-limit shocks of global banks' trading desks—the central intermediaries in the foreign exchange market. We also exploit supervisory information on global banks' net position in different currencies, thereby enabling us to draw a detailed picture of their FX trading activity and its impact on exchange rates.

Our empirical results are consistent with a model of currency intermediation under risk constraints. Theory predicts that in response to tighter limits and a net dollar demand shift, banks reduce their net positions, increase bid–ask spreads, and intermediate fewer dollars, such that the exchange rate adjusts to ensure market clearing in the new equilibrium.

In contrast to the results from previous studies that focus on constraints on the holders of ultimate risk, our key finding highlights the role of market makers, which may hold only a small share of the total foreign exchange risk. Yet, due to these market makers' central intermediation function, shocks to their risk limits have sizable implications for currency pricing.

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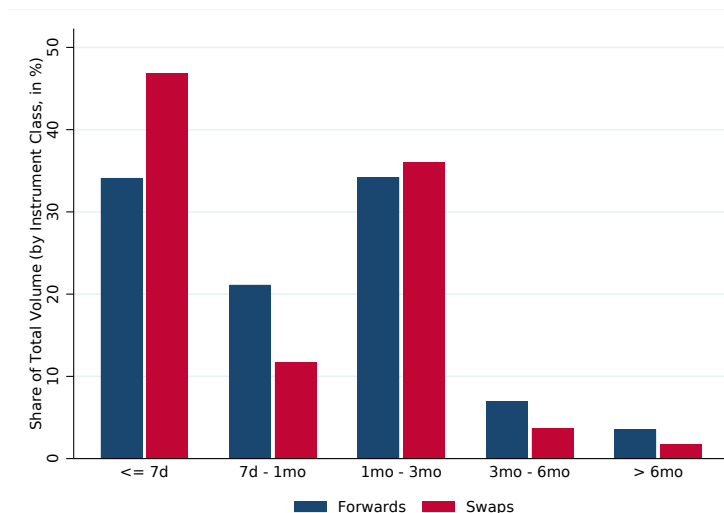
Online Appendix

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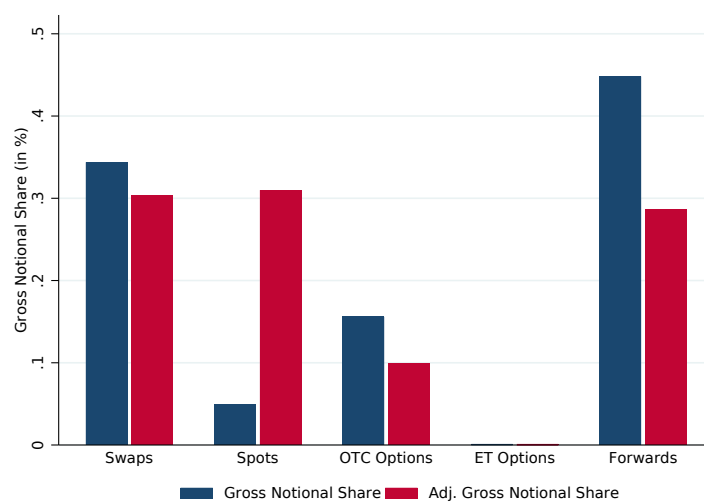
A Additional Figures

Figure A.1: Maturity Breakdown of Derivatives and Adjustments of Notionals Values



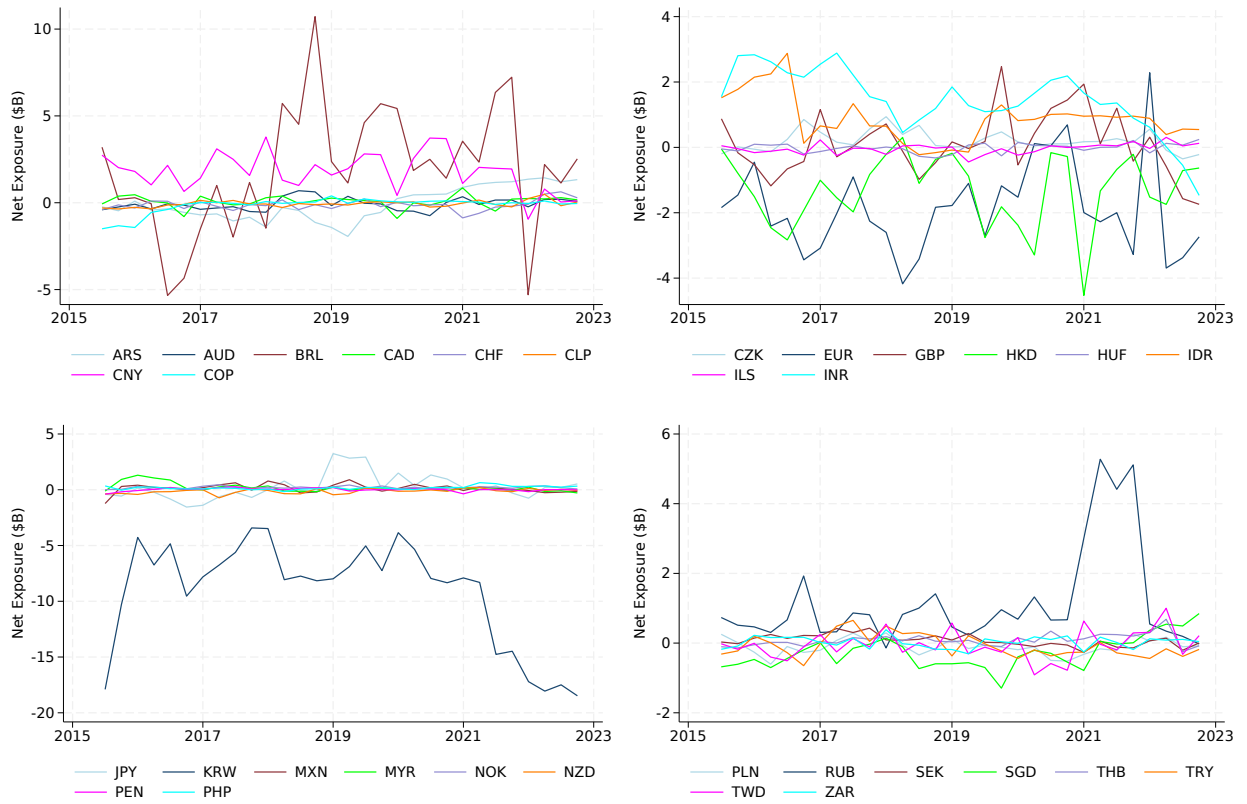
Notes: The figure shows the within-instrument (forwards, swaps) share of daily average turnover in 2019 for every given maturity class as reported by the Bank for International Settlements. To adjust the end-of-quarter gross notional values, we proceed as follows: Under the assumption that the maturities within each maturity class follow a uniform distribution, we estimate the average maturity in each bucket as the midpoint between the start and end dates (for example, the midpoint for the class "over 7 days and up to 1 month" is $(30d - 8d)/2 = 11d$). For the class ">6mo," we assume a midpoint of 360 days. Finally, we estimate the maturity for each instrument class as the turnover-weighted average of midpoints. We then divide the length of one quarter (90 days) by the maturity estimates. Multiplying the result with end-of-quarter gross notionals allows us to estimate gross notional values for the entire quarter. *Sources:* BIS, authors' calculations.

Figure A.2: Gross Notional Shares vs. Adj. Gross Notional Shares



Notes: The figure shows the share of total gross notional volume for each instrument class in 2019:Q1. Adjusted gross notionals are calculated following the steps outlined in the footnote for figure A.1. The shares for adjusted and unadjusted gross notionals are calculated by dividing by the adjusted/unadjusted total gross notional, across banks and instruments. *Sources:* BIS, authors' calculations.

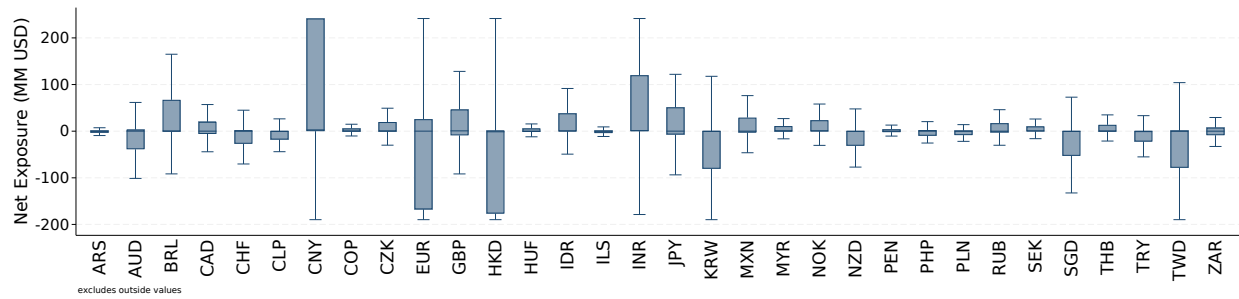
Figure A.3: Net Positions over Time, by Currency



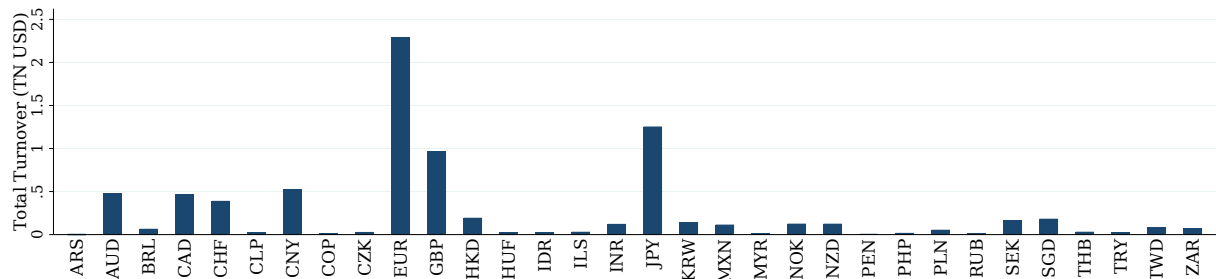
Notes: This figure shows the sum of net positions across all banks, by currency, for each quarter. Positive values correspond to long positions to foreign currency. *Sources:* FR Y-14Q, authors' calculations.

Figure A.4: Net Positions and Turnover, by Currency

(a) Net Positions

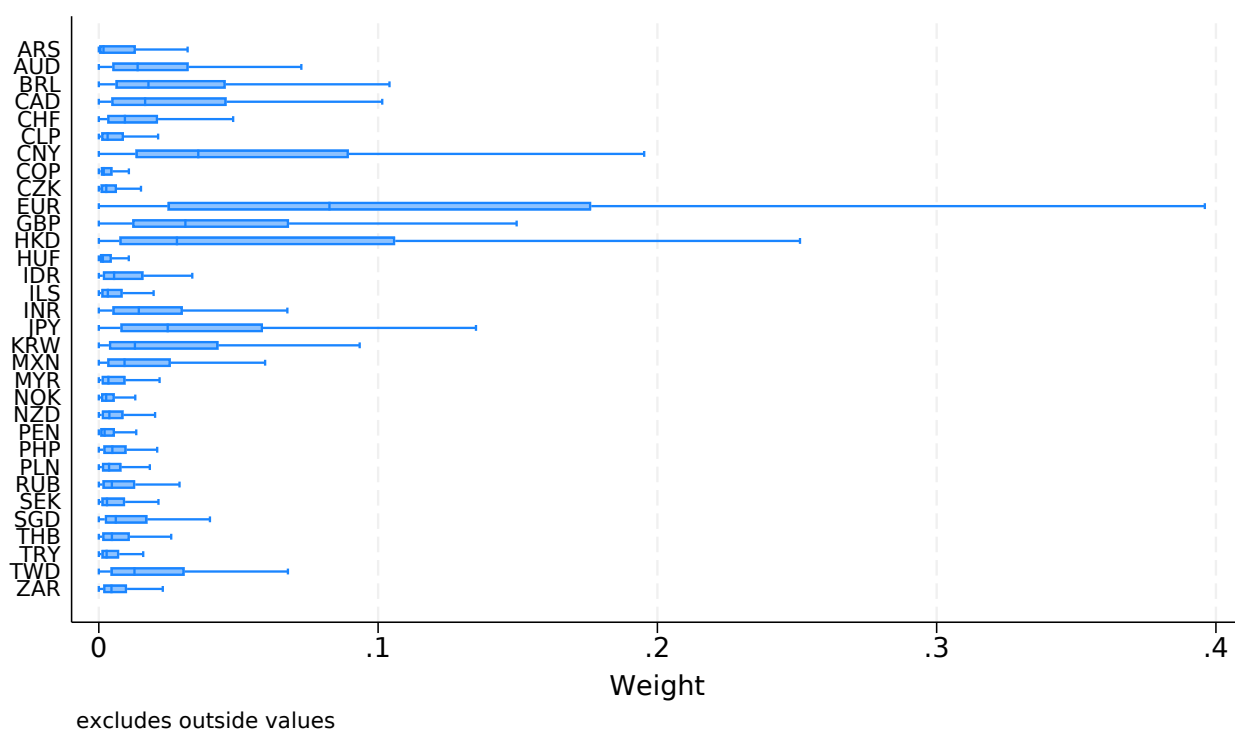


(b) Global Turnover



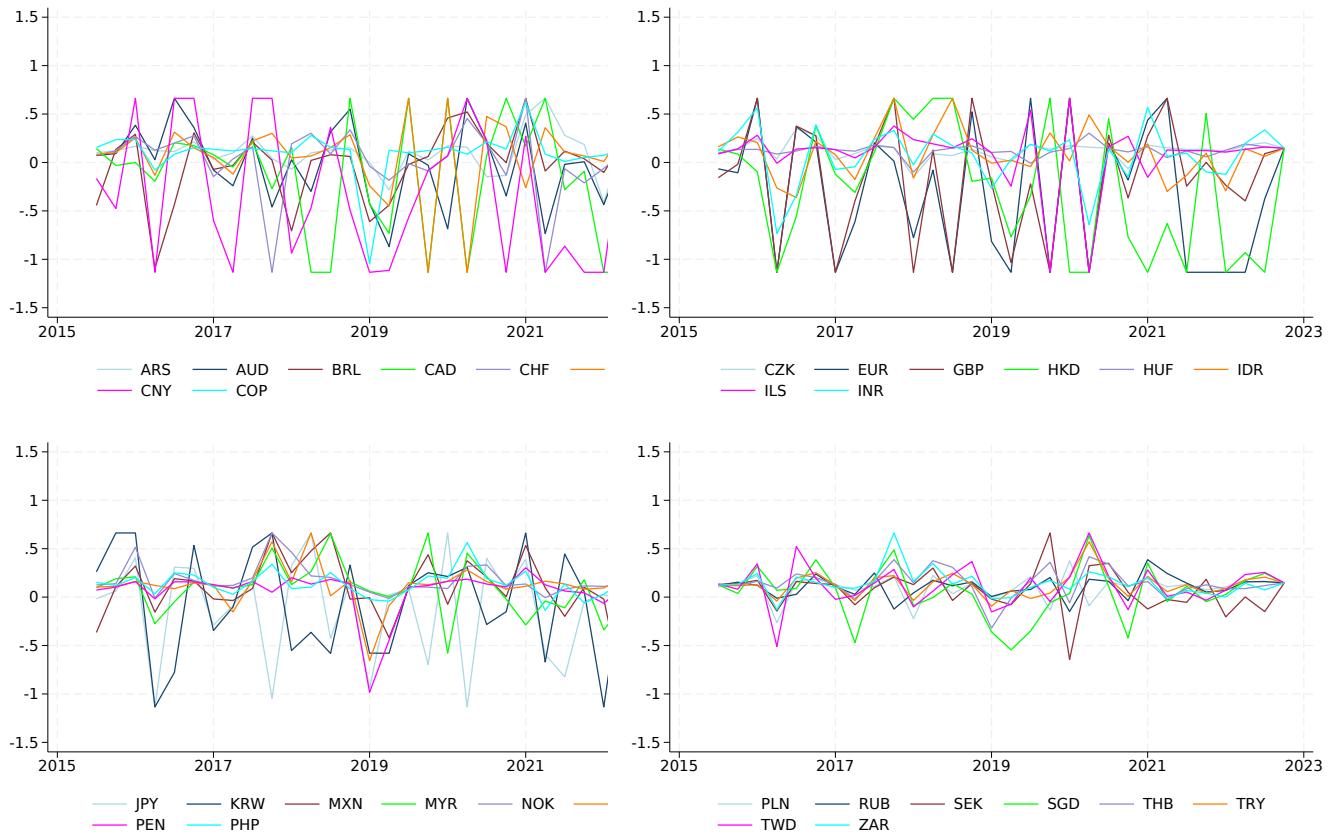
Notes: Panel (a) shows box plots summarizing the distribution of net positions for each currency at the bank-quarter level from FR Y-14Q. The underlying data are winsorized at the 5th and 95th percentiles. Panel (b) shows the daily average turnover by currency in April 2022, in trillions of US dollars from the Triennial Central Bank Survey of foreign exchange and over-the-counter (OTC) derivatives markets. *Sources:* FR Y-14Q, BIS, authors' calculations.

Figure A.5: Weights Used in Shock Aggregation



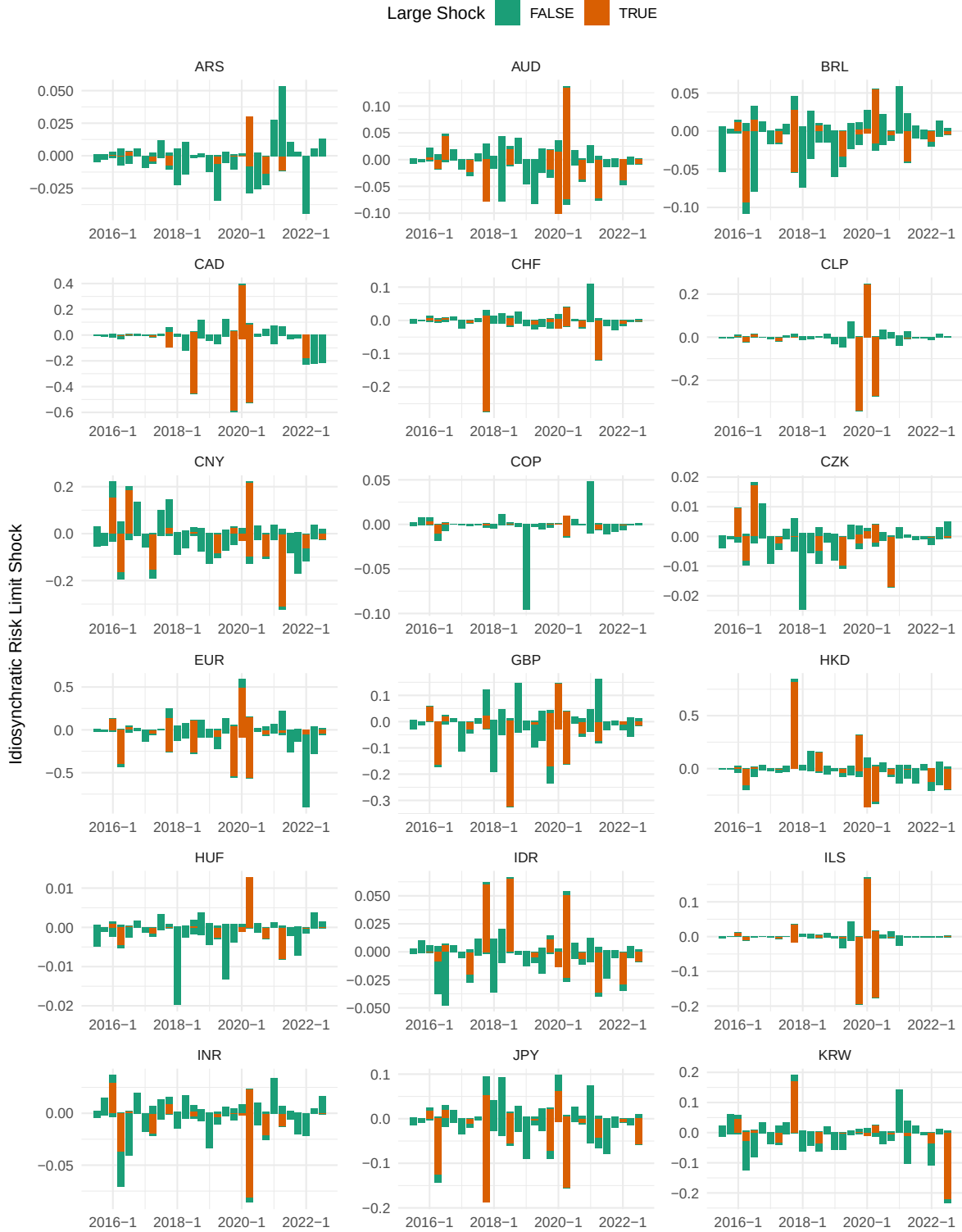
Notes: Distribution of weights used in the aggregation from bank-quarter limit innovations to currency-quarter shocks. *Sources:* FR VV-1, FR Y-14Q, Bloomberg, authors' calculations.

Figure A.6: Limit Shocks over Time



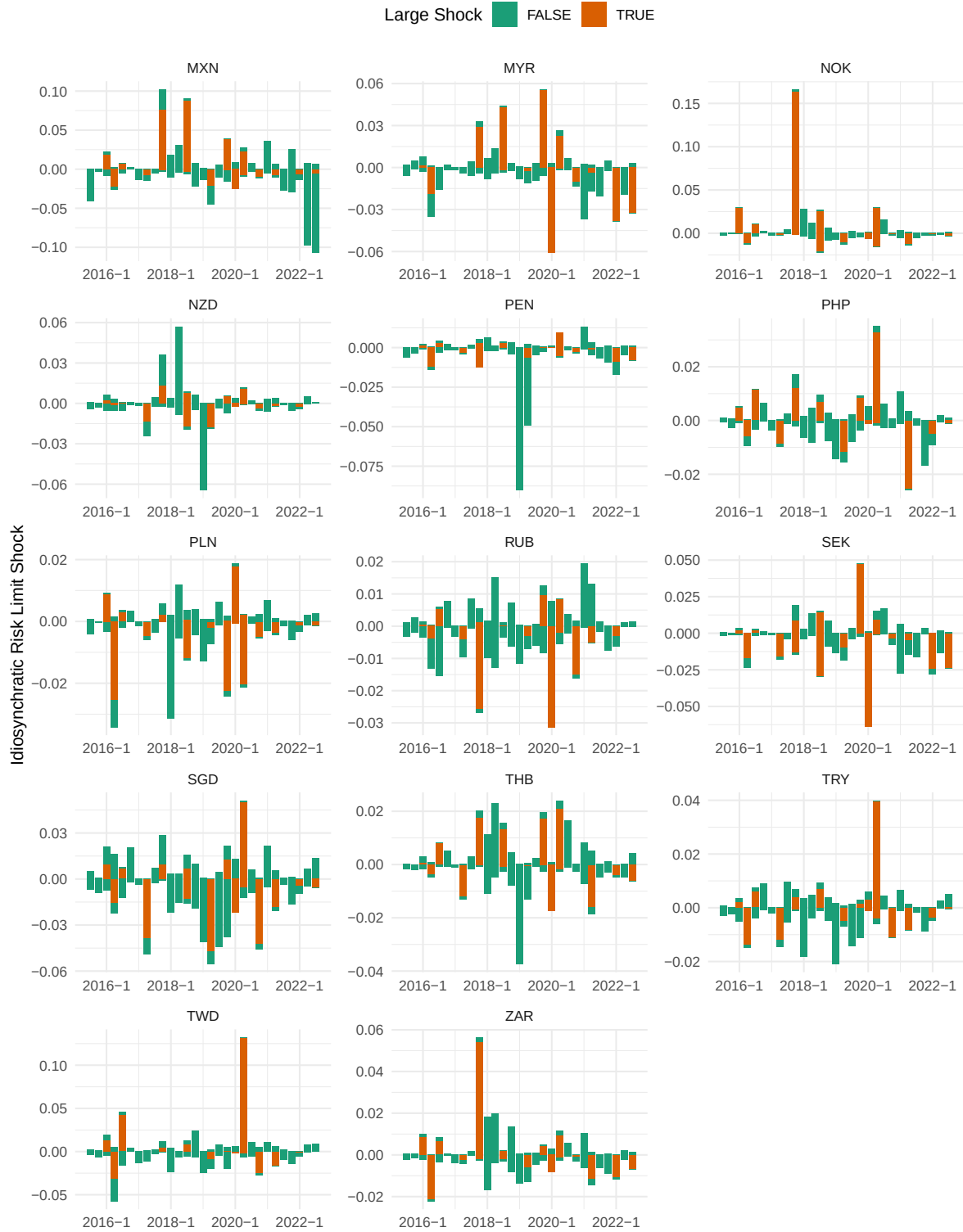
Notes: Baseline risk limit shocks over time identified from idiosyncratic dealer-level limit changes. *Sources:* FR VV-1, FR Y-14Q, authors' calculations.

Figure A.7: How the 20 Largest Bank-quarter Shocks Impact the Currency-level Shock



Notes: Each panel represents the variation of the currency-level risk-limit shock as defined in equation (40). The red bars represent the contribution of the 20 largest bank-quarter risk-limit shocks as defined in equation 39 to the currency-level shock variation. The green bars represent all the residual variation. *Sources:* FR VV-1, FR Y-14Q, authors' calculations.

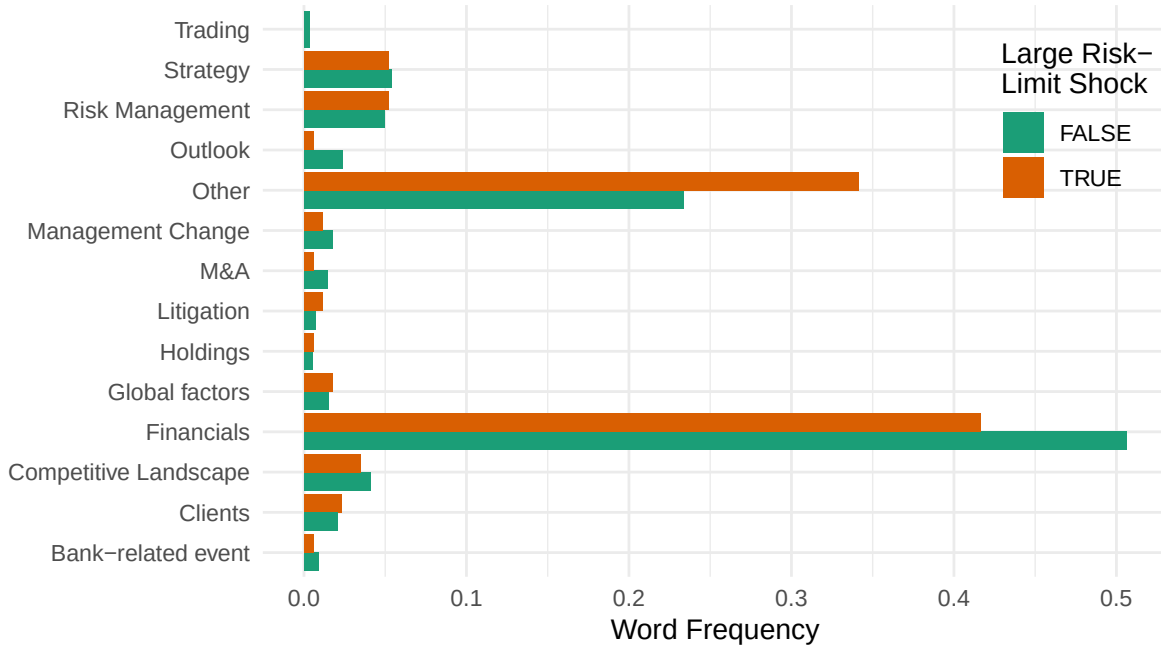
Figure A.8: How the 20 Largest Bank-quarter Shocks Impact the Currency-level Shock



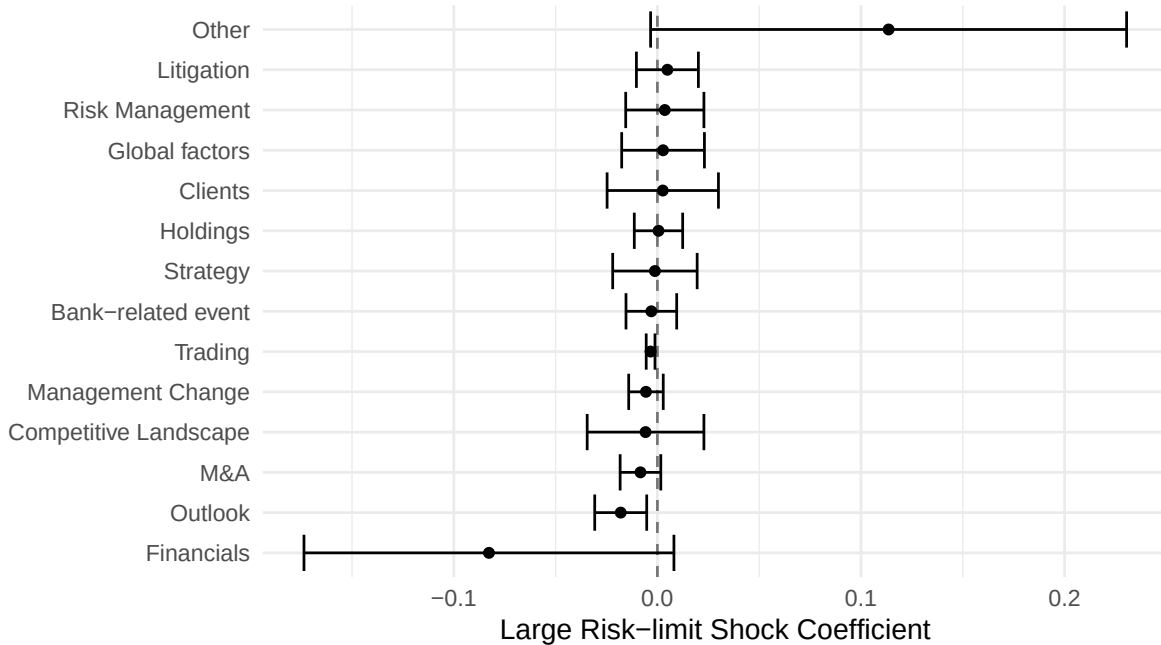
Notes: Each panel represents the variation of the currency-level risk-limit shock as defined in equation (40). The red bars represent the contribution of the 20 largest bank-quarter risk-limit shocks as defined in equation 39 to the currency-level shock variation. The green bars represent all the residual variation. *Sources:* FR VV-1, FR Y-14Q, authors' calculations.

Figure A.10: Comparison of Topic Frequency for a Large vs. Small Risk-limit Shock Sample

(a) Frequency of Topics in Earnings Calls for Large vs. Small Risk-limit Shocks

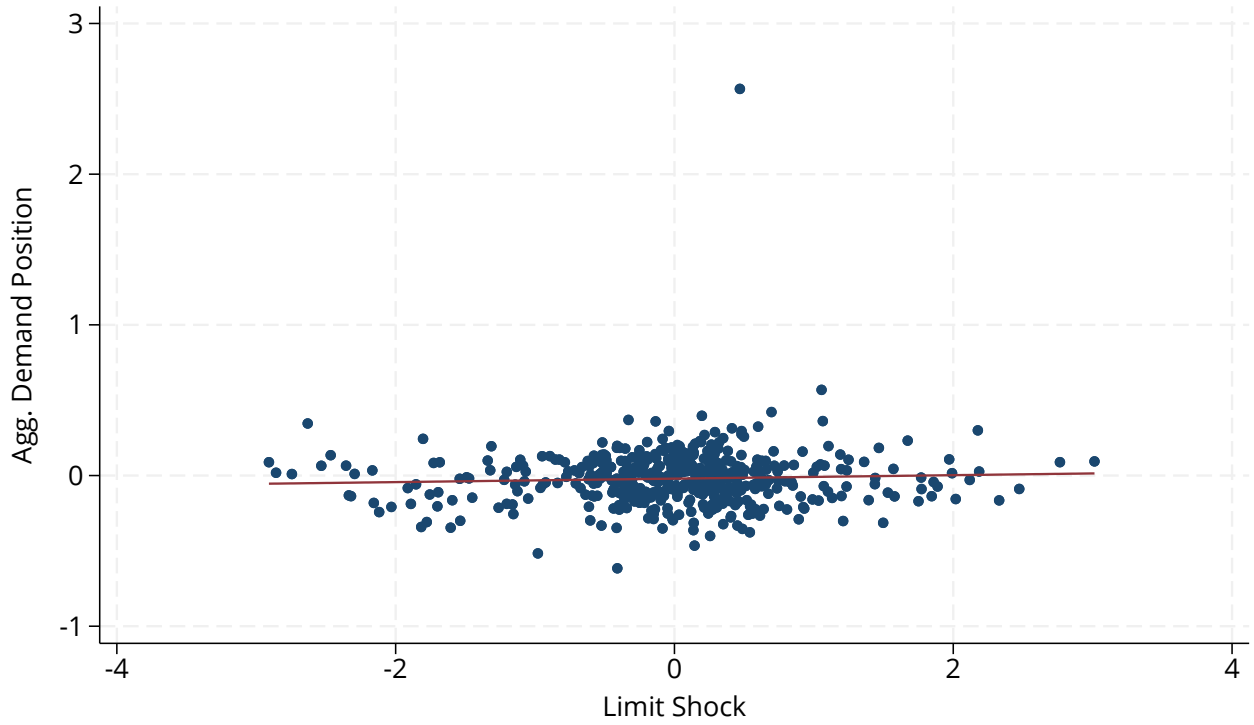


(b) Mean Testing of Topic Frequencies for Large vs. Small Risk-limit Shocks



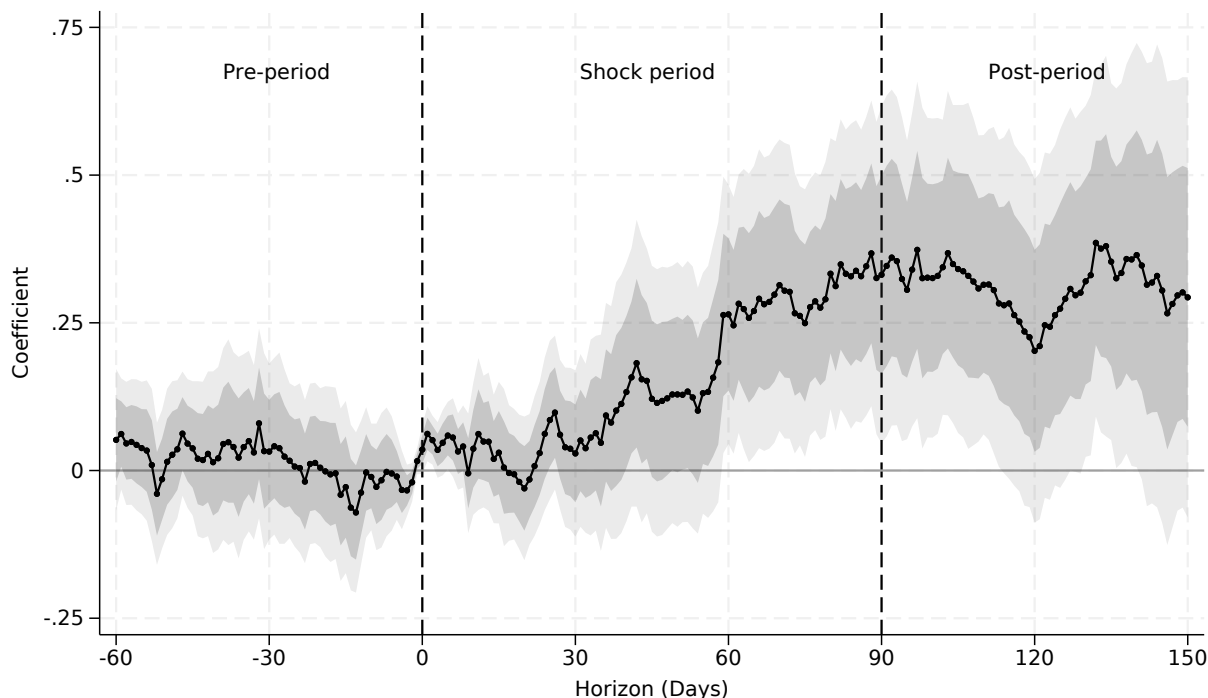
Notes: Panel (a) shows the frequency with which 14 categories of topics appear in the 11 banks' earnings calls from 2015:Q3 through 2021:Q4. The red bars represent the topic frequencies in earnings calls for bank-quarters corresponding to the top-20 risk-limit shocks as defined in equation (39). The green bars represent the topic frequencies for all other earnings calls. The clustering of each topic within the 14 main categories was implemented by a large language model, with the summarized topics in Figure A.9 provided as input. Panel (b) tests whether the difference in topic frequency between the two samples is significant. The point estimates and confidence interval for each topic category c are estimated from the following logit model: $1(\text{word} = c)_{bt} = \alpha + \beta(\text{Large Shock Dummy})_{bt} + \epsilon_{bt}$. Standard errors are clustered by bank. *Sources:* FR VV-1, FR Y-14Q, authors' calculations.

Figure A.11: Relationship between Limit Shocks over Demand Shifter



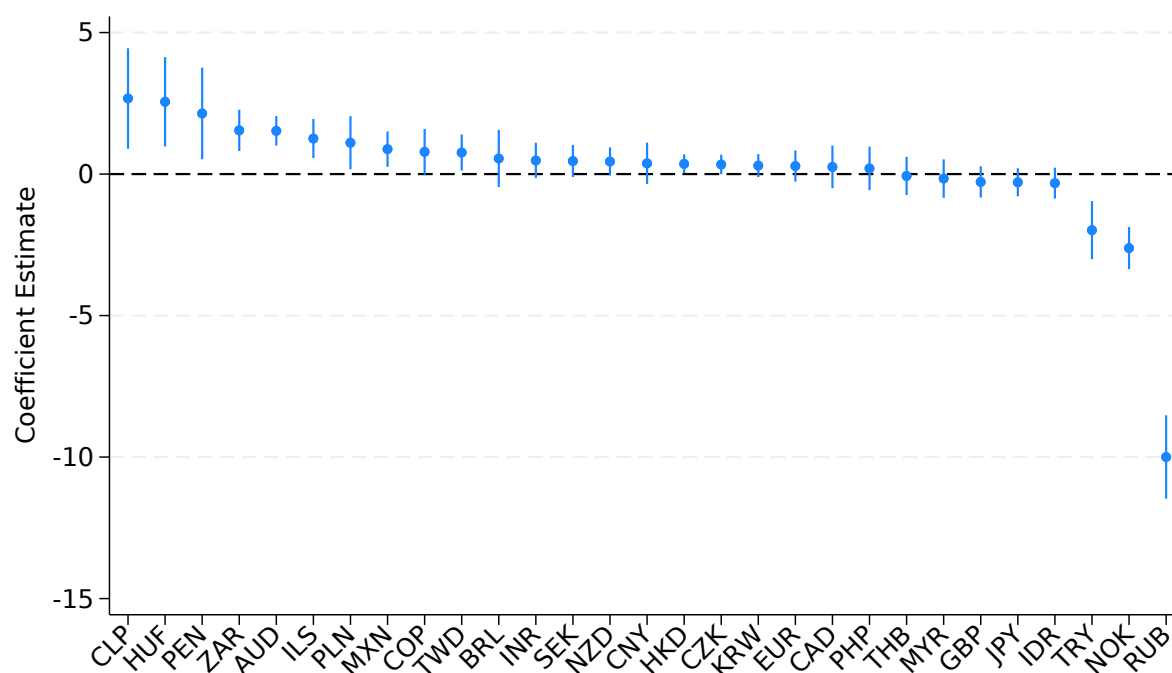
Notes: The binned scatter plot between Limit Shocks and aggregate demand position shows that the two variables are uncorrelated. The binned scatter plot absorbs date and currency fixed effects and controls for four lags of the exchange rate. *Sources:* FR VV-1, FR Y-14Q, Bloomberg, authors' calculations.

Figure A.12: Dynamic (Daily) Exchange Rate Response to Limit Shock and Aggregate Demand Shock Interaction



Notes: This figure depicts the dynamic response of the the exchange rate to the interaction of a Limit Shock and a Aggregate Demand Shock happening during quarter t . Responses are reported at several daily horizons (on the horizontal axis). Horizon 0 indicates the first day of the quarter during which the shocks occurred, assuming a 90-day quarter. Formally, we estimate equation (46) for the exchange rate change as the dependent variable using our dataset at the quarterly frequency and focus on the estimate of β , as before. However, we now shift the dependent variable (exchange rate change) from the first day of the quarter by τ days back or forward, with $\tau = -60, \dots, 150$. That is, $\tau = 90$ corresponds to the quarter end of the shock quarter, the pre-period corresponds to $\tau < 0$, and the post-period, or the period after the quarter in which we define the shock, corresponds to $\tau > 90$. Thus, the estimate reported at $\tau = 90$ corresponds to the estimate shown in Table 5, column 2 (see the caption of this table for additional information). For each τ , estimated coefficients are presented as black dots. The dark gray area represents a standard error band, and the light gray area represents the 90 percent confidence interval. Robust standard errors are clustered at the currency level. *Sources:* FR VV-1, FR Y-14Q, Bloomberg, authors' calculations.

Figure A.13: Effects of Limit Shock Interaction with CDS innovation on Exchange Rate Change, by Currency



Notes: This figure depicts currency-specific estimates of the coefficient estimate on the interaction term between Limit Shock and CDS innovations. The dependent variable is the currency return. The regression equation is $\Delta e_{c,t} = \beta_{1,c} \text{Limit Shock}_{c,t} \times \Delta \text{CDS}_{c,t} + \alpha_i + \alpha_t + \gamma' X_{c,t} + u_{c,t}$. Limit Shock is standardized by currency to have mean zero and unit variance within each currency. Point estimates are shown as dots, and the bars represent 95 percent confidence intervals. *Sources:* FR VV-1, FR Y-14Q, Bloomberg, authors' calculations.

B Additional Tables

Table B.1: US Dealers' Share in Total Gross Notional, by Currency, in Percentages

Currency/Year	2010	2013	2016	2019	2022
AUD	15.7	17.9	18.9	14.1	20.8
BRL	34.8	43.8	47.7	27.4	36.3
CAD	32.0	34.4	36.3	28.3	32.8
CHF	18.7	17.7	17.0	15.5	20.4
CNY	7.1	5.8	9.5	8.3	12.4
EUR	19.1	17.6	19.4	18.2	18.8
GBP	16.5	17.4	22.3	19.2	19.1
HUF	10.5	11.9	18.0	17.0	18.5
JPY	15.0	19.7	18.4	13.5	17.1
KRW	4.6	7.0	9.4	10.8	16.2
MXN	36.5	44.8	43.6	33.3	41.0
NOK	10.7	16.6	21.7	19.5	21.7
NZD	16.3	22.0	18.7	15.2	20.2
PLN	10.8	12.8	17.5	18.7	16.8
RUB	2.4	7.0	7.5	11.8	8.0
SEK	8.6	16.6	21.6	15.6	14.6
SGD	7.4	15.3	13.4	9.5	13.2
TRY	1.8	10.6	14.2	13.7	12.1
TWD	6.7	8.3	6.8	8.3	6.6
USD	18.2	19.2	19.2	16.6	19.6
ZAR	7.2	16.4	17.0	14.7	20.1

Note: Based on BIS Triennial FX Data. *Sources:* BIS, authors' calculations.

Table B.2: Persistence of Weights

	Weights	
	(1)	(2)
Lagged Weight	0.717*** (0.008)	0.628*** (0.009)
Currency FE	No	Yes
Time FE	No	Yes
Bank FE	No	Yes
R^2	0.512	0.538
R^2 within	0.512	0.394
N	8,049	8,049

Note: Autoregressive coefficients of the weights used in the aggregation of bank-quarter Limit Shocks to currency-quarter Limit Shocks. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. *Sources:* FR VV-1, FR Y-14Q, Bloomberg, authors' calculations.

Table B.3: Correlation between Bank Limit Shocks and Earnings Calls Text Characteristics

Dependent Variables: Model:	Text Similarity			Text Sentiment		
	(1)	(2)	(3)	(4)	(5)	(6)
<i>Variables</i>						
Raw Limit	0.1822 (0.1001)			2.168 (1.513)		
Bank Limit Shock		-0.0872 (0.1371)			-1.231 (1.123)	
Large Bank Shock Dummy			-0.1518 (0.2336)			-1.242 (1.314)
<i>Fixed-effects</i>						
Bank	Yes	Yes	Yes	Yes	Yes	Yes
Date	Yes	Yes	Yes	Yes	Yes	Yes
<i>Fit statistics</i>						
Observations	213	213	213	213	213	213
R ²	0.52450	0.50406	0.50366	0.46005	0.41947	0.41694
Within R ²	0.04340	0.00227	0.00147	0.07505	0.00553	0.00120

Notes: The table shows the correlation between measures of banks' earnings calls similarity and sentiment with our measure of risk-limit shocks. We find no significant correlation, in line with Limit Shocks being orthogonal to bank- or market-level conditions. Each earnings call has a unique similarity score representing the average similarity between the earnings call's text and all other earnings calls' texts within the same quarter. Text similarity is defined as the cosine similarity score between two texts. Text sentiment is defined as the net number of positive minus negative sentences over the total number of sentences in each earnings call. The sentiment is assigned according to FinBERT: a sentiment analysis model pre-trained on financial texts. The "Raw limit" variable contains the raw sum of maximum VaR limits of all desks for each bank quarter in the sample. The Bank Limit Shock variable is defined as in equation (39). The Large Shock Dummy is one only for the 20 largest Limit Shocks. Standard errors are two-way clustered by bank and quarter.*** p<0.01, ** p<0.05, * p<0.1. *Sources:* FR VV-1, FR Y-14Q, authors' calculations.

Table B.4: Table 5 Using Alternative Limit Shocks

	Δ FX Rate				Δ Limit	
					IV	
	(1)	(2)	(3)	(4)	2nd Stage (5)	1st Stage (6)
Limit Shock	0.374** (0.147)		0.341** (0.142)			-0.066** (0.027)
Δ Limit		-0.031 (0.262)		-0.011 (0.134)	-5.149** (2.286)	
Dep. Variable Lags	No	No	Yes	Yes	Yes	Yes
Time FE	No	No	Yes	Yes	Yes	Yes
Currency FE	No	No	Yes	Yes	Yes	Yes
R^2 within (partial)	0.007	0.000	0.007	0.000	0.007	0.039
Anderson-Rubin CI					(-11.1, -2.28)	
Eff. F-Stat						10.47
N	928	928	800	800	800	800

Notes: Same as Table 5 but using alternative Limit Shocks, where we purge three common factors (instead of one in our baseline) from desk-level limit innovations using principal component analysis. *** p<0.01, ** p<0.05, * p<0.1. *Sources:* FR VV-1, FR Y-14Q, Bloomberg, authors' calculations.

Table B.5: Table 7 using Implied Volatility as Demand Shock

	Δ FX Rate			
	(1)	(2)	(3)	(4)
Limit Shock	0.045 (0.115)	0.032 (0.096)	-0.022 (0.088)	-0.028 (0.099)
Agg. Demand Position		1.597*** (0.328)		
Limit Shock $\times$ Agg. Demand Position		0.412*** (0.062)		
Agg. Demand Shock			1.339*** (0.390)	
Limit Shock $\times$ Agg. Demand Shock			0.333*** (0.045)	
Agg. Demand Shock ^{PCA}				1.193*** (0.392)
Limit Shock $\times$ Agg. Demand Shock ^{PCA}				0.267*** (0.086)
Dep. Variable Lags	Yes	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes
Currency FE	Yes	Yes	Yes	Yes
R^2	0.470	0.529	0.514	0.514
R^2 within	0.000	0.01	0.007	0.005
N	800	800	800	800

Notes: Same as Table 7 but with use of Implied Volatility as Aggregate Demand Position/Shock. *** p<0.01, ** p<0.05, * p<0.1. *Sources:* FR VV-1, FR Y-14Q, Bloomberg, authors' calculations.

Table B.6: Effect of Limit Shocks on Interest Rate Differentials

	Δ Interest Diff. 1m	Δ Interest Diff. 1m	Δ Interest Diff. 3m	Δ Interest Diff. 3m	Δ Interest Diff. 6m	Δ Interest Diff. 6m
	(1)	(2)	(3)	(4)	(5)	(6)
Limit Shock	0.031 (0.028)	0.021 (0.025)	0.031 (0.034)	0.031 (0.025)	0.027 (0.032)	0.036 (0.026)
Dep. Variable Lags	Yes	Yes	Yes	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes	Yes	Yes
Currency FE	Yes	Yes	Yes	Yes	Yes	Yes
R ²	0.237	0.247	0.256	0.253	0.257	0.263
Within R ²	0.155	0.179	0.165	0.193	0.172	0.210
N	740	740	740	740	655	655

Notes: This table shows the effect of shocks to banks' risk limits on interest rate differentials. In columns (1) and (2), the dependent variable is the change in the interest differential between a country's one-month XIBOR rate and the US one-month LIBOR. In columns (3) and (4), the dependent variable is the change in the interest differential between a country's three-month XIBOR rate and the US three-month LIBOR. In columns (5) and (6), the dependent variable is the change in the interest differential between a country's six-month XIBOR rate and the US six-month LIBOR. The independent variable Limit Shock is constructed as described in the main text, and they are standardized to have zero mean and unit variance in each regression sample. Controls include four lags of the dependent variable. Quarter fixed effects and currency fixed effects are included. Robust standard errors are two-way clustered at the currency and quarter levels and are shown in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. *Sources:* FR VV-1, FR Y-14Q, Bloomberg, authors' calculations.

Table B.7: Response of Bank-currency-level Positions to Own Bank Limit Shocks

	$\Delta \text{ Log Abs Exposure}$		
	(1)	(2)	(3)
Bank Limit Shock	-0.072** (0.026)	-0.072*** (0.025)	-0.147*** (0.052)
Currency*Time FE	Yes	Yes	Yes
Currency*Bank FE	No	Yes	Yes
Bank*Time FE	No	No	Yes
Within R ²	0.831	0.131	0.119
N	12179	12128	12128

Notes: This table shows the response of banks' net position (absolute value) to Bank Limit Shocks. The dependent variable is the change in the logarithm of the absolute value of the delta position, at the bank-currency-quarter level. The key independent variable is Bank Limit Shock, which is the idiosyncratic limit change at the bank-quarter level. Fixed effects are included or not as indicated. The regression includes four quarterly lags of the dependent variable. Standard errors are two-way clustered at the currency and quarter levels. *** p<0.01, ** p<0.05, * p<0.1. *Sources:* FR VV-1, FR Y-14Q, Bloomberg, authors' calculations.

Table B.8: Response of Bank-currency-level Positions to Other Banks' Bank Limit Shocks

	$\Delta \text{ Log Abs Exposure}$		
	(1)	(2)	(3)
Bank Limit Shock	0.143** (0.054)	0.159*** (0.050)	0.179*** (0.064)
Currency*Time FE	Yes	Yes	Yes
Currency*Bank FE	No	Yes	Yes
Bank*Time FE	No	No	Yes
Within R ²	0.829	0.132	0.121
N	11547	11500	11500

Notes: This table shows the response of banks' net position (absolute value) to Bank Limit Shocks of other banks. The dependent variable is the change in the logarithm of the absolute value of the delta position, at the bank-currency-quarter level. The key independent variable is Bank Limit Shock, which is, as before, the aggregated idiosyncratic limit changes, but now aggregated only for other banks, excluding the bank under consideration. Fixed effects are included or not as indicated. The regression includes four quarterly lags of the dependent variable. Standard errors are two-way clustered at the currency and quarter levels. *** p<0.01, ** p<0.05, * p<0.1. *Sources:* FR VV-1, FR Y-14Q, Bloomberg, authors' calculations.

Table B.9: Response of Bank-currency-level Positions to Limit Shocks

	$\Delta \text{ Log Abs Exposure}$		
	(1)	(2)	(3)
Bank Limit Shock	-0.072** (0.026)	-0.072*** (0.025)	-0.147*** (0.052)
Currency*Time FE	Yes	Yes	Yes
Currency*Bank FE	No	Yes	Yes
Bank*Time FE	No	No	Yes
Within R ²	0.831	0.131	0.119
N	12179	12128	12128

Notes: This table shows the response of banks' net position (absolute value) to Bank Limit Shocks on the bank. The dependent variable is the change in the logarithm of the absolute value of the delta position, at the bank-currency-quarter level. The key independent variable is Bank Limit Shock, which is the idiosyncratic limit change at the bank-quarter level. Fixed effects are included or not as indicated. The regression includes four quarterly lags of the dependent variable. Standard errors are two-way clustered at the currency and quarter levels. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. *Sources:* FR VV-1, FR Y-14Q, Bloomberg, authors' calculations.

C Proofs for the main body of the paper

C.1 Derivation of Demand Equations

Households

We assume that the real sector of the economy follows the structure and notation of Gabaix and Maggiori (2015). The following model allows for high tractability – since households decision making is not the focus of this model – while allowing for intuitive extensions. US households solve the following problem:

$$\max_{\{C_{NT,t}, C_{H,t}, C_{F,t}\}_{t=0,1}} \log C_0 + \beta \mathbb{E}_0 \log C_1, \quad (53)$$

$$\text{such that: } C_t := \left[C_{NT,t}^{\chi_t} C_{H,t}^{a_t} C_{F,t}^{\iota_t} \right]^{\frac{1}{\xi_t + a_t + \iota_t}}, \quad (54)$$

$$\text{and subject to: } \sum_{t=0}^1 R^{-t} (Y_{NT,t} + P_{H,t} Y_{H,t}) = \sum_{t=0}^1 R^{-t} (C_{NT,t} + P_{H,t} C_{H,t} + P_{F,t} C_{F,t}). \quad (55)$$

Assume that the endowment process follows $Y_{NT,t} = S_t \chi_t$, where S_t is an aggregate intermediation (gross) spread charged by banks and that will be introduced in the next section with more detail. Adding the first order conditions of this problem and the market clearing for non-tradables $Y_{NT,t} = C_{NT,t}$ implies that the dollar value of US import demand follows the exogenous preference variable ι_t :

$$P_{F,t} C_{F,t} = S_t \iota_t.$$

This expression is virtually similar to the one in Gabaix and Maggiori (2015), except we explicitly consider ι_t as determining the ex-spread (mid-exchange rate) dollar expenditure by US households.⁴⁸ Similarly, the foreign household solves the following maximization problem:

$$\max_{\{C_{NT,t}^*, C_{H,t}^*, C_{F,t}^*\}_{t=0,1}} \log C_0^* + \beta \mathbb{E}_0 \log C_1^*, \quad (56)$$

$$\text{such that: } C_t^* := \left[(C_{NT,t}^*)^{\chi_t^*} (C_{H,t}^*)^{\xi_t^*} (C_{F,t}^*)^{a_t^*} \right]^{\frac{1}{\xi_t^* + a_t^* + \xi_t^*}}, \quad (57)$$

$$\text{and subject to: } \sum_{t=0}^1 R^{*-t} (Y_{NT,t}^* + P_{F,t}^* Y_{F,t}^*) = \sum_{t=0}^1 R^{*-t} (C_{NT,t}^* + P_{H,t}^* C_{H,t}^* + P_{F,t}^* C_{F,t}^*). \quad (58)$$

⁴⁸This is just a convention that does not change the qualitative findings of our model.

Assuming that the foreign endowment process follows $Y_{NT,t}^* = S_t \chi_t^*$, in combination with the first order conditions of the above problem and the market clearing for non-tradables $Y_{NT,t}^* = C_{NT,t}^*$ implies that the foreign-currency value of US exports follows the exogenous preference variable ξ_t :

$$P_{H,t}^* C_{H,t}^* = S_t \xi_t.$$

US Importer

A US importer trades with a heterogeneous set of FX dealers to obtain foreign currency assets used to buy foreign goods, which are then supplied to US households. The US importer effectively acts as a foreign currency buyer, trading with all FX dealers under a CES production function with elasticity of substitution σ and a dealer-specific quality parameter q_i . Using the duality principle (and avoiding the time subscript, for now) we can write the US importer problem as:

$$\min_{\{x_i^t\}} \int_0^1 \frac{s_i x_i^{t,*}}{\mathcal{E}} di, \quad (59)$$

$$\text{subject to: } \left[\int_0^1 (q_i x_i^{t,*})^{\frac{\sigma-1}{\sigma}} di \right]^{\frac{\sigma}{\sigma-1}} \geq P_F^* \cdot C_F. \quad (60)$$

Where:

- $x_i^{t,*}$ is the amount of foreign currency demanded from dealer i , specified in foreign currency terms.
- s_i is the (gross) spread charged by dealer i
- $s_i/\mathcal{E}$ is the effective price of the foreign currency assets, or the cum-spread exchange rate level, that is if the US importer wants to buy x_i^t in foreign currency, they will give the dealer $s_i/\mathcal{E} \cdot x_i^t$ dollars.
- q_i is the quality parameter for dealer i
- σ is the elasticity of substitution between dealers
- P_F^* is the ex-spread price of foreign goods (expressed in foreign currency term)
- C_F is the aggregate quantity of foreign goods demanded by the home country

The importer problem implies the following optimal foreign currency demand function:

$$x_i^{t,*} = q_i^{\sigma-1} \left(\frac{s_i}{S} \right)^{-\sigma} (P_F^* \cdot C_F). \quad (61)$$

With aggregate spread defined as:

$$S^{1-\sigma} = \int_0^1 q_i^{\sigma-1} s_i^{1-\sigma} di.$$

Importers act in a fully competitive markets, therefore, their total costs equal their revenues:

$$P^F = \frac{\text{Total Costs}}{C^F} = \frac{\int_0^1 \frac{s_i x_i^{\ell_i^*}}{\mathcal{E}} di}{C^F} = \frac{P_F^* \cdot S}{\mathcal{E}}. \quad (62)$$

Substitute equation (62) and the equilibrium condition from the household problem $S_t \ell_t = P_{F,t} C_{F,t}$ to obtain:

$$\boxed{x_{it}^{\ell} = \left(\frac{s_{it}}{S_t} \right)^{-\sigma} q_i^{\sigma-1} \ell_t} \quad (63)$$

where:

$$\boxed{S_t = \left(\int_0^1 q_i^{\sigma-1} s_i^{1-\sigma} di \right)^{\frac{1}{1-\sigma}}} \quad (64)$$

Foreign-country Importer

In the foreign country there is an importer that acts exactly as the US importer except in this case it is a foreign importer buying US goods in a competitive environment. The foreign country importer trades with the same dealers as the US importer and has identical CES preferences as the US importer. The equivalent demand solution is:

$$\boxed{x_{it}^{\xi} = \left(\frac{s_{it}}{S_t} \right)^{-\sigma} q_i^{\sigma-1} \frac{\xi_t}{\mathcal{E}_t}}$$

where x_{it}^{ξ} is the amount of dollars demanded from dealer i , and specified in dollar terms.

Financiers

Similarly to Gabaix and Maggiori (2015) we admit a representative financier that is separate from the dealer bank and solves the problem:

$$\max_{X^F} V = \mathbb{E}_0 \left[\frac{\mathcal{E}_0}{S_0 S_1 \mathcal{E}_1} - 1 \right] X_0^F, \quad (65)$$

such that:

$$V \geq \Gamma \Phi (X_0^F)^2 \mathcal{E}_0, \quad (66)$$

$$\text{where: } \Phi = \mathbb{E} \left[\frac{\mathcal{E}_0 - S_0 S_1 \mathcal{E}_1}{S_0 S_1 (\mathcal{E}_0 - \mathcal{E}_1)} \right]. \quad (67)$$

First, note how we assume returns R , R^* , and discount factor β equal 1. Second, note how the financier also has to pay the cum-spread exchange rate at both time 0 and time 1. At time 0, the financier is a net supplier of dollars. That is, they borrow in dollars, supply them to the rest of the world, and hold foreign currency. To sell 1 dollar to the rest of the world, they receive $\mathcal{E}_0/S_0$ back from the dealers. At time 1, they sell back their $\mathcal{E}_0/S_0$ FX holdings to the dealers, at cum-spread price $\frac{1}{S_1 \mathcal{E}_1}$. Beside the addition of the spread, we depart from Gabaix and Maggiori (2015) by adding Φ as a measure of net-to-gross efficiency of the dealers market into the dealer constraint:

$$\Pi_{\text{gross}} = \frac{\mathcal{E}_0}{\mathcal{E}_1} - 1, \quad (68)$$

$$\Pi_{\text{net}} = \frac{\mathcal{E}_0}{S_0 S_1 \mathcal{E}_1} - 1, \quad (69)$$

$$\Phi = \frac{\Pi_{\text{net}}}{\Pi_{\text{gross}}}. \quad (70)$$

Φ corrects for the spread-related inefficiency to simplify the solution.⁴⁹ With Φ we are effectively adjusting the risk aversion to only include the volatility of the mid-point exchange rate and not the cum-spread volatility.

The aggregate solution then coincides with Gabaix and Maggiori (2015) (they define $\mathcal{E}$ in dollars per units of foreign currency, as opposed to foreign currency units per dollar):

$$X_0^F = \frac{1}{\Gamma} \mathbb{E}_0 \left[\frac{1}{\mathcal{E}_1} - \frac{1}{\mathcal{E}_0} \right].$$

Assume for now a scenario of expected foreign currency appreciation, so $X_0^F \geq 0$ can be interpreted as supply of dollars in exchange of foreign currency assets. We also assume that, after deciding the aggregate quantity to invest, financiers have CES preferences over dealers i

⁴⁹If we did not adjust by Φ , spread would lower the Sharpe ratio of the aggregate financier, shrinking their positions, increasing foreign currency depreciation, and increasing spread. That is, the testable dynamics in our model would be amplified, after a limit risk shock.

identical to importers:

$$X_0^F = \left[\int (q_i x_{it}^F)^{\frac{\sigma-1}{\sigma}} di \right]^{\frac{\sigma}{\sigma-1}} \quad \text{for } t \in \{0, 1\}.$$

Note how we allow the financiers' demand for dealer i change across time, as long as the aggregate demand remains the same at X_0^F .

Financiers then solve the problem:

$$\min_{\{x_{it}^f\}} \int_0^1 s_{it} x_{it}^F di, \quad (71)$$

$$\text{subject to: } \left[\int_0^1 (q_i x_{it}^F)^{\frac{\sigma-1}{\sigma}} di \right]^{\frac{\sigma}{\sigma-1}} \leq X_0^F. \quad (72)$$

Which gives the solution:

$$x_{it}^f = \left(\frac{s_{it}}{S_t} \right)^{-\sigma} q_i^{\sigma-1} \frac{1}{\Gamma} \mathbb{E}_0 \left[\frac{1}{\mathcal{E}_1} - \frac{1}{\mathcal{E}_0} \right]$$

C.2 Proof of Proposition 1

As in the main body of the text, dealers act in a monopolistically competitive market and choose spread s_{it} and position δ_i given their profit:

$$\max_{\{s_{it}\}_{t=0,1}, \delta_i} \sum_{t=0}^1 [(s_{it} - 1)x_{it} - \frac{c_i}{2} x_{it}^2] + \mathbb{E}_0 [\mathcal{E}_0 / \mathcal{E}_1 - 1] \delta_i, \quad (73)$$

$$\text{such that: } \mathbb{E}_0 [\mathcal{E}_0 / \mathcal{E}_1 - 1] \delta_i \mathcal{E}_0 \geq \gamma_i (\mathcal{E}_0 \delta_i)^2. \quad (74)$$

Each dealer bank faces (ex-spread) demand functions for dollars and foreign currencies derived above and defined as:

$$x_{it}^\iota = \left(\frac{s_{it}}{S_t} \right)^{-\sigma} q_i^{\sigma-1} \iota_t, \quad (75)$$

$$x_{it}^\xi = \left(\frac{s_{it}}{S_t} \right)^{-\sigma} q_i^{\sigma-1} \frac{\xi_t}{\mathcal{E}_t}, \quad (76)$$

$$x_{it}^f = \left(\frac{s_{it}}{S_t} \right)^{-\sigma} q_i^{\sigma-1} \frac{1}{\Gamma} \mathbb{E}_0 \left[\frac{1}{\mathcal{E}_1} - \frac{1}{\mathcal{E}_0} \right], \quad (77)$$

$$x_{it} = x_{it}^\iota + x_{it}^\xi + x_{it}^f. \quad (78)$$

We assume that all demands are positive, that is we first provide a proof under an expected foreign currency appreciation (and therefore a trade surplus from the US standpoint). We will later relax this assumption and provide more general results.

Individual dealers market clearings are:

$$s_{i0}(x_{i0}^s - x_{i0}^b/\mathcal{E}_0 - x_{i0}^F) - \delta_i = \delta_{i0}^D \quad \forall i, \quad (79)$$

$$s_{i1}(x_{i1}^s - x_{i1}^b/\mathcal{E}_1 + x_{i1}^F) + \delta_i = \delta_{i1}^D \quad \forall i. \quad (80)$$

$$(81)$$

Finally, the aggregate dealer market clearings are:

$$\int \delta_{i0}^D di = 0, \quad (82)$$

$$\int \delta_{i1}^D di = 0. \quad (83)$$

$$(84)$$

Dealer's δ_i FOC: $\delta_i = \frac{1}{\gamma_i} \mathbb{E}_0 \left[\frac{1}{\mathcal{E}_{t+1}} - \frac{1}{\mathcal{E}_t} \right]$

Dealer's s_{it} FOC: $s_{it} = \frac{\sigma}{\sigma - 1} (1 + c_i x_{it})$

where: $x_{it} = \left(\frac{s_{it}}{S_t} \right)^{-\sigma} q_i^{\sigma-1} \Theta_t$

$\Theta_t = \iota_t + \frac{\xi_t}{\mathcal{E}_t} + \frac{1}{\Gamma} \mathbb{E}_0 \left[\frac{1}{\mathcal{E}_1} - \frac{1}{\mathcal{E}_0} \right]$

(85)

Solving for the exchange rate

Expanding the market clearing gives:

$$\int (s_{i0}(x_{i0}^s - x_{i0}^b/\mathcal{E}_0 - x_{i0}^F) - \delta_i) di = S_0(\xi_0/\mathcal{E}_0 - \iota_0 - \frac{1}{\Gamma} \mathbb{E}_0 \left[\frac{1}{\mathcal{E}_1} - \frac{1}{\mathcal{E}_0} \right]) - \overbrace{\int \frac{1}{\gamma_i} di}^{1/\gamma} \mathbb{E}_0 \left[\frac{1}{\mathcal{E}_1} - \frac{1}{\mathcal{E}_0} \right] = 0,$$

$$\int (s_{i1}(x_{i1}^s - x_{i1}^b/\mathcal{E}_1 + x_{i1}^F) + \delta_i) di = S_1(\xi_1/\mathcal{E}_1 - \iota_1 + \frac{1}{\Gamma} \mathbb{E}_0 \left[\frac{1}{\mathcal{E}_1} - \frac{1}{\mathcal{E}_0} \right]) + \int \frac{1}{\gamma_i} di \mathbb{E}_0 \left[\frac{1}{\mathcal{E}_1} - \frac{1}{\mathcal{E}_0} \right] = 0.$$

Rearranging the time-0 aggregate market clearing:

$$\frac{1}{\mathcal{E}_0} = \frac{S_0 \iota_0 + \left(\frac{1}{\gamma} + \frac{S_0}{\Gamma} \right) \mathbb{E}_0 \left[\frac{1}{\mathcal{E}_1} \right]}{S_0 \xi_0 + 1/\gamma + S_0/\Gamma}. \quad (86)$$

Rearranging the time-1 aggregate market clearing (holding for all state realizations at $t = 1$):

$$S_1 \frac{\xi_1}{\mathcal{E}_1} = S_1 \iota_1 - \left(\frac{1}{\gamma} + \frac{S_1}{\Gamma} \right) \mathbb{E}_0 \left[\frac{1}{\mathcal{E}_1} - \frac{1}{\mathcal{E}_0} \right], \quad (87)$$

$$\frac{1}{\mathcal{E}_1} = \frac{\iota_1}{\xi_1} - \frac{1}{S_1 \xi_1} \left(\frac{1}{\gamma} + \frac{S_1}{\Gamma} \right) \mathbb{E}_0 \left[\frac{1}{\mathcal{E}_1} - \frac{1}{\mathcal{E}_0} \right]. \quad (88)$$

Taking expectations of (88) delivers a fixed point solution for $\mathbb{E}_0[1/\mathcal{E}_1]$:

$$\mathbb{E}_0 \left[\frac{1}{\mathcal{E}_1} \right] = \mathbb{E}_0 \left[\frac{\iota_1}{\xi_1} \right] - \mathbb{E}_0 \left[\frac{1}{S_1 \xi_1} \left(\frac{1}{\gamma} + \frac{S_1}{\Gamma} \right) \right] \mathbb{E}_0 \left[\frac{1}{\mathcal{E}_1} - \frac{1}{\mathcal{E}_0} \right]. \quad (89)$$

Rearrange to isolate $\mathbb{E}_0[1/\mathcal{E}_1]$ on the left:

$$\left(1 + \mathbb{E}_0 \left[\frac{1}{S_1 \xi_1} \left(\frac{1}{\gamma} + \frac{S_1}{\Gamma} \right) \right] \right) \mathbb{E}_0 \left[\frac{1}{\mathcal{E}_1} \right] = \mathbb{E}_0 \left[\frac{\iota_1}{\xi_1} \right] + \mathbb{E}_0 \left[\frac{1}{S_1 \xi_1} \left(\frac{1}{\gamma} + \frac{S_1}{\Gamma} \right) \right] \frac{1}{\mathcal{E}_0}. \quad (90)$$

Substitute $\mathbb{E}_0[1/\mathcal{E}_1]$ from (90) into (86). This delivers a closed form for $1/\mathcal{E}_0$:

$$\frac{1}{\mathcal{E}_0} = \frac{\left(1 + \mathbb{E}_0 \left[\frac{1}{S_1 \xi_1} \left(\frac{1}{\gamma} + \frac{S_1}{\Gamma} \right) \right] \right) S_0 \iota_0 + \left(\frac{1}{\gamma} + \frac{S_0}{\Gamma} \right) \mathbb{E}_0 \left[\frac{\iota_1}{\xi_1} \right]}{\left(1 + \mathbb{E}_0 \left[\frac{1}{S_1 \xi_1} \left(\frac{1}{\gamma} + \frac{S_1}{\Gamma} \right) \right] \right) \left(S_0 \xi_0 + \frac{1}{\gamma} + \frac{S_0}{\Gamma} \right) - \left(\frac{1}{\gamma} + \frac{S_0}{\Gamma} \right) \mathbb{E}_0 \left[\frac{1}{S_1 \xi_1} \left(\frac{1}{\gamma} + \frac{S_1}{\Gamma} \right) \right]}. \quad (91)$$

We can further simplify this solution. Define, for notational convenience:

$$\mathcal{K} \equiv \mathbb{E}_0 \left[\frac{1}{S_1 \xi_1} \left(\frac{1}{\gamma} + \frac{S_1}{\Gamma} \right) \right], \quad \mathcal{A}_0 \equiv \left(\frac{1}{\gamma} + \frac{S_0}{\Gamma} \right). \quad (92)$$

The denominator in (91) is

$$D = (1 + \mathcal{K}) (S_0 \xi_0 + \mathcal{A}_0) - \mathcal{A}_0 \mathcal{K}. \quad (93)$$

Expand the product:

$$\begin{aligned}
D &= (S_0\xi_0 + \mathcal{A}_0) + \mathcal{K} (S_0\xi_0 + \mathcal{A}_0) - \mathcal{A}_0 \mathcal{K}, \\
&= (S_0\xi_0 + \mathcal{A}_0) + \mathcal{K}S_0\xi_0 + \mathcal{K}\mathcal{A}_0 - \mathcal{A}_0 \mathcal{K}, \\
&= (S_0\xi_0 + \mathcal{A}_0) + \mathcal{K}S_0\xi_0.
\end{aligned} \tag{94}$$

Substituting back the definitions in (92) yields the simplified denominator:

$$D = \left(S_0\xi_0 + \frac{1}{\gamma} + \frac{S_0}{\Gamma} \right) + S_0\xi_0 \mathbb{E}_0 \left[\frac{1}{S_1\xi_1} \left(\frac{1}{\gamma} + \frac{S_1}{\Gamma} \right) \right]. \tag{95}$$

Replacing the denominator in (91) by (95) gives

$$\frac{1}{\mathcal{E}_0} = \frac{\left(1 + \mathbb{E}_0 \left[\frac{1}{S_1\xi_1} \left(\frac{1}{\gamma} + \frac{S_1}{\Gamma} \right) \right] \right) S_0\iota_0 + \left(\frac{1}{\gamma} + \frac{S_0}{\Gamma} \right) \mathbb{E}_0 \left[\frac{\iota_1}{\xi_1} \right]}{\left(S_0\xi_0 + \frac{1}{\gamma} + \frac{S_0}{\Gamma} \right) + S_0\xi_0 \mathbb{E}_0 \left[\frac{1}{S_1\xi_1} \left(\frac{1}{\gamma} + \frac{S_1}{\Gamma} \right) \right]}. \tag{96}$$

Define $\tilde{\gamma}$, as (half) the harmonic mean of dealers' aggregate constraints and financiers' aggregate constraints:

$$\tilde{\gamma} \equiv \frac{1}{\frac{1}{\gamma} + \frac{S_0+S_1}{2\Gamma}}.$$

Note that, if we assume $\xi_0 = \xi_1 = S_0 = S_1 = 1$, the equation above collapses to the same as in Gabaix and Maggiori (2015):

$$\frac{1}{\mathcal{E}_0} = \frac{(1 + \tilde{\gamma})\iota_0 + \mathbb{E}_0[\iota_1]}{2 + \tilde{\gamma}}.$$

Given $1/\mathcal{E}_0$, obtain $\mathbb{E}_0[1/\mathcal{E}_1]$ from (90):

$$\mathbb{E}_0 \left[\frac{1}{\mathcal{E}_1} \right] = \frac{\mathbb{E}_0 \left[\frac{\iota_1}{\xi_1} \right] + \mathbb{E}_0 \left[\frac{1}{S_1\xi_1} \left(\frac{1}{\gamma} + \frac{S_1}{\Gamma} \right) \right] \frac{1}{\mathcal{E}_0}}{1 + \mathbb{E}_0 \left[\frac{1}{S_1\xi_1} \left(\frac{1}{\gamma} + \frac{S_1}{\Gamma} \right) \right]}. \tag{97}$$

Finally, the state-by-state expression for $1/\mathcal{E}_1$ is (88), rewritten using $\mathbb{E}_0[1/\mathcal{E}_1 - 1/\mathcal{E}_0] = \mathbb{E}_0[1/\mathcal{E}_1] - 1/\mathcal{E}_0$:

$$\frac{1}{\mathcal{E}_1} = \frac{\iota_1}{\xi_1} - \frac{1}{S_1\xi_1} \left(\frac{1}{\gamma} + \frac{S_1}{\Gamma} \right) \left(\mathbb{E}_0 \left[\frac{1}{\mathcal{E}_1} \right] - \frac{1}{\mathcal{E}_0} \right). \tag{98}$$

Final System of Equations (assuming US trade surplus, for now)

$$\frac{1}{\mathcal{E}_0} = \frac{\left(1 + \mathbb{E}_0 \left[\frac{1}{S_1 \xi_1} \left(\frac{1}{\gamma} + \frac{S_1}{\Gamma} \right) \right] \right) S_0 \iota_0 + \left(\frac{1}{\gamma} + \frac{S_0}{\Gamma} \right) \mathbb{E}_0 \left[\frac{\iota_1}{\xi_1} \right]}{\left(S_0 \xi_0 + \frac{1}{\gamma} + \frac{S_0}{\Gamma} \right) + S_0 \xi_0 \mathbb{E}_0 \left[\frac{1}{S_1 \xi_1} \left(\frac{1}{\gamma} + \frac{S_1}{\Gamma} \right) \right]} \quad (99)$$

$$\mathbb{E}_0 \left[\frac{1}{\mathcal{E}_1} \right] = \frac{\mathbb{E}_0 \left[\frac{\iota_1}{\xi_1} \right] + \mathbb{E}_0 \left[\frac{1}{S_1 \xi_1} \left(\frac{1}{\gamma} + \frac{S_1}{\Gamma} \right) \right] \frac{1}{\mathcal{E}_0}}{1 + \mathbb{E}_0 \left[\frac{1}{S_1 \xi_1} \left(\frac{1}{\gamma} + \frac{S_1}{\Gamma} \right) \right]} \quad (100)$$

$$\frac{1}{\mathcal{E}_1} = \frac{\iota_1}{\xi_1} - \frac{1}{S_1 \xi_1} \left(\frac{1}{\gamma} + \frac{S_1}{\Gamma} \right) \left(\mathbb{E}_0 \left[\frac{1}{\mathcal{E}_1} \right] - \frac{1}{\mathcal{E}_0} \right) \quad (101)$$

$$S_t^{1-\sigma} = \int_0^1 q_i^{\sigma-1} s_{it}^{1-\sigma} di \quad (102)$$

$$s_{it} = \frac{\sigma}{\sigma-1} (1 + c_i x_{it}) \quad (103)$$

$$x_{it} = \left(\frac{s_{it}}{S_t} \right)^{-\sigma} \Theta_t \quad (104)$$

$$\Theta_t = \iota_t + \frac{\xi_t}{\mathcal{E}_t} + \frac{1}{\Gamma} \mathbb{E}_0 \left[\frac{1}{\mathcal{E}_1} - \frac{1}{\mathcal{E}_0} \right] \quad (105)$$

Log-linearization:

We linearize the equilibrium around a deterministic steady state.

Steady State under Balanced Trade

$$\bar{\mathcal{E}} = \frac{\bar{\xi}}{\bar{\iota}}$$

$$\bar{s}_i = \frac{\sigma}{\sigma-1} (1 + 2c_i \bar{\iota})$$

Log-linearization of aggregate spread S_t :

Taking log deviations of x_{it} around steady state gives

$$\hat{x}_{it} = -\sigma(\hat{s}_{it} - \hat{S}_t) + \hat{\Theta}_t. \quad (106)$$

Log-linearize $s_{it} = \frac{\sigma}{\sigma-1} (1 + c_i x_{it})$.

Differentiating,

$$ds_{it} = \frac{\sigma}{\sigma-1} c_i dx_{it}.$$

Convert to log deviations:

$$\hat{s}_{it} \equiv \frac{ds_{it}}{\bar{s}_i} = \frac{\frac{\sigma}{\sigma-1} c_i \bar{x}_i}{\bar{s}_i} \hat{x}_{it} = \underbrace{\frac{c_i \bar{x}_i}{1 + c_i \bar{x}_i}}_{D_i} \hat{x}_{it}. \quad (107)$$

Substitute (106) into (107):

$$\hat{s}_{it} = D_i \left[-\sigma(\hat{s}_{it} - \hat{S}_t) + \hat{\Theta}_t \right].$$

Rearrange:

$$\hat{s}_{it} = \frac{D_i}{1 + \sigma D_i} \left(\sigma \hat{S}_t + \hat{\Theta}_t \right). \quad (108)$$

Log-linearizing the S_t aggregator gives:

$$\hat{S}_t = \int_0^1 \omega_i \hat{s}_{it} di, \quad \text{where market share } \omega_i \equiv \frac{q_i^{\sigma-1} \bar{s}_i^{1-\sigma}}{\int_0^1 q_j^{\sigma-1} \bar{s}_j^{1-\sigma} dj} \quad (109)$$

Substitute (108) into (109):

$$\hat{S}_t = \int_0^1 \omega_i \frac{D_i}{1 + \sigma D_i} di \left(\sigma \hat{S}_t + \hat{\Theta}_t \right).$$

Define

$$\psi_\sigma \equiv \int_0^1 \omega_i \frac{D_i}{1 + \sigma D_i} di. \quad (110)$$

Then

$$\hat{S}_t = \frac{\psi_\sigma}{1 - \sigma \psi_\sigma} \hat{\Theta}_t. \quad (111)$$

Log-linearize Θ_t first by differentiating it:

$$d\Theta_t = d\iota_t + d\left(\frac{\xi_t}{\mathcal{E}_t}\right) + \frac{1}{\Gamma} d\left(\mathbb{E}_0\left[\frac{1}{\mathcal{E}_1} - \frac{1}{\mathcal{E}_0}\right]\right).$$

Given that $d(\mathbb{E}_0[Z]) = \mathbb{E}_0[dZ]$, we obtain:

$$d\left(\mathbb{E}_0\left[\frac{1}{\mathcal{E}_1} - \frac{1}{\mathcal{E}_0}\right]\right) = -\frac{1}{\bar{\mathcal{E}}} \left(\mathbb{E}_0[\hat{\mathcal{E}}_1] - \hat{\mathcal{E}}_0\right).$$

Putting these pieces together:

$$d\Theta_t = \bar{\iota} \hat{\iota}_t + \frac{\bar{\xi}}{\bar{\mathcal{E}}} (\hat{\xi}_t - \hat{\mathcal{E}}_t) - \frac{1}{\Gamma \bar{\mathcal{E}}} \left(\mathbb{E}_0[\hat{\mathcal{E}}_1] - \hat{\mathcal{E}}_0\right). \quad (112)$$

Convert to log-deviations:

$$\hat{\Theta}_t = \frac{1}{2} \left[\hat{\iota}_t + \hat{\xi}_t - \hat{\mathcal{E}}_t + \frac{1}{\Gamma \bar{\xi}} \hat{\mathcal{E}}_0 - \frac{1}{\Gamma \bar{\xi}} \mathbb{E}_0[\hat{\mathcal{E}}_1] \right]. \quad (113)$$

Combine (111) with (113):

$$\hat{S}_t = \frac{\psi_\sigma}{1 - \sigma\psi_\sigma} \cdot \frac{1}{2} \left[\hat{\iota}_t + \hat{\xi}_t - \hat{\mathcal{E}}_t + \frac{1}{\Gamma \bar{\xi}} \hat{\mathcal{E}}_0 - \frac{1}{\Gamma \bar{\xi}} \mathbb{E}_0[\hat{\mathcal{E}}_1] \right], \quad t \in \{0, 1\}. \quad (114)$$

Using certainty equivalence under log-approximation

$$\boxed{\hat{S}_t = \frac{\psi_\sigma}{1 - \sigma\psi_\sigma} \cdot \frac{1}{2} \left[\hat{\iota}_t + \hat{\xi}_t - \hat{\mathcal{E}}_t + \frac{1}{\Gamma \bar{\xi}} \hat{\mathcal{E}}_0 - \frac{1}{\Gamma \bar{\xi}} \hat{\mathcal{E}}_1 \right], \quad t \in \{0, 1\}} \quad (115)$$

Log-linerization of the exchange rate $\mathcal{E}_0$

$$\frac{1}{\mathcal{E}_0} = \frac{N}{D},$$

where

$$N \equiv \left(1 + \mathbb{E}_0\left[\frac{1}{S_1 \xi_1} \left(\frac{1}{\gamma} + \frac{S_1}{\Gamma}\right)\right]\right) S_0 \iota_0 + \left(\frac{1}{\gamma} + \frac{S_0}{\Gamma}\right) \mathbb{E}_0\left[\frac{\iota_1}{\xi_1}\right], \quad (116)$$

$$D \equiv \left(1 + \mathbb{E}_0\left[\frac{1}{S_1 \xi_1} \left(\frac{1}{\gamma} + \frac{S_1}{\Gamma}\right)\right]\right) S_0 \xi_0 + \left(\frac{1}{\gamma} + \frac{S_0}{\Gamma}\right). \quad (117)$$

Hence at steady state,

$$\bar{N} = \left(1 + \overline{\mathbb{E}_0\left[\frac{1}{S_1 \xi_1} \left(\frac{1}{\gamma} + \frac{S_1}{\Gamma}\right)\right]}\right) \bar{S} \bar{\iota} + \frac{1}{\bar{\gamma}} \overline{\mathbb{E}_0\left[\frac{\iota_1}{\xi_1}\right]}, \quad (118)$$

$$\bar{D} = \left(1 + \overline{\mathbb{E}_0\left[\frac{1}{S_1 \xi_1} \left(\frac{1}{\gamma} + \frac{S_1}{\Gamma}\right)\right]}\right) \bar{S} \bar{\xi} + \frac{1}{\bar{\gamma}}. \quad (119)$$

Then:

$$\hat{\mathcal{E}}_0 = \frac{dD}{\bar{D}} - \frac{dN}{\bar{N}}$$

Compute each first-order contribution at the steady state. When doing so, remember that $\tilde{\gamma} = \frac{1}{\frac{1}{\gamma} + \frac{S_0 + S_1}{2\Gamma}}$, implying that $\frac{\partial^2 \tilde{\gamma}}{\partial S_t} = 1/\Gamma$

Differentiate (116)–(117):

$$\begin{aligned} dN = & \left(1 + \mathbb{E}_0\left[\frac{1}{S_1 \xi_1} \left(\frac{1}{\gamma} + \frac{S_1}{\Gamma}\right)\right]\right) (\iota_0 dS_0 + S_0 d\iota_0) + S_0 \iota_0 d\left(\mathbb{E}_0\left[\frac{1}{S_1 \xi_1} \left(\frac{1}{\gamma} + \frac{S_1}{\Gamma}\right)\right]\right) \\ & + \mathbb{E}_0\left[\frac{\iota_1}{\xi_1}\right] d\left(\frac{1}{\gamma} + \frac{S_0}{\Gamma}\right) + \left(\frac{1}{\gamma} + \frac{S_0}{\Gamma}\right) d\left(\mathbb{E}_0\left[\frac{\iota_1}{\xi_1}\right]\right), \end{aligned} \quad (120)$$

$$dD = \left(1 + \mathbb{E}_0\left[\frac{1}{S_1 \xi_1} \left(\frac{1}{\gamma} + \frac{S_1}{\Gamma}\right)\right]\right) (\xi_0 dS_0 + S_0 d\xi_0) + S_0 \xi_0 d\left(\mathbb{E}_0\left[\frac{1}{S_1 \xi_1} \left(\frac{1}{\gamma} + \frac{S_1}{\Gamma}\right)\right]\right) + d\left(\frac{1}{\gamma} + \frac{S_0}{\Gamma}\right). \quad (121)$$

Evaluate (120)–(121) at the symmetric steady state ($S_0 = \bar{S}$, $\xi_0 = \bar{\xi}$, $\iota_0 = \bar{\iota}$, $(1/\gamma + S_0/\Gamma) = 1/\tilde{\gamma}$, and $d(1/\gamma + S_0/\Gamma) = (1/\Gamma)dS_0$), then combine them using $\bar{N} = (\bar{\iota}/\bar{\xi})\bar{D}$ to write

$$\frac{dN}{\bar{N}} = \frac{\bar{\xi}}{\bar{\iota}} \frac{dN}{\bar{D}}.$$

All terms involving dS_0 and $d\left(\mathbb{E}_0\left[\frac{1}{S_1 \xi_1} \left(\frac{1}{\gamma} + \frac{S_1}{\Gamma}\right)\right]\right)$ cancel between $dD/\bar{D}$ and $dN/\bar{N}$, leaving

$$\hat{\mathcal{E}}_0 = \frac{\left(1 + \overline{\mathbb{E}_0\left[\frac{1}{S_1 \xi_1} \left(\frac{1}{\gamma} + \frac{S_1}{\Gamma}\right)\right]}\right) \bar{S} \bar{\xi} (\hat{\xi}_0 - \hat{\iota}_0) - \frac{1}{\tilde{\gamma}} \widehat{\mathbb{E}_0\left[\frac{\iota_1}{\xi_1}\right]}}{\left(1 + \overline{\mathbb{E}_0\left[\frac{1}{S_1 \xi_1} \left(\frac{1}{\gamma} + \frac{S_1}{\Gamma}\right)\right]}\right) \bar{S} \bar{\xi} + \frac{1}{\tilde{\gamma}}}}. \quad (122)$$

Evaluating:

$$\overline{\mathbb{E}_0\left[\frac{1}{S_1 \xi_1} \left(\frac{1}{\gamma} + \frac{S_1}{\Gamma}\right)\right]} = \frac{1}{\bar{S} \bar{\xi}} \left(\frac{1}{\gamma} + \frac{\bar{S}}{\Gamma}\right) = \frac{1}{\bar{S} \bar{\xi}} \cdot \frac{1}{\tilde{\gamma}},$$

and using the certainty equivalence approximation implied by the log linearized form:

$$\widehat{\mathbb{E}_0\left[\frac{\iota_1}{\xi_1}\right]} \approx \hat{\iota}_1 - \hat{\xi}_1,$$

the solution becomes:

$$\boxed{\hat{\mathcal{E}}_0 = \frac{(1 + \bar{S} \bar{\xi} \tilde{\gamma})(\hat{\xi}_0 - \hat{\iota}_0) + \hat{\xi}_1 - \hat{\iota}_1}{2 + \bar{S} \bar{\xi} \tilde{\gamma}}} \quad (123)$$

Relaxing the sign restriction on the solutions

We derived the solutions in (115) and (123) under US trade surplus and expected foreign currency appreciation. How do solutions change under US trade deficit?

A generalized way to rewrite the financiers' problem in equation (65) is to define the excess return as:

$$\mu^F = \mathbb{E}_0 \left[\frac{\mathcal{E}_0}{S_0 S_1 \mathcal{E}_1} - 1 \right]$$

Define now $\tilde{X}_0^F$ as the volume demand of the financier, and $\zeta \in \{-1, +1\}$ as the direction of demand. Under $\zeta = 1$ the dealer is short dollar and long foreign currency. Then the general problem becomes:

$$\max_{\tilde{X}_0^F \geq 0, \zeta \in \{-1, +1\}} V = \zeta \mu \tilde{X}_0^F \quad (124)$$

such that equations (66)-(67) hold. This guarantees that the CES volume demand for dealers' services and financiers' profits are always positive, regardless of the sign of the appreciation. The aggregate financier's solution then becomes:

$$\tilde{X}_0^F = \begin{cases} \frac{1}{\Gamma} \mathbb{E}_0 \left[\frac{1}{\mathcal{E}_0} - \frac{1}{\mathcal{E}_1} \right] & \mu^F \geq 0 \\ -\frac{1}{\Gamma} \mathbb{E}_0 \left[\frac{1}{\mathcal{E}_0} - \frac{1}{\mathcal{E}_1} \right] & \mu^F < 0 \end{cases} \quad (125)$$

The dealers' problems similarly generalizes to:

$$\max_{\{s_{it}\}_{t=0,1}, \tilde{\delta}_i \geq 0, \zeta_i \in \{-1, +1\}} \sum_{t=0}^1 [(s_{it} - 1)x_{it} - \frac{c_i}{2}x_{it}^2] + \mathbb{E}_0 [\mathcal{E}_0/\mathcal{E}_1 - 1] \zeta_i \tilde{\delta}_i \quad (126)$$

$$\text{such that: } \mathbb{E}_0 [\mathcal{E}_0/\mathcal{E}_1 - 1] \zeta_i \tilde{\delta}_i \mathcal{E}_0 \geq \gamma_i (\mathcal{E}_0 \delta_i)^2 \quad (127)$$

With the solution for $\tilde{\delta}_i$ given by:

$$\tilde{\delta}_i = \begin{cases} \frac{1}{\gamma_i} \mathbb{E}_0 \left[\frac{1}{\mathcal{E}_0} - \frac{1}{\mathcal{E}_1} \right] & \text{if } \mathbb{E}_0 [\mathcal{E}_0/\mathcal{E}_1 - 1] \geq 0 \\ -\frac{1}{\gamma_i} \mathbb{E}_0 \left[\frac{1}{\mathcal{E}_0} - \frac{1}{\mathcal{E}_1} \right] & \text{if } \mathbb{E}_0 [\mathcal{E}_0/\mathcal{E}_1 - 1] < 0 \end{cases} \quad (128)$$

Market clearing:

$$s_{i0}(x_{i0}^s - x_{i0}^b/\mathcal{E}_0 - \zeta \tilde{x}_{i0}^F) - \zeta_i \tilde{\delta}_i = \delta_{i0}^D \quad \forall i \quad (129)$$

$$s_{i1}(x_{i1}^s - x_{i1}^b/\mathcal{E}_1 + \zeta \tilde{x}_{i1}^F) + \zeta_i \tilde{\delta}_i = \delta_{i1}^D \quad \forall i \quad (130)$$

$$(131)$$

The market clearing flows now account for the signed financiers flows $\zeta \tilde{X}_0^F$ and the dealers flows $\zeta_i \tilde{\delta}_i$. Note that finding solution for $\delta_i = \zeta_i \tilde{\delta}_i$ and $X_0^F = \zeta \tilde{X}_0^F$ is equivalent as solving the generalized problem solved above. This means that the solutions for the exchange rate variable do not change.

There is only one equation that meaningfully changes under equations (125) and (128), and that is the spread equation. Because we now distinguish between volume of financiers' and dealers' intermediation from net positions, aggregate intermediation demand becomes:

$$\Theta_t = \iota_t + \xi_t/\mathcal{E}_t + \frac{1}{\Gamma} |\mathbb{E}_0 \left[\frac{1}{\mathcal{E}_1} - \frac{1}{\mathcal{E}_0} \right]| \quad (132)$$

We run a piece-wise linearization for the case $\zeta = -1$ and $\zeta = +1$ to avoid the kink in the absolute value term. All terms in the linearization of 132 remain the same as in the derivation of the log-linearized expression (113), except:

$$d \left(|\mathbb{E}_0 \left[\frac{1}{\mathcal{E}_1} - \frac{1}{\mathcal{E}_0} \right]| \right) = \zeta \left(\mathbb{E}_0 \left[\frac{1}{\mathcal{E}_1} - \frac{1}{\mathcal{E}_0} \right] \right)$$

Leading to:

$$\hat{\Theta}_t = \frac{1}{2} \left[\hat{\iota}_t + \hat{\xi}_t - \hat{\mathcal{E}}_t + \frac{\zeta}{\Gamma \bar{\xi}} \hat{\mathcal{E}}_0 - \frac{\zeta}{\Gamma \bar{\xi}} \mathbb{E}_0[\hat{\mathcal{E}}_1] \right] \quad (133)$$

And

$$\hat{S}_t = \frac{\psi_\sigma}{1 - \sigma \psi_\sigma} \cdot \frac{1}{2} \left[\hat{\iota}_t + \hat{\xi}_t - \hat{\mathcal{E}}_t + \frac{\zeta}{\Gamma \bar{\xi}} \hat{\mathcal{E}}_0 - \frac{\zeta}{\Gamma \bar{\xi}} \hat{\mathcal{E}}_1 \right], \quad t \in \{0, 1\} \quad (134)$$

Log-linearized final system of equations (assuming $\hat{\xi}_1 = \hat{\iota}_1 = \hat{\mathcal{E}}_1 = \hat{S}_1 = 0$)

$$\hat{S}_0 = \frac{\psi_\sigma}{1 - \sigma \psi_\sigma} \cdot \frac{1}{2} \cdot \left[\hat{\iota}_0 + \hat{\xi}_0 - \hat{\mathcal{E}}_0 + \frac{\zeta}{\Gamma \bar{\xi}} \hat{\mathcal{E}}_0 \right] \quad (135)$$

$$\hat{\mathcal{E}}_0 = \frac{(1 + \bar{S} \bar{\xi} \bar{\gamma})(\hat{\xi}_0 - \hat{\iota}_0)}{2 + \bar{S} \bar{\xi} \bar{\gamma}} \quad (136)$$

$$d\delta_i = \frac{1}{\gamma_i} \frac{\hat{\mathcal{E}}_0}{\bar{\mathcal{E}}} \quad (137)$$

$$d\delta = \frac{1}{\gamma} \frac{\hat{\mathcal{E}}_0}{\bar{\mathcal{E}}} \quad (138)$$

C.3 Proof of Corollaries 1 and 2

Note that $\tilde{\gamma}$ in the system of equations above is not a function of endogenous variables, but it is evaluated at the steady state. That is $\tilde{\gamma} = \frac{1}{\frac{1}{\gamma} + \frac{\bar{S}}{\Gamma}}$ and $1/\gamma = \int 1/\gamma_i di$. Therefore:

$$\frac{\partial \tilde{\gamma}}{\partial \gamma_i} = \frac{\partial (\int \gamma_i^{-1} di + (\Gamma/\bar{S})^{-1})^{-1}}{\partial \gamma_i} = (\gamma^{-1} + (\Gamma/\bar{S})^{-1})^{-2} \gamma_i^{-2} = \left(\frac{\gamma_i^{-1}}{\gamma^{-1} + (\Gamma/\bar{S})^{-1}} \right)^2 \quad (139)$$

$$\frac{\partial \hat{\mathcal{E}}}{\partial \gamma_i} = \frac{\partial \hat{\mathcal{E}}}{\partial \tilde{\gamma}} \cdot \frac{\partial \tilde{\gamma}}{\partial \gamma_i} = \frac{\bar{S}\bar{\xi}}{(2 + \bar{S}\bar{\xi}\tilde{\gamma})^2} (\hat{\xi}_0 - \hat{\iota}_0) \left(\frac{\gamma_i^{-1}}{\gamma^{-1} + (\Gamma/\bar{S})^{-1}} \right)^2 \quad (140)$$

This also implies that:

$$\frac{\partial |\hat{\mathcal{E}}_0|}{\partial \gamma_i} = \frac{\partial |\hat{\mathcal{E}}_0|}{\partial \tilde{\gamma}} \frac{\partial \tilde{\gamma}}{\partial \gamma_i} = \frac{\bar{S}\bar{\xi}}{(2 + \bar{S}\bar{\xi}\tilde{\gamma})^2} |\hat{\xi}_0 - \hat{\iota}_0| \left(\frac{\gamma_i^{-1}}{\gamma^{-1} + (\Gamma/\bar{S})^{-1}} \right)^2 \quad (141)$$

Where the last equality follows from the fact that $\hat{\xi}_0 - \hat{\iota}_0$ is an exogenous shock in this model and after applying piecewise derivation around zero.

$$\frac{\partial \delta_i}{\partial \gamma_i} = \frac{1}{\bar{\mathcal{E}}} (-1) \frac{1}{\gamma_i^2} \frac{\partial \hat{\mathcal{E}}_0}{\partial \gamma_i} = -\frac{1}{\bar{\mathcal{E}}} \frac{1}{\gamma_i^2} \frac{\bar{S}\bar{\xi}}{(2 + \bar{S}\bar{\xi}\tilde{\gamma})^2} (\hat{\xi}_0 - \hat{\iota}_0) \left(\frac{\gamma_i^{-1}}{\gamma^{-1} + (\Gamma/\bar{S})^{-1}} \right)^2 \quad (142)$$

Which also implies that:

$$\frac{\partial |\delta_i|}{\partial \gamma_i} = -\frac{1}{\bar{\mathcal{E}}} \frac{1}{\gamma_i^2} \frac{\bar{S}\bar{\xi}}{(2 + \bar{S}\bar{\xi}\tilde{\gamma})^2} |\hat{\xi}_0 - \hat{\iota}_0| \left(\frac{\gamma_i^{-1}}{\gamma^{-1} + (\Gamma/\bar{S})^{-1}} \right)^2 \quad (143)$$

Similarly:

$$\frac{\partial \hat{S}_0}{\partial \gamma_i} = \frac{\partial \hat{S}_0}{\partial \tilde{\gamma}} \cdot \frac{\partial \tilde{\gamma}}{\partial \gamma_i} \propto \left(\frac{\zeta}{\Gamma\bar{\xi}} - 1 \right) (\hat{\xi}_0 - \hat{\iota}_0) \left(\frac{\gamma_i^{-1}}{\gamma^{-1} + (\Gamma/\bar{S})^{-1}} \right)^2 \quad (144)$$

Note how $\zeta = -1$ under US trade deficit (or $\hat{\xi}_0 - \hat{\iota}_0 < 0$) and $\zeta = +1$ under US trade surplus (or $\hat{\xi}_0 - \hat{\iota}_0 > 0$). Therefore,

$$\frac{\partial \hat{S}_0}{\partial \gamma_i} \propto \begin{cases} \left(\frac{1}{\Gamma\bar{\xi}} - 1 \right) (\hat{\xi}_0 - \hat{\iota}_0) & \text{if } \hat{\xi}_0 - \hat{\iota}_0 > 0 \\ \left(-\frac{1}{\Gamma\bar{\xi}} - 1 \right) (\hat{\xi}_0 - \hat{\iota}_0) & \text{if } \hat{\xi}_0 - \hat{\iota}_0 < 0 \end{cases}$$

Therefore

$$\frac{\partial \hat{S}_0}{\partial \gamma_i} > 0 \quad \text{if} \quad \frac{1}{\Gamma \xi} - 1 > 0$$

Proof of Proposition 2

US, foreign importers, and financiers' demands are the same as in the main model derivation. Dealers now solve:

$$\max_{\{s_{it}\}_{t=0,1}, \delta_i} \sum_{t=0}^1 [(s_{it} - 1)x_{it} - \frac{c_i}{2}x_{it}^2] + \mathbb{E}_0^i [\mathcal{E}_0/\mathcal{E}_1 - 1] \delta_i \quad (145)$$

$$\text{such that:} \quad \mathbb{E}_0^i [\mathcal{E}_0/\mathcal{E}_1 - 1] \delta_i \mathcal{E}_0 \geq \gamma_i (\mathcal{E}_0 \delta_i)^2 \quad (146)$$

$$\text{where:} \quad \mathbb{E}_0^i \left[\frac{1}{\mathcal{E}_1} \right] = \mathbb{E}_0 \left[\frac{1}{\mathcal{E}_1} \right] + \nu_i \quad (147)$$

The new solution for δ_i is:

$$\delta_i = \frac{1}{\gamma_i} \left[\left(\frac{1}{\mathcal{E}_{t+1}} - \frac{1}{\mathcal{E}_t} \right) + \nu_i \right] \quad (148)$$

Note that our proposition does not necessarily rely on heterogeneous expectations as a micro foundation. What matters is that the dealers problem delivers a solution isomorphic to (148). For instance, the following mean-variance problem with additional marginal cost ν_i delivers the same solution:

$$\pi_i = \sum_{t=0}^1 [(s_{it} - 1)x_{it} - \frac{c}{2}T_{it}^2] + \mathbb{E}_0 [\mathcal{E}_0/\mathcal{E}_1 - 1] \delta_i - \frac{\gamma_i}{2} \delta_i^2 \mathcal{E}_0 - \nu_i \delta_i$$

The solution for spread determination does not change so we will focus only on the solution for the exchange rate.

Define the aggregates

$$\frac{1}{\gamma} \equiv \int_0^1 \frac{1}{\gamma_i} di, \quad \tilde{\nu} \equiv \int_0^1 \frac{\nu_i}{\gamma_i} di.$$

Integrating (148) across dealers gives

$$\int_0^1 \delta_i di = \frac{1}{\gamma} \mathbb{E}_0 \left[\frac{1}{\mathcal{E}_1} - \frac{1}{\mathcal{E}_0} \right] + \tilde{\nu}. \quad (149)$$

Expanding the market clearing gives:

$$0 = S_0 \left(\frac{\xi_0}{\mathcal{E}_0} - \iota_0 - \frac{1}{\Gamma} \mathbb{E}_0 \left[\frac{1}{\mathcal{E}_1} - \frac{1}{\mathcal{E}_0} \right] \right) - \frac{1}{\gamma} \mathbb{E}_0 \left[\frac{1}{\mathcal{E}_1} - \frac{1}{\mathcal{E}_0} \right] - \tilde{\nu}, \quad (150)$$

$$0 = S_1 \left(\frac{\xi_1}{\mathcal{E}_1} - \iota_1 + \frac{1}{\Gamma} \mathbb{E}_0 \left[\frac{1}{\mathcal{E}_1} - \frac{1}{\mathcal{E}_0} \right] \right) + \frac{1}{\gamma} \mathbb{E}_0 \left[\frac{1}{\mathcal{E}_1} - \frac{1}{\mathcal{E}_0} \right] + \tilde{\nu}. \quad (151)$$

Following similar passages as the proof of Proposition 1 we obtain the closed form solution:

$$\frac{1}{\mathcal{E}_0} = \frac{\left(1 + \mathbb{E}_0 \left[\frac{1}{S_1 \xi_1} \left(\frac{1}{\gamma} + \frac{S_1}{\Gamma} \right) \right] \right) S_0 \iota_0 + \left(\frac{1}{\gamma} + \frac{S_0}{\Gamma} \right) \mathbb{E}_0 \left[\frac{\iota_1}{\xi_1} \right] + \left(1 + \mathbb{E}_0 \left[\frac{1}{S_1 \xi_1} \left(\frac{1}{\gamma} + \frac{S_1}{\Gamma} \right) \right] - \left(\frac{1}{\gamma} + \frac{S_0}{\Gamma} \right) \mathbb{E}_0 \left[\frac{1}{S_1 \xi_1} \right] \right) \tilde{\nu}}{\left(S_0 \xi_0 + \frac{1}{\gamma} + \frac{S_0}{\Gamma} \right) + S_0 \xi_0 \mathbb{E}_0 \left[\frac{1}{S_1 \xi_1} \left(\frac{1}{\gamma} + \frac{S_1}{\Gamma} \right) \right]} \quad (152)$$

We linearize around the balanced-trade deterministic steady state $\bar{\mathcal{E}} = \bar{\xi}/\bar{\iota}$, $\bar{S}_0 = \bar{S}_1 = \bar{S}$, and $\tilde{\nu} = 0$. Proceeding exactly as in the rational-expectations proof, the only additional first-order term comes from the level wedge $d\tilde{\nu}$. The resulting log-linearized exchange rate is

$$\hat{\mathcal{E}}_0 = \frac{(1 + \bar{S}\bar{\xi}\bar{\gamma})(\hat{\xi}_0 - \hat{\iota}_0) + \hat{\xi}_1 - \hat{\iota}_1 - \bar{\gamma}\bar{\mathcal{E}}d\tilde{\nu}}{2 + \bar{S}\bar{\xi}\bar{\gamma}}, \quad \bar{\gamma} \equiv \frac{1}{\frac{1}{\gamma} + \frac{\bar{S}}{\Gamma}}. \quad (153)$$

Under the simplifying assumptions $\hat{\xi}_1 = \hat{\iota}_1 = 0$ this reduces to

$$\hat{\mathcal{E}}_0 = \frac{(1 + \bar{S}\bar{\xi}\bar{\gamma})(\hat{\xi}_0 - \hat{\iota}_0) - \bar{\gamma}\bar{\mathcal{E}}d\tilde{\nu}}{2 + \bar{S}\bar{\xi}\bar{\gamma}}. \quad (154)$$

Log-linearized final system of equations (assuming $\hat{\xi}_1 = \hat{\iota}_1 = \hat{\mathcal{E}}_1 = \hat{S}_1 = 0$)

$$\hat{S}_0 = \frac{\psi_\sigma}{1 - \sigma\psi_\sigma} \cdot \frac{1}{2} \cdot \left[\hat{\iota}_0 + \hat{\xi}_0 - \hat{\mathcal{E}}_0 + \frac{\zeta}{\Gamma\bar{\xi}}\hat{\mathcal{E}}_0 \right] \quad (155)$$

$$\hat{\mathcal{E}}_0 = \frac{(1 + \bar{S}\bar{\xi}\bar{\gamma})(\hat{\xi}_0 - \hat{\iota}_0) + \hat{\xi}_1 - \hat{\iota}_1 - \bar{\gamma}\bar{\mathcal{E}}d\tilde{\nu}}{2 + \bar{S}\bar{\xi}\bar{\gamma}} \quad (156)$$

$$d\delta_i = \frac{1}{\gamma_i} \frac{\hat{\mathcal{E}}_0}{\bar{\mathcal{E}}} + \frac{d\nu_i}{\gamma_i} \quad (157)$$

$$d\delta = \frac{\hat{\mathcal{E}}_0}{\bar{\mathcal{E}}} \int \gamma_i^{-1} di + \int \frac{d\nu_i}{\gamma_i} di = \frac{1}{\gamma} \frac{\hat{\mathcal{E}}_0}{\bar{\mathcal{E}}} + d\tilde{\nu} \quad (158)$$

Derivatives with respect to γ_i

$$\frac{\partial \tilde{\gamma}}{\partial \gamma_i} = \left(\frac{\gamma_i^{-1}}{\gamma^{-1} + \bar{S}/\Gamma} \right)^2 = \left(\frac{\tilde{\gamma}}{\gamma_i} \right)^2 \quad (159)$$

$$\frac{\partial \tilde{\nu}}{\partial \gamma_i} = \frac{\partial \int \nu_i \gamma_i^{-1} di}{\partial \gamma_i} = -\nu_i / \gamma_i^2 \quad (160)$$

$$\frac{\partial d\tilde{\nu}}{\partial \gamma_i} = \frac{\partial (\int d\nu_i / \gamma_i di)}{\partial \gamma_i} = -d\nu_i / \gamma_i^2 \quad (161)$$

For simplicity, assume no shocks at time 1:

$$\frac{\partial \hat{\mathcal{E}}_0}{\partial \gamma_i} = \frac{\partial \hat{\mathcal{E}}_0}{\partial \tilde{\gamma}} \cdot \frac{\partial \tilde{\gamma}}{\partial \gamma_i} + \frac{\partial \hat{\mathcal{E}}_0}{\partial d\tilde{\nu}} \cdot \frac{\partial d\tilde{\nu}}{\partial \gamma_i} = \frac{1}{(2 + \bar{S}\bar{\xi}\tilde{\gamma})^2} \left[\bar{S}\bar{\xi}(\hat{\xi}_0 - \hat{\iota}_0) - 2\bar{\mathcal{E}}d\tilde{\nu} \right] \left(\frac{\gamma_i^{-1}}{\gamma^{-1} + (\Gamma/\bar{S})^{-1}} \right)^2 + \frac{\tilde{\gamma}\bar{\mathcal{E}}}{2 + \bar{S}\bar{\xi}\tilde{\gamma}} \frac{1}{\gamma_i^2} d\nu_i \quad (162)$$

$$= \frac{1}{(2 + \bar{S}\bar{\xi}\tilde{\gamma})^2} \left[\left(\frac{\tilde{\gamma}}{\gamma_i} \right)^2 (\bar{S}\bar{\xi}(\hat{\xi}_0 - \hat{\iota}_0) - 2\bar{\mathcal{E}}d\tilde{\nu}) + \left(\frac{2\tilde{\gamma} + \bar{S}\bar{\xi}\tilde{\gamma}^2}{\gamma_i^2} \right) \bar{\mathcal{E}}d\nu_i \right] \quad (163)$$

D Aggregate Model

D.1 Illustration of Aggregate Dynamics

In this section, we provide intuition on the aggregate dynamics of the model in the main body of the text by collapsing dealer heterogeneity and assuming a representative monopolist dealer with $\gamma_i = \gamma$ and $c_i = c$. The dealer supplies foreign currency, in equilibrium, according to a (linearized) supply function which is virtually the same as in the benchmark model:

$$d\delta = \frac{1}{\gamma} \frac{1}{\bar{\mathcal{E}}_0} \hat{\mathcal{E}}_0. \quad (164)$$

The dealer sets a spread S_t on top of the foreign price that US households pay for the foreign goods and to financiers a la Gabaix and Maggiori (2015). US households have CES preferences between foreign and domestic goods. The financiers' problem behave as in Gabaix and Maggiori (2015), except they transact in cum-spread FX prices. Assuming no surprises relative to period-1 expectations, the linearized demand and supply changes at $t = 0$ are:

$$dX_t^\xi = \hat{\xi}_t - \sigma(\hat{\mathcal{E}}_t - \hat{S}_t) \quad (165)$$

$$dX^f = \frac{1}{\Gamma} \frac{\bar{S}_0}{\bar{\mathcal{E}}_0} (\hat{\mathcal{E}}_0 - \hat{S}_0) \quad (166)$$

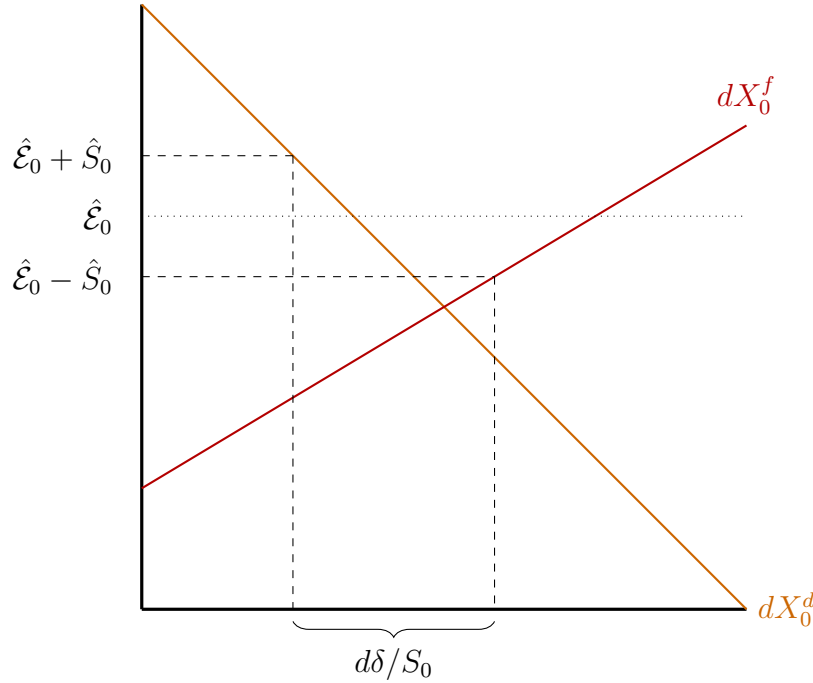
where dX denotes level changes and hats denote log deviations from the expansion point.⁵⁰

Because (as in the main model) demands are expressed in *ex-spread value* terms, the market-clearing condition at $t = 0$ is

$$dX_0^\xi - dX^f - d\left(\frac{\delta}{S_0}\right) = 0. \quad (167)$$

The last term, $d(\delta/S_0)$, is the *ex-spread (wedge-exclusive) value* of the dealer's position—the quantity of intermediation the dealer provides once the monopolist's wedge is factored out. Appendix D.2 shows that the equilibrium (aggregate) dynamics remain consistent with Propositions 1–2 of the main text.

Figure D.1: Foreign Exchange Market Equilibrium



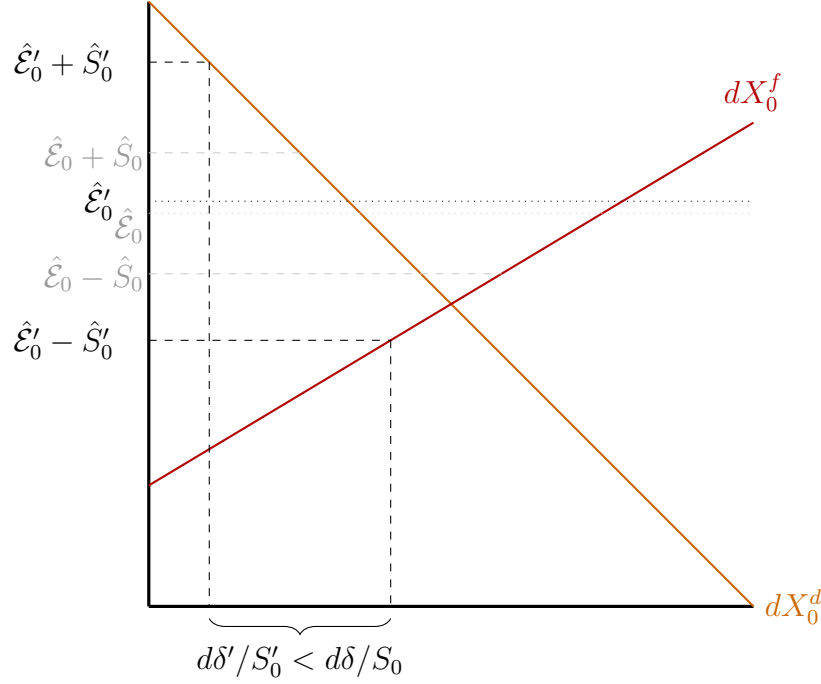
Note: Equilibrium for positive dollar demand shift in full model. Note that, in general, the equilibrium exchange rate $\hat{\mathcal{E}}$ (midpoint between bid and ask) in our model is not equal to the equilibrium exchange rates that clears $dX_0^d = dX_0^f$.

Figure D.1 illustrates the (linearized) equilibrium around a positive dollar demand shock $\bar{X}_0^\xi > 0$. In equilibrium, the mid-point exchange rate is higher than under a scenario without the dealer bank. This is because the bank adds an inefficiency in the model. Under the assumption that demand is more elastic than supply (equivalent to the $1/\Gamma > \bar{\xi}$ condition

⁵⁰For this example, we assume an expansion point different from the steady state of the model. At this expansion point, US exports, financiers demand and dealer's position are positive and there is a positive expected depreciation both in mid-price terms ($1/\bar{\mathcal{E}}_1 - 1/\bar{\mathcal{E}}_0 > 0$) and at-spread terms ($\frac{1}{\bar{S}_1\bar{\mathcal{E}}_1} - \frac{\bar{S}_0}{\bar{\mathcal{E}}_0} > 0$)

in corollary 1), this leads to the mid point exchange rate being higher than the constrained efficient equilibrium. Similarly to the financiers in Gabaix and Maggiori (2015) or Itskhoki and Mukhin (2021a), the bank also accommodates a demand growth for dollars in equilibrium by selling dollars and holding long the foreign currency position, $d\delta > 0$.

Figure D.2: Effect of Tighter Risk Limits (increase from γ to γ')



Note: Comparative statics showing the effect of tighter risk limits on the currency market equilibrium. After the increase in γ , conditional on positive dollar demand ($\bar{X}_0^d > 0$), the foreign currency depreciates more, spread increases, and banks' net position decreases, leading to an overall reduction in dollar intermediation.

We next turn to the central focus of our analysis: the effects of changes in trading desks' risk limits on the foreign exchange market. Figure D.2 illustrates the impact of an increase in γ on the previous equilibrium. When the bank is more constrained (γ increases to $\gamma' > \gamma$), the bank provides fewer dollars in equilibrium, the total supply of dollars decreases, and the foreign currency depreciates ($\hat{\mathcal{E}}'_0 > 0$). Higher flows and depreciations will also lead to an increased spread. Taking together equilibrium adjustments to the spread and exchange rate, both the buy and sell prices increase, leading to a larger supply of dollars by the financiers, yet the increase does not make up for the reduction in the bank's position, leading to an overall decline in dollar supply (movement up the demand curve).

D.2 Proofs and Derivation of Aggregate Model

Consumers Problem Assume that the static utility function of US and foreign consumers is a general CES aggregator:

$$C_t = \left(\chi_t C_{NT,t}^{\frac{\sigma-1}{\sigma}} + a_t C_{H,t}^{\frac{\sigma-1}{\sigma}} + \iota_t C_{F,t}^{\frac{\sigma-1}{\sigma}} \right)^{\frac{\sigma}{\sigma-1}} \quad (168)$$

$$C_t^* = \left(\chi_t^* C_{NT,t}^{*\frac{\sigma-1}{\sigma}} + \xi_t C_{H,t}^{*\frac{\sigma-1}{\sigma}} + a_t^* C_{F,t}^{*\frac{\sigma-1}{\sigma}} \right)^{\frac{\sigma}{\sigma-1}} \quad (169)$$

Assume that a representative monopolist charges a gross spread S_t on both US importers of foreign goods and Foreign importers of US goods.

Then the demand of US importers and foreign importers is:

$$C_{F,t} = \iota_t C_t \left(\frac{P_{F,t}^* S_t}{P_t \mathcal{E}_t} \right)^{-\sigma} \quad (170)$$

$$C_{H,t}^* = \xi_t C_t^* \left(\frac{P_{H,t} \mathcal{E}_t S_t}{P_t^*} \right)^{-\sigma} \quad (171)$$

Financiers Problem At time 0, the aggregate financier pays $S_0 X_0^f$ dollars to the dealers and receives $\mathcal{E}_0 X_0^f$ units of foreign currency. Note how X_0^f is defined in dollars, at ex-spread value. At time 1 the financier sells $\mathcal{E}_0 X_0^f$ foreign currency units to the dealer and receives $\frac{\mathcal{E}_0 X_0^f}{\mathcal{E}_1 S_1}$. Then, for every unit of (ex-spread valued) dollar X^f provided at time zero, the profit of the financier is:

$$\max_{X^f} V = \mathbb{E}_0 \left[\frac{\mathcal{E}_0}{S_1 \mathcal{E}_1} - S_0 \right] X^f = \mathbb{E}_0 \left[\underbrace{\frac{\mathcal{E}_0}{S_0 S_1 \mathcal{E}_1} - 1}_{\text{Net R per \$ inv.}} \right] \underbrace{(S_0 X^f)}_{\text{Dollars invested}} = \mathbb{E}_0 \left[\frac{\mathcal{E}_0}{S_0 S_1 \mathcal{E}_1} - 1 \right] \tilde{X}^f$$

such that:

$$V \geq \Gamma \mathcal{E}_0 (X^f)^2 \quad (172)$$

Then nominal demand and supply functions for foreign currency (and expressed in dollar

terms) are:

$$X_t^\xi = P_{H,t} C_{H,t}^* = P_{H,t} \xi_t C_t^* \left(\frac{P_{H,t} \mathcal{E}_t S_t}{P_t^*} \right)^{-\sigma} \quad (173)$$

$$X_t^\iota = \frac{1}{\mathcal{E}_t} P_{F,t}^* C_{F,t} = \frac{1}{\mathcal{E}_t} P_{F,t}^* \iota_t C_t \left(\frac{P_{F,t}^* S_t}{P_t \mathcal{E}_t} \right)^{-\sigma} \quad (174)$$

$$X^f = \frac{1}{\Gamma} \mathbb{E} \left(\frac{1}{S_1 \mathcal{E}_1} - \frac{S_0}{\mathcal{E}_0} \right) \quad (175)$$

Representative Dealer's Problem

$$\max_{S_t, \delta} \{ \pi_i = \sum_{t=0}^1 \mathbb{E}_0 [(S_t - 1)(X_t^\xi + X_t^\iota + X_t^f) - \frac{C}{2}(X_t^\xi + X_t^\iota + X_t^f)^2] + \mathbb{E}_0 [\mathcal{E}_0 / \mathcal{E}_1 - 1] \delta \} \quad (176)$$

such that:

$$\mathbb{E}_0 [\mathcal{E}_0 / \mathcal{E}_1 - 1] \delta \geq \gamma \delta^2 \mathcal{E}_0$$

Solution for S_t :

$$S_t = \frac{\sigma}{\sigma - 1} (1 + C(X_t^\xi + X_t^\iota + X_t^f)) \quad (177)$$

Solution for δ :

$$\delta = \frac{1}{\gamma} \mathbb{E} \left[\frac{1}{\mathcal{E}_1} - \frac{1}{\mathcal{E}_0} \right] \quad (178)$$

Market clearing:

$$S_0(X_0^\xi - X_0^\iota - X^f) - \frac{1}{\gamma} \left[\frac{1}{\mathcal{E}_1} - \frac{1}{\mathcal{E}_0} \right] = 0 \quad (179)$$

$$S_1(X_1^\xi - X_1^\iota + X^f) + \frac{1}{\gamma} \left[\frac{1}{\mathcal{E}_1} - \frac{1}{\mathcal{E}_0} \right] = 0 \quad (180)$$

Log-linearization

We aim at representing the model derivatives conditional a positive supply of dollars from the financier and dealers (US trade surplus). Therefore we take the expansion point for the

log-linearized solution at:

$$\bar{S}_t = \frac{\sigma}{\sigma - 1} (1 + C(\bar{X}_t^\xi + \bar{X}_t^f)) > 1, \quad \text{always} \quad (181)$$

$$\bar{X}^f = \frac{1}{\Gamma} \mathbb{E} \left(\frac{1}{\bar{S}_1 \bar{\mathcal{E}}_1} - \frac{\bar{S}_0}{\bar{\mathcal{E}}_0} \right) > 0 \quad \text{iff} \quad \bar{\mathcal{E}}_0 > \bar{S}_0 \bar{S}_1 \bar{\mathcal{E}}_1 \quad (182)$$

$$\bar{\delta} = \frac{1}{\gamma} \left(\frac{1}{\bar{\mathcal{E}}_1} - \frac{1}{\bar{\mathcal{E}}_0} \right) > 0 \quad \text{iff} \quad \bar{\mathcal{E}}_0 > \bar{\mathcal{E}}_1 \quad (183)$$

$$\bar{X}_t^\xi = \bar{\xi} \bar{P}_H \bar{C}^* (\bar{P}_H \bar{\mathcal{E}}_t \bar{S}_t / \bar{P}^*)^{-\sigma} > 0 \quad (184)$$

We assume that $\bar{\mathcal{E}}_1 < 1$ and $\bar{S}_1 > 1$ and $\bar{S}_1 \bar{\mathcal{E}}_1 = 1$ just to simplify the solution.

Log-linearization of exchange rate Log-linearize $C_{F,0}$ assuming constant $P_{F,t}^*$, and P_t , and C_t :

$$C_{F,t} \approx \bar{C}_F + \bar{C}_F \hat{c}_t - \sigma \bar{C}_F (\hat{s}_t - \hat{e}_t) \quad (185)$$

$$\hat{C}_{F,t} \approx \hat{c}_t - \sigma (\hat{s}_t - \hat{e}_t) \quad (186)$$

Similarly:

$$\hat{C}_{H,t}^* \approx \hat{c}_t^* - \sigma (\hat{s}_t + \hat{e}_t) \quad (187)$$

Define (spread included) net exports:

$$NX_t = S_t P_{H,t} C_{H,t}^* - S_t \frac{1}{\mathcal{E}_t} P_{F,t}^* C_{F,t} \quad (188)$$

$$\hat{N}X_t = \frac{1}{\bar{N}\bar{X}} \left[\underbrace{\bar{S} \bar{P}_H \bar{C}_H^*}_{\bar{\alpha}_x} (\hat{C}_{H,0}^* + \hat{S}_0) - \underbrace{\bar{P}_F^* \bar{S} / \bar{\mathcal{E}} \bar{C}_F}_{\bar{\alpha}_m} (\hat{C}_{F,0} + \hat{S}_0 - \hat{e}_0) \right] \quad (189)$$

$$\hat{N}X_t = \frac{1}{\bar{N}\bar{X}} \left[\tilde{\alpha}_x \hat{c}_t^* - \tilde{\alpha}_m \hat{c}_t - (\sigma - 1) (\tilde{\alpha}_x - \tilde{\alpha}_m) \hat{s}_t - (\tilde{\alpha}_x \sigma + \tilde{\alpha}_m (\sigma - 1)) \hat{e}_t \right] \quad (190)$$

Log-linearize ex-sread and cum-spread financier demand function:

$$\hat{X}^f = \frac{1}{\bar{X}^f} \frac{1}{\Gamma} \frac{\bar{S}}{\bar{\mathcal{E}}} (\hat{e}_0 - \hat{S}_0) \quad (191)$$

$$S_0 \hat{X}^f = \frac{1}{\bar{S} \bar{X}^f} \left[\frac{\bar{S}}{\bar{\mathcal{E}}} \frac{1}{\Gamma} \left((\bar{\mathcal{E}} - 2\bar{S}) \hat{S}_0 + \bar{S} \hat{e}_0 \right) \right] \quad (192)$$

Substitute all variables into the log-linearized market clearing equation at time zero, and assume no shocks at time 1:

$$\tilde{\alpha}_x \hat{\xi}_0 - \tilde{\alpha}_m \hat{l}_0 - (\sigma - 1)(\tilde{\alpha}_x - \tilde{\alpha}_m) \hat{S}_0 - (\tilde{\alpha}_x \sigma + \tilde{\alpha}_m (\sigma - 1)) \hat{\mathcal{E}}_0 - \left[\frac{\bar{S}}{\bar{\mathcal{E}}} \frac{1}{\bar{\Gamma}} \left((\bar{\mathcal{E}} - 2\bar{S}) \hat{S}_0 + \bar{S} \hat{\mathcal{E}}_0 \right) \right] - \frac{1}{\gamma} \frac{1}{\bar{\mathcal{E}}} \hat{\mathcal{E}}_0 = 0 \quad (193)$$

Then:

$$\hat{\mathcal{E}}_0 = \frac{\tilde{\alpha}_x \hat{\xi}_0 - \tilde{\alpha}_m \hat{l}_0 + (-(\sigma - 1)(\tilde{\alpha}_x - \tilde{\alpha}_m) + \frac{\bar{S}}{\bar{\mathcal{E}}\bar{\Gamma}}(2\bar{S} - \bar{\mathcal{E}})) \hat{S}_0}{\tilde{\alpha}_m (\sigma - 1) + \tilde{\alpha}_x \sigma + \frac{\bar{S}^2}{\bar{\mathcal{E}}\bar{\Gamma}} + \frac{1}{\bar{\mathcal{E}}\gamma}} \quad (194)$$

Log-linearization of spread

$$S_t = \frac{\sigma}{\sigma - 1} (1 + C(X_t^\xi + X_t^\iota + X_t^f)) = \frac{\sigma}{\sigma - 1} (1 + C X_t) \quad (195)$$

$$\hat{X}_t = \underbrace{\bar{P}_H \bar{C}_H^* / \bar{X}}_{\frac{\tilde{\alpha}_x}{\bar{S}\bar{X}}} (\hat{\xi}_t - \sigma(\hat{s}_t + \hat{e}_t)) + \underbrace{\bar{P}_F^* / \bar{\mathcal{E}} \bar{C}_F / \bar{X}}_{\frac{\tilde{\alpha}_m}{\bar{S}\bar{X}}} (\hat{l}_t - \sigma(\hat{s}_t - \hat{e}_t) - \hat{e}_t) + \frac{1}{\bar{X}} \frac{1}{\bar{\Gamma}} \frac{\bar{S}}{\bar{\mathcal{E}}} (\hat{e}_0 - \hat{S}_0) \quad (196)$$

$$\hat{X}_0 = \frac{1}{\bar{X}} \{ \tilde{\alpha}_x \hat{\xi}_0 + \tilde{\alpha}_m \hat{l}_0 - \sigma(\tilde{\alpha}_x + \tilde{\alpha}_m) \hat{S}_0 - (\tilde{\alpha}_x \sigma - \tilde{\alpha}_m (1 - \sigma)) \hat{e}_0 + \frac{1}{\bar{\Gamma}} \frac{\bar{S}}{\bar{\mathcal{E}}} (\hat{e}_0 - \hat{S}_0) \} \quad (197)$$

$$\hat{S}_0 = \frac{C \bar{X}}{1 + C \bar{X}} \hat{X}_0 \quad (198)$$

$$\hat{S}_0 = \underbrace{\frac{C}{1 + C \bar{X}}}_{\phi} \left[\tilde{\alpha}_x \hat{\xi}_0 + \tilde{\alpha}_m \hat{l}_0 - \sigma(\tilde{\alpha}_x + \tilde{\alpha}_m) \hat{S}_0 - (\tilde{\alpha}_x \sigma - \tilde{\alpha}_m (1 - \sigma)) \hat{e}_0 + \frac{1}{\bar{\Gamma}} \frac{\bar{S}}{\bar{\mathcal{E}}} (\hat{e}_0 - \hat{S}_0) \right] \quad (199)$$

Final system of equations:

$$\hat{S}_0 = \frac{\tilde{\alpha}_x \hat{\xi}_0 + \tilde{\alpha}_m \hat{l}_0 + (-\tilde{\alpha}_x \sigma + \tilde{\alpha}_m (1 - \sigma) + \frac{1}{\bar{\Gamma}} \frac{\bar{S}}{\bar{\mathcal{E}}}) \hat{e}_0}{1/\phi + \sigma(\tilde{\alpha}_x + \tilde{\alpha}_m) + \frac{1}{\bar{\Gamma}} \frac{\bar{S}}{\bar{\mathcal{E}}}} \quad (200)$$

$$\hat{\mathcal{E}}_0 = \frac{\tilde{\alpha}_x \hat{\xi}_0 - \tilde{\alpha}_m \hat{\iota}_0 + (-(\sigma - 1)(\tilde{\alpha}_x - \tilde{\alpha}_m) + \frac{\bar{S}}{\bar{\mathcal{E}}\Gamma}(2\bar{S} - \bar{\mathcal{E}}))\hat{S}_0}{\tilde{\alpha}_m(\sigma - 1) + \tilde{\alpha}_x\sigma + \frac{\bar{S}^2}{\bar{\mathcal{E}}\Gamma} + \frac{1}{\bar{\mathcal{E}}\gamma}} \quad (201)$$

To simplify the solution, assume $\alpha_m = 0$, as in the main text and define $\alpha_j = \tilde{\alpha}/\bar{S}_0$:

$$\hat{S}_0 = \frac{\alpha_x \hat{\xi}_0 + (-\alpha_x\sigma + \frac{1}{\Gamma}\frac{1}{\bar{\mathcal{E}}})\hat{e}_0}{\frac{1}{\bar{S}\phi} + \sigma\alpha_x + \frac{1}{\Gamma}\frac{1}{\bar{\mathcal{E}}}} \quad (202)$$

$$\hat{\mathcal{E}}_0 = \frac{\alpha_x \hat{\xi}_0 + (-(\sigma - 1)\alpha_x + \frac{1}{\bar{\mathcal{E}}\Gamma}(2\bar{S} - \bar{\mathcal{E}}))\hat{S}_0}{\alpha_x\sigma + \frac{\bar{S}}{\bar{\mathcal{E}}\Gamma} + \frac{1}{\bar{S}\bar{\mathcal{E}}\gamma}} \quad (203)$$

Note how, compared to the solution in equation (136) the exchange rate solution depends on spread movements too, on top of net flows. This is a general feature of our general solution in equation (99). However, the log-linearized heterogeneous model does not have this feature as it is evaluated around a balanced trade steady state, where spread movements have a negligible effect on the exchange rate. Second, as long as the following conditions are satisfied:

$$\alpha_x\sigma < \frac{1}{\Gamma}\frac{1}{\bar{\mathcal{E}}} \quad (204)$$

$$(\sigma - 1)\alpha_x < \frac{1}{\bar{\mathcal{E}}\Gamma}(2\bar{S} - \bar{\mathcal{E}}) \quad (205)$$

Then all derivatives signs in proposition 1 are the same as in this model.

E Narrative Analysis

E.1 Summarization via Large Language Model

To summarize the most important topics discussed by the banks in our sample we proceed as follows:

1. We retrieve all earnings calls from 2015:Q3 through 2021:Q4 of all banks in our sample. We consider the full text of the earnings calls, including speeches and Q&A sessions. We purge only the operator messages.
2. We download the open-source pre-trained 9B-parameter large language model (LLM) called Faro-Yi-9B-200K. One of the advantages of this model is its large context window, which is necessary for incorporating the entire body of the earnings calls into the prompt. For each earnings call, we provide the following prompt to the model:

```

LLM_SYSTEM_DIRECTIVE_SUMMARY = (
    "You are a financial analyst with a PhD in finance, specialized in
    the banking and trading sector, who is particularly attentive to
    major and/or unusual pieces of news related to large banks."
)
LLM_PROMPT_SUMMARY = (
    "Summarize the most important facts, decisions, or news discussed
    in the earnings call as it relates to the particular bank in the
    earnings call. Summarize in ten bullet points, ranked from most to
    least important to you. Use as few words as possible without
    sacrificing meaning. Put a bullet point title at the beginning.
    This is an example of how the output should look like: '1. *
    litigation*: the litigation with BlaBla has come to an end with
    only $1Billion losses.\n2. **fixed-income**: our fixed-income
    trading division has made abnormal profits...' and so on for all
    bullet points. Do not make up any information and be truthful to
    what is contained in the earnings call. The text of the earnings
    call is: <...>"
)

```

We manually verify the quality of the LLM output for a small sample of earnings calls. The 10 words contained between the characters ‘**’ of the LLM output are shown in the word cloud in Figure A.9.

3. The same LLM model generates the topic clusters associated with the previous topic’s summary and shown in Figure A.10. The clusters are manually selected based on the topics frequently discussed in each earnings calls. We also select the topics to ensure that any macro-related or bank-related news stands out. This is because our aim in this section is to verify whether risk limits move with institutional or global events.

For each earnings call, we provide the following prompt to the model:

```

LLM_SYSTEM_DIRECTIVE_LABEL = (
    "You are a financial analyst with a PhD in finance, specialized in
    the banking and trading sector, who has a deep understanding of the
    meaning and implication of financial jargon"
)

```

LLM_PROMPT_LABEL = (

"Categorize the words that appear in the input text under new umbrella cluster words based on their meaning. The input text is a summary, in bullet points, of what a bank has mostly talked about during an earnings call. The input text has the following repeated format: 'number *word-to-categorize*: summary of what that word implies and means in the text'. You should use both the *word-to-categorize* and the sentence that comes after to assign each word to a category. Do not make up any information and assign each of the terms between '*' according to the rules I give you. There are only 15 categories: Management Change, Global factors, Strategy, Competitive Landscape, Risk Management, Financials, M&A, Clients, Exchange Rate Market, Bank-related event, Litigation, Regulation, Trading, Holdings, Outlook. You must pick one between them, do not create one yourself. The ****categories****, and the rules for assignment are as follows:\n****Management Change****: anything related to the bank's own change of management or transition, not only for the CEO but of any kind of lower management position.\n****Global factors****: any talk of global event like COVID, economic crises, geo-political events, trade-related events, political events in foreign countries and similar.\n****Strategy****: any talk of business strategy like entering/exiting a new market, cutting costs, etc.\n****Competitive Landscape****: any talk of competitors or the position of the bank compared to competitors.

\n****Risk Management****: any talk of change in risk position of the bank, note that if the bank simply presents some risk metric, that is not an explicit change and it should instead fall into financials.\n****Financials****: any talk about standard financial metrics like earnings, dividends, ROE, revenues, etc. If, together with the financial metric, there is also an explanation of why the financial indicator has fallen or increased, then see if you can assign to another category according to the explanation.\n****M&A****: any talk of M&A of the bank in the earnings call.\n****Clients****: any talk about change in clients landscape.\n****Exchange Rate Market****:

any talk about FX, FOREX, exchange market, or exchange-related risk.\n**Bank-related event**: any event that affected the earnings call's bank specifically and that does not fall into any other category, for instance bankruptcy, very large, surprising, or sudden losses/gains, anniversaries, scandals etc.\n**Litigation**: talk of specific litigations or litigation issue.\n**Regulation**: any talk about regulatory compliance.\n**Trading**: any event or change specifically related to the trading activity of the bank, this should not include standard presentations of the trading activity of the bank, that should instead fall into financials.\n**Holdings**: any presentation of the asset portfolio of the bank, either bonds, or equities, or derivatives. Returns on holdings should go on financials but any info on composition of portfolio should go here instead.\n**Outlook**: talk about the future in a very general way, without giving information that could help categorizing the record in any other category.\n\nThe following is an example of input and output that I want (the following example input does not necessarily span the full set of categories above):
Input: '1. **CEO transition**: New CEO James takes over with John Adams and Rita Pavone on his team.\n2.

Competitive landscape: We aim at being number one, two, or three in all businesses.\n3. **Institutional Securities** (ISG): Revenues of \$3.2 billion, 39percent compensation ratio for the quarter, 40percent for the year.\n4. **Investment Banking**: Top global M&A advisor in 2009, advised four of the top 10 announced transactions.\n5. **Equity Sales and Trading**: Revenue of \$722 million, lower than expected due to reduced client activity.\n6. **Fixed Income and other sales and trading**: Revenue of \$963 million, losses of \$380 million from credit spreads.\n7. **Commodities**: Lower revenues due to lower volatility.\n8. **Value-at-Risk (VaR)**: Total VaR increased to \$187 million from \$168 million.\n9. **Global Wealth Management (GWM)**: Revenues up 4percent sequentially, net profit of \$231 million, pretax margin of 7%.\n10. **Impairment**: The bank credit conditions look benign, and they expect impairment to be low in Q1 and high in Q4.\n11. **Cost flexibility**


```

**: The bank is prepared to flex costs where appropriate to deal
with revenue pressure.\n12. **Leadership changes**: The bank
announced leadership changes to help execute their plans in the
global consumer banking and payments businesses.\n13**Funding**:
Basel III net stable funding ratio at 100%, Swiss regulatory
leverage ratio at 4.7%.\n14. **Litigation**: Litigation with RMBS
repurchase claims transferred out CHF 2 billion into litigation.'
Output: '1. **Management change**\n2. **Competitive Landscape**\n3.
**Financials**\n4. **Competitive Landscape**\n5. **Financials**\n6.
**Financials**\n7. **Global factors and events**\n8. **Risk
Management**\n9. **Financials**\n10. **Outlook**\n11. **Strategy**\n
12. **Management Change**.\n13. **Financials**\n14. **Litigation
**'. Given this information, generate the output text from the
input text I provided in the beginning. This is the text to
categorize: <...>"
)

```

Sometimes, the model does not categorize the words according to the instructions in the prompt. Given that these cases do look peculiar, we manually force such cases to fall into an “Other” category.

E.2 Sentiment and Similarity Analysis

Table B.3 contains earnings calls’ similarity and sentiment scores. To compute the similarity score between each earnings call within the same quarter, we compute the cosine similarity score between each combination of texts. To do so, we use the cosine similarity functionality of the sentence-transformer model paraphrase-MiniLM-L6-v2. We define text sentiment as the net number of positive minus negative sentences over the total number of sentences in each earnings calls. The sentence sentiment is assigned according to the FinBERT model, a sentiment analysis model pre-trained on financial texts.

